

Hecke–Eisenstein Extensions and Modular Apéry Systems

A Three-Layer Classification of Dirichlet L -Value Realizations

Version 8 — 18 August 2026

Abstract

We propose a structural classification of a broad class of modular Apéry phenomena. The basic object has only three layers. The first is a pure elliptic local system on an orientation-resolved modular cover, with basic form

$$\widetilde{\mathbb{V}}_{r,\chi} = \mathrm{Sym}^{r-1} \widetilde{\mathbb{H}} \otimes \mathbb{L}_\chi,$$

where $\mathbb{L}_\chi$ is a rank-one character line; rational quotient realizations are finite direct images of this system, and the character orientation is encoded by the resulting deck/normalizer action rather than by a naive extra rank factor. The second is a mixed Eisenstein/polylogarithmic extension of the relevant resolved or descended local system whose distinguished regulator is an opposite-parity Dirichlet value $L(r, \chi)$. The third is an *adelic* Apéry realization: a marked de Rham/cyclic section, scalar or vector companion descent, a coefficient coordinate together with a common complex continuation leaf, an arithmetic lattice, and finite-place Frobenius/free-depth data. Recurrences, continued fractions, CM identities, arithmetic-holonomy spaces, and irrationality estimates are coordinate shadows of this three-layer object.

Within this framework we prove general finite interpolation theorems for critical-period annihilation, explicit prime-power projectors, Fricke/Weyl rank formulas, affine fixed-regulator fibres, Hecke/isogeny functoriality under Eichler integration, a cohomological criterion for scalar or vector companion descent, and a pullback-defect formula separating transport of an extension from re-canonicalization of the rigid exact source after a ramified coordinate change. We further combine the quantitative arithmetic-holonomy theorem of Calegari–Dimitrov–Tang with a Frobenius/free-depth filtration of the coefficient lattice: if the j th LCM layer is free on a subspace of dimension u_j inside an m -dimensional holonomy space, its holonomy penalty is reduced by the exact quadratic amount $b_j u_j^2 / m^2$. This yields a Frobenius-filtered adelic holonomy score that replaces the earlier scalar “analytic gain minus denominator entropy” heuristic. We place the linear critical-kernel part in the context of Franke’s exact weak-function theory of Eisenstein series with prescribed critical-value zeros: our barycentric oldform projectors become explicit Hecke-generated slices of broader critical fibres, while the principal-character regulator sector motivates a boundary-completed extension of that picture. We also isolate a Tate/regulator dichotomy: in the complementary parity the distinguished special-value component is already Tate, and the extension layer loses the rigidity it has on the regulator side. A universal central Tate projector gives explicit scalar controls for $\zeta(2m)$, including level-16 and level-64 realizations of $\zeta(2)$ and $\zeta(4)$. The theory naturally separates character-orientation monodromy, higher symmetric-power extension complexity, and direct-image rank created when a quotient coordinate forgets modular/isogeny sheets.

We then assemble an applications portfolio. The framework recovers the Beukers $\zeta(3)$ source; organizes the canonical modular Zagier sporadic companions into Eisenstein extensions; explains why several Ramanujan–Machine identities lie in common modular/normal-function orbits; yields finite uniqueness and no-go theorems; produces prime-isogeny completion families for $\zeta(3)$; proves intrinsic rank-two orientation monodromy in the quintic Catalan system; and, in the project calculations motivating this paper, predicts a hidden level-12 $\zeta(7)$ parent whose 2- and 3-isogeny completions recover the level-24 system and produce a level-36 system. Franke’s level-25 weight-four critical-null form also supplies explicit published deformation directions for the η extension. A complete level-100 boundary test

carried out in the programme shows that this apparent freedom collapses after symmetry and cusp data are restored: on the three-dimensional oldspace the coarse scalar critical kernel has dimension 3, the orientation-resolved critical kernel is the unique line $(1, -10, 16)$, and the full boundary kernel is zero. The surviving critical-null vector is W_{100} -anti-Fricke but intrinsically two-orientation, and the Franke and η residues lie on disjoint cusp orbits, so the fixed-regulator, fixed-boundary fibre meets the Franke deformation family only at the original η point. The $L(3, \chi_5)$ application closes the conjugate-fold geometry of the original η sporadic family exactly: its two branch constants are $\Lambda/2 \pm 2i\pi^3/(125\sqrt{5})$, and a centered Hadamard realization cancels the non-Tate cross branch while leaving only Tate side monodromy. An exact cubic determinant then proves that no scalar combination of the natural quadratic tensor rows can cancel all three nearest branch orbits, forcing a genuinely larger vector/holonomic realization. A higher-rank nontrivial-character stress test at $\beta(4) = L(4, \chi_{-4})$ yields a unique minimal level-24 two-orientation source with target $\beta(4)/6$, an independent Sym^3 isogeny companion, a rational Galois-trace Apéry pair, and a degree-four modular direct image whose natural rational Picard–Fuchs rank is 16. This example corrects a naive “symmetric-power rank times orientation rank” rule and makes direct image part of Layer A. A new external stress test comes from Shvets’s Belyi-pullback and symmetric-cube work. The eleven integral Belyi rows compress to six algebraic finite-monodromy objects and five elliptic objects on only two Gauss progenitors; one row is exactly a quadratic pullback of A036917 and another is only a contiguous de Rham section of the Legendre system. At Layer B, the Legendre mother has canonical source $(1 - V_2)(1 - 16V_2)E_4$, exactly the programme’s #11 $\zeta(3)$ parent, whose 3-isogeny completion is the Domb source. Shvets’s neighboring signature-three symmetric cube has canonical source $E_{5, \chi_{-3}, 1}$ and therefore regulator $L(4, \chi_{-3})$; the Hecke/Fricke machinery then produces a fully purified level-24 orientation doublet and a rational depth-four Apéry realization with target $7L(4, \chi_{-3})/32$. Shvets independently proves a mod- p^4 supercongruence for the same homogeneous symmetric-cube coefficients, giving a clean finite-place check of the adelic Layer-C organization. As an external arithmetic benchmark, the $L(2, \chi_{-3})$ case identifies the two Eisenstein orientations of the classification with the exact modular sources used by Calegari–Dimitrov–Tang in their proof of the $\mathbb{Q}$ -linear independence of $1, \zeta(2), L(2, \chi_{-3})$. No new irrationality theorem is claimed. The purpose is instead to identify the objects that modular Apéry constructions are classifying and to show that the resulting theory explains, predicts, and obstructs concrete examples.

Version and status

This is an eighth foundational draft intended to organize several completed calculations and working papers into one conceptual theory. It deliberately distinguishes three levels of evidence.

- **LITERATURE**: established results from the theory of Eisenstein series, elliptic polylogarithms, Eisenstein symbols, modular curves, and independently published proofs of specific Ramanujan-Machine identities.
- **PROGRAMME THEOREM/CERTIFICATE**: exact algebraic identities, modular/Sturm certificates, finite-rank classifications, and explicit constructions established in the accompanying project papers and certificates. These results require a normal publication audit before novelty claims are made.
- **OPEN**: conjectural generalizations, classification problems, or Diophantine applications not yet proved.

The paper now incorporates Franke’s published exact theory of Eisenstein series with prescribed vanishing critical values. We explicitly credit that work for the broad nonprincipal-character critical-kernel/Vandermonde layer; our barycentric oldform theorem is presented as a closed-form Hecke slice of that broader theory. The proposed boundary-completed Franke sequence is **OPEN**, and is not attributed to Franke. The level-100 Franke– η worked example is a programme calculation based on the exact cusp-constant, character, and normalizer formulas

recorded in the accompanying research note: it proves, within the displayed three-oldform deformation space, the strict kernel dimensions $3 \rightarrow 1 \rightarrow 0$ and the disjoint-support boundary rigidity statement. These calculations are not attributed to Franke.

The adelic Layer-C synthesis uses two distinct evidence levels. The quantitative arithmetic-holonomy inequality and its denominator-array penalty are [LITERATURE](#), taken from Calegari–Dimitrov–Tang. The Frobenius/free-depth filtration is a new organizational definition, and Theorem 10.2 is a formal corollary once an all- n free-depth flag satisfying the CDT denominator hypotheses has been proved. The existence of syntomic Eisenstein realizations is [LITERATURE](#); the explicit identification of the off-diagonal Frobenius extension coordinate with normalized p -adic Dirichlet L -values, and the full p -adic Hecke–Eisenstein functoriality statement, are [OPEN](#). No such open p -adic identification is used in the proved archimedean applications. The Shvets cross-pollination section combines two evidence levels. Shvets’s eleven-row Belyi scan, Sym^2 pullback framework, signature-three symmetric-cube modular dictionary, and $\text{mod-}p^4$ supercongruence are [LITERATURE](#). The compression of the eleven rows into six algebraic and five elliptic realizations, the exact identification of the Legendre mother with the programme’s #11 $\zeta(3)$ parent, the pullback-defect interpretation, the fully purified level-24 $L(4, \chi_{-3})$ doublet, and the rational trace Apéry pair are [PROGRAMME THEOREM/CERTIFICATE](#) calculations based on the accompanying Shvets treatment. The strongest common-cover/isogeny identification among all three signature-three Belyi presentations and a publication audit of the remaining cusp-phase calculation are not silently assumed.

The paper does *not* claim irrationality of Catalan’s constant, $\zeta(5)$, $\zeta(7)$, $L(3, \chi_5)$, or any other individual special value whose irrationality is currently open. The $\beta(4)$ section incorporates a programme research dossier whose source, Fricke, companion, arithmetic, and direct-image statements carry explicit evidence labels; its degree-four elimination and rank-16 cyclicity remain certificate-level until the indicated modular-function proof is written. The $L(3, \chi_5)$ section incorporates a programme dossier in which the half-translation, elliptic fold constants, centered Hadamard branch matrix, three-branch determinant obstruction, Wronskian, and singularity-preserving minimality are exact; full irreducibility of the order-six Hadamard module and the optimal prime-by-prime denominator lattice remain open. The $L(2, \chi_{-3})$ section uses the independently proved Calegari–Dimitrov–Tang theorem only as an external benchmark; our contribution there is the exact source-level identification and classification-theoretic interpretation. Its principal claim is structural: a substantial class of modular Apéry phenomena admits a common three-layer description, and this description already has nontrivial explanatory, predictive, and obstruction-theoretic applications.

Contents

Version and status	2
1 Introduction	6
1.1 Main structural results	9
1.2 Applications as a test of utility	9
2 The three-layer object	10
2.1 Layer A: the pure local system and its direct images	10
2.2 Layer B: the mixed extension	11
2.3 Layer C: the adelic Apéry realization	12
2.4 The conceptual pipeline	12
3 Parity dichotomy and opposite-parity Eisenstein extensions	13
3.1 The parity checkerboard	13

3.2	Two primitive orientations	13
3.3	Weyl orientation	14
4	Hecke projectors, interpolation, and Fricke folding	14
4.1	Oldforms as finite Hecke correspondences	14
4.2	Prime-power projectors	15
4.3	Depth and isogeny functoriality	16
4.4	Mellin-degree shifts by differential conjugation	16
4.5	A paired-character Fricke count: the $\beta(4)$ model	17
4.6	Fricke folding	17
4.7	The minimal Fricke sign switch	18
5	Franke critical kernels and Hecke–Eisenstein purification	18
5.1	Franke’s critical-coordinate exact sequence	18
5.2	Critical coordinates are diagonal for degeneracy	19
5.3	Critical fibres rather than period-killing recipes	19
5.4	Where the regulator sector begins: boundary completion	20
5.5	Critical-invisible directions and affine extension geometry	21
5.6	A published deformation direction for the $L(3, \chi_5)$ extension	21
5.7	Hierarchy after the Franke synthesis	22
6	Worked example: the level-100 Franke–η boundary test	22
6.1	Diamond decomposition of Franke’s form	22
6.2	The unique orientation-resolved critical-null line	23
6.3	The complete level-100 cusp map	24
6.4	Boundary support separation from the η source	24
6.5	Normalizer action and vector necessity	25
6.6	What the worked example adds to the theory	26
7	The Tate side: central projectors and even-zeta controls	26
7.1	The Tate-side collapse	26
7.2	A universal central Tate projector	27
7.3	$\zeta(2)$: the level-16 control	27
7.4	$\zeta(4)$: a Fricke obstruction and the weight-8 cure	28
7.5	What the controls teach the three-layer theory	29
8	Affine extension geometry and boundary purification	29
8.1	Critical fibres and regulator fibres	29
8.2	Fixed-regulator fibres	30
8.3	The boundary map	31
8.4	Critical–orientation–boundary kernel filtration	31
8.5	Completion cubes	32
8.6	Cusp boundary versus realization-fold functionals	32
9	Companion descent, exactness, and vector-valued realizations	32
9.1	Companion descent as a cohomological kernel	32
9.2	The rigid exactness locus	33
9.3	Marked de Rham sections and scalar presentations	34
9.4	Extension transport versus coordinate re-canonicalization	34
9.5	Holomorphic and meromorphic exact-source strata	35
9.6	Purification depends on the arithmetic task	35
9.7	Centered tensor reduction and scalarization obstruction	35

9.8	Scalar versus vector descent	36
9.9	Orientation multiplicity versus direct-image rank	36
9.10	Sources of non-scalar complexity	37
9.11	Vectoriality can lower arithmetic cost	37
10	Adelic Layer C: arithmetic holonomy and Frobenius filtration	37
10.1	The arithmetic-holonomy penalty	38
10.2	Frobenius-free depth	38
10.3	The Cooper weight drop as a first arithmetic check	40
10.4	The Shvets symmetric cube as an independent depth-four check	40
10.5	Local Frobenius of the extension	40
10.6	Isogeny and almost-everywhere Frobenius	41
10.7	Adelic Apéry realization	41
10.8	Known p -adic Apéry phenomena	42
11	Applications portfolio	42
11.1	Application A: Beukers and the prototype $\zeta(3)$ projector	42
11.2	Application B: the modular Zagier sporadic universe	42
11.3	Application C: a finite $\zeta(3)$ extension classification	43
11.4	Application D: Ramanujan-Machine identities	43
11.5	Application E: new prime-isogeny $\zeta(3)$ completions	43
11.6	Application F: Catalan and intrinsic vector-valuedness	44
11.7	Application G: $\zeta(5)$ as the first multi-completion problem	44
11.8	Application H: the hidden $\zeta(7)$ parent	44
11.9	Application I: Tate controls $\zeta(2)$ and $\zeta(4)$	45
11.10	Application J: the known-irrational $L(2, \chi_{-3})$ benchmark	45
11.11	Application K: $\beta(4)$, the first higher-rank character stress test	45
11.12	Application L: $L(3, \chi_5)$ and exact conjugate-fold geometry	46
11.13	Application M: the level-100 Franke- η boundary test	46
11.14	Application N: adelic Layer C and the Cooper weight drop	46
11.15	Application O: Shvets scan compression and a new $L(4, \chi_{-3})$ realization	47
11.16	Application P: no-go theorems	47
11.17	What this portfolio demonstrates	48
12	A known-irrational regulator benchmark: $L(2, \chi_{-3})$	48
12.1	Primitive orientations and Zagier sources	48
12.2	A hidden fully purified source	49
12.3	Exactness on the level-12 elliptic host	49
12.4	CDT use exactly the two predicted orientations	50
12.5	The arithmetic-holonomy realization	50
12.6	Task-dependent purification	51
12.7	Implication for Catalan	51
13	$\beta(4)$ at level 24: higher symmetric power, character orientation, and direct image	51
13.1	Prediction and the minimal purified source	51
13.2	Level-24 minimality	52
13.3	The exact orientation doublet and pure CM value	53
13.4	Independent Sym^3 companion: exactness is not universal	53
13.5	Rational trace and arithmetic realization	54
13.6	Why the rational Picard–Fuchs rank is 16	54
13.7	Lessons for the general classification	55

14	$L(3, \chi_5)$: elliptic folds, centered tensor descent, and a scalar no-go theorem	56
14.1	Half-translation to the level-five host	56
14.2	An exact elliptic fold/cusp formula	56
14.3	Exact branch constants	57
14.4	Centered Hadamard normal form	58
14.5	Exact three-branch scalarization obstruction	58
14.6	Order-six module and arithmetic lattice	59
14.7	The level-100 Franke deformation test	60
14.8	What the example changes in the general theory	60
15	Shvets Belyi pullbacks and symmetric-cube cross-pollination	61
15.1	Layer-A compression of the eleven-row scan	61
15.2	The Legendre mother is exactly the project $\zeta(3)$ parent	62
15.3	Nonlinear pullback: transported extension versus canonical coordinate source	63
15.4	A meromorphic signature-three square source	63
15.5	A symmetric-cube regulator: $L(4, \chi_{-3})$	64
15.6	A new fully purified level-24 doublet	64
15.7	A rational vector-descended Apéry realization	65
15.8	An adelic validation: Sym^3 , depth four, and a mod- p^4 congruence	66
15.9	What the Shvets cross-pollination teaches the general theory	67
16	The parity checkerboard and a periodic table of tests	67
16.1	Tate versus regulator parity	67
16.2	A first periodic table	68
16.3	The controls are now part of the theorem	68
16.4	Regulator-side benchmark and next character tests	68
16.5	Each square now has an adelic Layer-C profile	69
17	Research programme and publication architecture	69
17.1	What is essentially linear and what remains geometric	69
17.2	Next additions after Version 8	70
17.3	Suggested paper series	71
17.4	Diophantine merit is an adelic Layer-C invariant	71
A	Further proof details and useful formulas	72
A.1	The reflected Dirichlet target	72
A.2	The completion cube as a Koszul face diagram	72
A.3	Fricke reciprocal polynomials	72
A.4	Integrality under mirror-map composition	72
B	A compact dictionary	72

1 Introduction

Apéry’s proof of the irrationality of $\zeta(3)$ made a recurrence famous. In the modular interpretation of Beukers, however, the recurrence is not the most primitive object. One has an elliptic period, an Eisenstein modular form, an Eichler integral carrying $\zeta(3)$, and a modular coordinate in which the resulting connection becomes an arithmetic recurrence. This suggests reversing the usual order of thought:

special value $\longrightarrow$ Eisenstein extension $\longrightarrow$ modular geometry $\longrightarrow$ recurrence.

The same reversal repeatedly appears in the examples motivating this paper. Catalan’s constant $G = L(2, \chi_{-4})$ is carried by a weight-three character Eisenstein normal function; the cubic Zagier sporadic systems have weight-four Eisenstein companions; $\zeta(5)$ naturally couples to a rank-five symmetric fourth power; and $\zeta(7)$ to a symmetric sixth power. Degeneracy maps $\tau \mapsto d\tau$ act simultaneously on the modular source, the elliptic geometry, the Eichler primitive, and the target L -value. Fricke and Atkin–Lehner operations exchange or diagonalize character orientations. Cusp-period annihilation becomes finite interpolation. Some extensions admit scalar companions; others are intrinsically vector-valued. Tensor/Hadamard realizations can also expose a more subtle phenomenon: after centering by the regulator, non-Tate branch data may collapse to Tate data even when an exact local determinant proves that complete scalar branch cancellation is impossible.

These phenomena point to a simpler classification than a list of recurrences. We propose that the fundamental datum consists of three layers:

$$\boxed{\text{pure local system } \mathbb{V} \longrightarrow \text{mixed extension } [\nu] \longrightarrow \text{Apéry realization } \mathcal{R}.} \quad (1)$$

For the opposite-parity Dirichlet values considered here, the pure layer is most naturally defined first on an orientation-resolved modular cover $\tilde{Y}$:

$$\tilde{\mathbb{V}}_{r,\chi} = \text{Sym}^{r-1} \tilde{\mathbb{H}} \otimes \mathbb{L}_\chi,$$

where $\tilde{\mathbb{H}}$ is the rank-two elliptic Gauss–Manin system and $\mathbb{L}_\chi$ is a rank-one character line. If an Apéry coordinate lives on a quotient $f: \tilde{Y} \rightarrow Y$, the rational local system in that coordinate is the direct image $f_* \tilde{\mathbb{V}}_{r,\chi}$. The finite character/orientation representation is encoded in this deck/normalizer descent data and must not, in general, be multiplied in as an independent rank factor. The extension layer is an Eisenstein/polylogarithmic class with regulator proportional to $L(r, \chi)$. The realization layer is adelic: it consists of a marked de Rham/cyclic section, companion descent, a coefficient coordinate and common complex continuation leaf, an arithmetic lattice, and finite-place Frobenius/free-depth data.

This formulation has several advantages.

First, it gives isogeny the correct role. Isogenies are not an optional decoration, but neither is an isogeny class a complete invariant. Isogenies act naturally on $\mathbb{H}$ and hence on every $\text{Sym}^{r-1} \mathbb{H}$; degeneracy/Hecke correspondences transport Eisenstein classes. But the same pure elliptic local system may support several inequivalent extensions with different regulator normalizations. Thus isogeny organizes the homogeneous skeleton, while the extension carries the special value.

Second, the construction naturally admits vector-valued and finite-direct-image situations. For nontrivial characters, Fricke/Weyl can exchange two Eisenstein orientations. In the quintic Catalan example the resulting orientation representation is an irreducible two-dimensional dihedral representation, so no global scalar normal function exists. In higher odd-zeta problems, vectoriality can instead arise from the size of $\text{Sym}^{r-1} \mathbb{H}$ and from the need to cancel several extension directions simultaneously. The $\beta(4)$ stress test adds a third phenomenon: a rational quotient coordinate can forget several modular/isogeny sheets, so that the descended Picard–Fuchs rank is the rank of a finite direct image rather than a product of a symmetric-power rank with a separately counted orientation rank.

Third, much of the source problem becomes exact finite linear algebra, but the level-100 Franke– η test shows that this linear stage itself has a necessary filtration. Scalar critical vanishing may come from cancellation between distinct diamond/orientation sectors; after resolving those sectors the critical kernel can shrink, and imposing the full cusp boundary can shrink it again. In the worked example the dimensions are exactly $3 \rightarrow 1 \rightarrow 0$. Thus one must resolve representation data before interpreting a critical kernel as an extension deformation. This statement now has a sharper literature context. Franke gives a general exact description of broad nonprincipal-character Eisenstein subspaces with prescribed critical L -value vanishing, including computable

weak-function kernels and Vandermonde dimension formulas [13]. Our oldform interpolation should therefore be viewed as an explicit Hecke-generated slice of a larger critical fibre. The distinctive problem studied here begins after that linear step: boundary completion for principal-character regulator sectors, isogeny/Fricke organization of the fibre, companion descent, branch geometry, and arithmetic realization.

A fourth structural advantage is exposed by Shvets's Belyi-pullback scan. A nonlinear algebraic pullback can preserve the pure differential local system and transport an existing normal-function extension while changing the rigid source obtained by declaring the new coordinate canonical. If $y = R(x)$, then the transported and re-canonicalized exact sources differ by the ramification factor $R'(x)$. Ramification over an ordinary point can therefore force the new coordinate source to become meromorphic even though the transported extension remains regular. The same external corpus shows that a scalar recurrence presentation also depends on a marked de Rham section: a contiguous derivative row can be a new recurrence without being a new Layer-A motive. Thus Layer C must remember both the cyclic section and the ramification defect of the chosen coordinate.

An oldform combination

$$C\Phi = \sum_d c_d \Phi(d\tau)$$

has Mellin eigenvalue

$$P_C(s) = \sum_d c_d d^{-s}.$$

Zeros of P_C annihilate unwanted critical/Tate components. On arithmetic-progressions of critical integers, the entire nullspace has a barycentric/Vandermonde description. Fricke symmetry folds the set of constraints and leads to sign-refined rank formulas.

A fifth advantage is that the parity checkerboard becomes a theorem rather than a mnemonic. On the opposite-parity squares, the special value is the regulator coordinate of a genuinely mixed extension. On the complementary Tate squares, exemplified by $\zeta(2)$ and $\zeta(4)$, the same Hecke and Fricke machinery still produces structured modular identities, but the target itself is already a Tate period and no uniquely preferred extension host survives. The even-zeta controls make this collapse explicit through universal central projectors.

A sixth advantage is that the theory can be compared with an irrationality proof that is already known. For $L(2, \chi_{-3})$, the two weight-three sources used by Calegari–Dimitrov–Tang are exactly the inner and outer Eisenstein orientations singled out here. Their arithmetic-holonomy argument therefore supplies an external benchmark showing that the extension data classified by the theory are not merely retrospective labels.

Finally, the framework clarifies what an irrationality proof would require. Modularity and period purity are only the first two layers. The archimedean and nonarchimedean parts of Layer C must be optimized simultaneously. Calegari–Dimitrov–Tang bound the dimension of a rational holonomy space in terms of a common multivalent continuation leaf and a denominator array [16, 17]. Frobenius/Lucas/Dwork divisibility can remove selected denominator layers on positive-dimensional subspaces of the same holonomy space. Combining these gives the exact filtered penalty

$$\tau_F = \sum_j b_j \left(1 - \frac{u_j^2}{m^2} \right),$$

so finite-place savings enter the holonomy inequality quadratically in the dimension of the improved subspace. The Diophantine problem is therefore adelic: maximize complex continuation while minimizing Frobenius-filtered denominator entropy. The failure of many beautiful modular constructions to cross this inequality is not a failure of the first two layers; it is information about the realization layer.

1.1 Main structural results

The principal general statements proved or formalized in this paper are:

- (i) the two-orientation Eisenstein period theorem for opposite-parity $L(r, \chi)$;
- (ii) a Franke–Hecke spectral reinterpretation of critical annihilation, identifying the oldform projectors as explicit Hecke-generated slices of broader critical-value kernels;
- (iii) a critical–orientation–boundary kernel filtration, together with a support-separation rigidity criterion for fixed-boundary affine fibres;
- (iv) barycentric classification of finite oldform period annihilators;
- (v) explicit Gaussian-binomial prime-power projectors;
- (vi) sign-refined Fricke rank and a minimal Fricke sign-switch construction;
- (vii) affine fixed-regulator and cusp-purified fibres;
- (viii) Hecke/isogeny functoriality under depth- r Eichler integration;
- (ix) a cohomological companion-descent criterion;
- (x) an invariant-orbit criterion for scalar versus vector-valued descent and a finite direct-image rank principle;
- (xi) a Tate/regulator parity dichotomy and a universal central Tate projector;
- (xii) a task-dependent purification principle, motivated by the $L(2, \chi_{-3})$ arithmetic-holonomy benchmark;
- (xiii) a Mellin-degree shift identity $D^a C D^{-a}$ for Hecke correspondences, highlighted by the Beukers $\zeta(3)$ to $\beta(4)$ shift;
- (xiv) a centered tensor/Hadamard reduction theorem and a cubic branch-determinant obstruction showing when target cancellation leaves only Tate monodromy but complete scalarization is impossible.
- (xv) an adelic Layer-C formulation and a Frobenius-filtered arithmetic-holonomy theorem, combining the CDT denominator-array bound with a nested free-depth flag;
- (xvi) a marked de Rham-section refinement of Layer C, a pullback transport/re-canonicalization theorem $\Phi_{\text{tr}} = R'(x)\Phi_x$, and a holomorphic/meromorphic exact-source stratification controlled by ramification defect.

1.2 Applications as a test of utility

A classification is useful only if it does more than redescribe its motivating examples. We therefore emphasize a portfolio of applications with three different logical roles:

explain known phenomena + classify/obstruct + predict new phenomena.
--

The applications include the canonical Zagier sporadic source theorem, Ramanujan-Machine examples, prime-isogeny $\zeta(3)$ completions, the vector-valued quintic Catalan system, the $\zeta(7)$ parent/completion phenomenon, the Tate controls $\zeta(2), \zeta(4)$, the exact comparison with the Calegari–Dimitrov–Tang $L(2, \chi_{-3})$ architecture, the level-24 $\beta(4)$ stress test, the complete level-100 Franke- η boundary deformation test, the $L(3, \chi_5)$ conjugate-fold/Hadamard application of

higher symmetric powers, character orientations, independent companions, and rational direct image, and the Shvets Belyi/symmetric-cube cross-pollination, which compresses eleven raw recurrences, identifies the Legendre mother with the #11 $\zeta(3)$ parent, and produces a new level-24 $L(4, \chi_{-3})$ orientation doublet and rational Apéry realization. The distinction between literature results and programme calculations is maintained throughout.

2 The three-layer object

2.1 Layer A: the pure local system and its direct images

Let $\tilde{Y}$ be an orientation-resolved modular cover carrying an elliptic family

$$\tilde{\pi} : \tilde{\mathcal{E}} \longrightarrow \tilde{Y}$$

and let

$$\tilde{\mathbb{H}} = (R^1 \tilde{\pi}_* \mathbb{Q})^\vee$$

denote its rank-two Gauss–Manin local system. For an integer $r \geq 2$ and a Dirichlet character χ , define the resolved homogeneous system

$$\tilde{\mathbb{V}}_{r,\chi} := \mathrm{Sym}^{r-1} \tilde{\mathbb{H}} \otimes \mathbb{L}_\chi, \quad (2)$$

where $\mathbb{L}_\chi$ is the relevant rank-one character line after adjoining the necessary cyclotomic and orientation data. The two placements of the inducing character in Eisenstein series are extension/source orientations; after descent they are encoded by the deck, Galois, and normalizer action rather than by a universally independent rank-two tensor factor.

The appearance of $\mathrm{Sym}^{r-1} \tilde{\mathbb{H}}$ is intrinsic to the elliptic-polylogarithm/Eisenstein-symbol formalism: the $(r-1)$ st elliptic Eisenstein class has top de Rham component of modular weight $r+1$ [8, 9]. On the resolved cover this produces the basic ladder

r	target type	homogeneous local system
2	$L(2, \chi)$	$\tilde{\mathbb{H}}$
3	$L(3, \chi)$	$\mathrm{Sym}^2 \tilde{\mathbb{H}}$
4	$L(4, \chi)$	$\mathrm{Sym}^3 \tilde{\mathbb{H}}$
5	$L(5, \chi)$	$\mathrm{Sym}^4 \tilde{\mathbb{H}}$
7	$L(7, \chi)$	$\mathrm{Sym}^6 \tilde{\mathbb{H}}$

Role of isogeny. An elliptic isogeny acts naturally on $\tilde{\mathbb{H}}$, hence functorially on every symmetric power. Standard degeneracy maps between modular curves are the modular realization of cyclic isogeny correspondences. Thus isogeny belongs to Layer A. It transports the homogeneous skeleton, but does not determine the extension class in Layer B.

Rational quotient coordinates require direct image. An Apéry coordinate need not separate the oriented modular sheets. Let

$$f : \tilde{Y} \longrightarrow Y$$

be a finite map of generic degree d through which the chosen rational coordinate factors. The local system seen over Y is then

$$\mathbb{V}_{r,\chi}^{\mathrm{rat}} := f_* \tilde{\mathbb{V}}_{r,\chi}. \quad (3)$$

On the generically unramified locus,

$$\mathrm{rank} \mathbb{V}_{r,\chi}^{\mathrm{rat}} = d \mathrm{rank} \tilde{\mathbb{V}}_{r,\chi} = dr.$$

Deck/orientation monodromy is part of the action on this direct image. In particular, one must not in general multiply dr by a separate “orientation rank.”

Proposition 2.1 (Direct-image rank principle). *Let $f : \tilde{Y} \rightarrow Y$ be finite of generic degree d and let $\mathbb{W}$ be a rank- n local system on the generically unramified locus of $\tilde{Y}$. Then $f_*\mathbb{W}$ has rank dn on the corresponding locus of Y . If a chosen section is cyclic for the descended connection, its minimal scalar Picard–Fuchs operator has order dn .*

Proof. Over a point of Y with d distinct preimages, the fibre of $f_*\mathbb{W}$ is the direct sum of the d rank- n fibres upstairs. This gives rank dn . A cyclic vector generates the entire differential module, so the associated scalar cyclic operator has order equal to its rank. $\square$

The level-24 $\beta(4)$ application in Section 13 is the first example in this paper where this distinction is essential: Sym^3 has rank 4 on the oriented elliptic cover, the rational coordinate has degree 4, and the resulting rational direct image has rank 16.

In what follows, $\mathbb{V}_{r,\chi}$ denotes the relevant pure system on the base where the extension is being considered: either the resolved system $\tilde{\mathbb{V}}_{r,\chi}$ or a specified finite direct image such as (3).

2.2 Layer B: the mixed extension

The special value is carried by an extension

$$0 \longrightarrow \mathbb{V}_{r,\chi} \longrightarrow \mathbb{E}_\nu \longrightarrow \mathbf{1} \longrightarrow 0, \quad (4)$$

with class

$$[\nu] \in \text{Ext}_Y^1(\mathbf{1}, \mathbb{V}_{r,\chi}).$$

Within the Eisenstein/polylogarithmic subspace, Dirichlet specialization sends this extension to a cyclotomic/Dirichlet motivic class whose regulator is proportional to $L(r, \chi)$ [7, 8].

This layer contains the data that appeared separately during the motivating calculations:

- the de Rham Eisenstein source Φ ;
- the finite critical-period vector;
- cusp residues and Bernoulli boundary data;
- character orientation and Atkin–Lehner fingerprints;
- the affine freedom to add target-zero Eisenstein classes.

They are invariants or representatives of one extension, not separate foundational objects.

A useful linear shadow of Layer B is the finite critical-coordinate map. Franke’s exact weak-function theory shows that in a broad nonprincipal-character sector the fibre of this map is already an explicitly computable affine space [13]. Hence even the complete finite critical vector is generally only a quotient of the extension data. The Hecke projectors of this paper choose explicit degeneracy-generated directions inside such fibres; principal-character regulator orientations then require the additional boundary/regularized endpoint data discussed in Section 5. In particular, Layer B should be thought of as

critical fibre + boundary/regulator refinement + normalizer/orientation data.

The order of these refinements matters. A scalar critical map can forget cancellations between distinct diamond, Galois, or Fricke orientation sectors. For a deformation space W , it is therefore useful to distinguish

$$K_{\text{sc}}(W) \supseteq K_{\text{or}}(W) \supseteq K_{\partial}(W),$$

where K_{sc} kills the chosen scalar critical coordinates, K_{or} kills the full symmetry-resolved critical vector, and $K_{\partial} = K_{\text{or}} \cap \ker \text{Res}$ also preserves the boundary data. Section 6 gives an exact worked example with dimensions 3, 1, 0. Thus “critical-invisible” is not synonymous with “extension-invisible”: representation resolution and cusp boundary are genuinely finer Layer-B coordinates.

2.3 Layer C: the adelic Apéry realization

An abstract extension is not yet an Apéry recurrence. The coarse realization data used in the earlier stages of the project were

$$\mathcal{R}_{\text{coarse}} = (\sigma_{\text{dR}}, a, t, \Lambda), \quad (5)$$

where σ_{dR} is the marked cyclic/de Rham section used to present the pure local system, a is a scalar or vector companion-descent morphism, t is the coefficient coordinate, and Λ is an integral lattice. Arithmetic holonomy and Frobenius theory show that this is only the visible shadow of the full realization.

The enriched Layer C is an *adelic Apéry realization*

$$\mathcal{R}_{\text{ad}} = (\sigma_{\text{dR}}, a, (t, \varphi), \Lambda, U^\bullet, \{\Phi_p, \kappa_p\}_{p \notin S}),$$

in the notation of Definition 10.7. Here σ_{dR} records the marked de Rham/cyclic section whose scalar or block presentation is expanded in t , φ specifies the common complex continuation leaf used by arithmetic holonomy, $U^\bullet$ is the Frobenius/free-depth filtration of the holonomy space, and the good-prime data record the pure Frobenius structure together with the local extension coordinate when available. The coarse quadruple (5) is recovered by forgetting these analytic and finite-place refinements. Conceptually one may write

$$\mathcal{R}_{\text{ad}} = (\mathcal{R}_\infty, \{\mathcal{R}_p\}_{p \notin S}),$$

where $\mathcal{R}_\infty$ records the continuation/holonomy geometry and $\mathcal{R}_p$ records the finite-place lattice and extension Frobenius. The same companion and coordinate tie these local realizations together.

The resulting Picard–Fuchs equation, recurrence, continued fraction, or conservative matrix field is a coordinate description of $(\mathbb{V}, [\nu], \mathcal{R}_{\text{ad}})$. The Shvets compression theorem in Section 15 shows why the marked section is essential: contiguous de Rham sections can change a raw recurrence without changing Layer A. The same application also shows that nonlinear algebraic pullback can transport an extension while moving the newly re-canonicalized exact source into a meromorphic principal-part stratum; this distinction is formalized in Proposition 9.4. Layer C is closed under useful realization operations such as finite direct sums, tensor/Hadamard products, Galois traces, regulator-centering, and Frobenius-adapted changes of lattice. These operations can change complex branch geometry and nonarchimedean denominator cost without changing which Layer-B regulator is being tested.

Definition 2.2 (Hecke–Eisenstein Apéry object). *A Hecke–Eisenstein Apéry object is a triple*

$$(\mathbb{V}, [\nu], \mathcal{R}_{\text{ad}})$$

consisting of a pure modular local system, a Hecke-compatible Eisenstein/polylogarithmic extension, and an adelic arithmetic companion realization.

Remark 2.3 (Why the three layers matter). Two recurrences can represent the same pure local system but different extension classes. Conversely, one extension can admit several realizations, including scalar and vector-valued ones. This is why recurrence equivalence, isogeny equivalence, and equality of special values are distinct notions.

2.4 The conceptual pipeline

The complete structural pipeline may be written

$$\begin{array}{l} \text{elliptic family } \mathcal{E}/Y \longrightarrow \widetilde{\mathbb{V}}_{r,\chi} \longrightarrow f_* \widetilde{\mathbb{V}}_{r,\chi} \longrightarrow [\nu] \\ \longrightarrow \text{critical fibre} \\ \longrightarrow \text{Hecke/boundary purification} \\ \longrightarrow \text{companion descent} \longrightarrow \mathcal{R}_{\text{ad}} \longrightarrow \text{Apéry system.} \end{array}$$

The special value is attached to the extension; the recurrence belongs to the realization.

3 Parity dichotomy and opposite-parity Eisenstein extensions

3.1 The parity checkerboard

Let χ be primitive of conductor f and let $r \geq 2$. The regime of primary interest is

$$\chi(-1) = (-1)^{r+1}. \quad (6)$$

This is the opposite-parity Dirichlet regime. It contains odd zeta values and even beta values, including

$$\zeta(3), \zeta(5), \zeta(7), \dots, \quad \beta(2) = G, \beta(4), \dots$$

The complementary parity contains the classical critical/Tate values, algebraic up to the expected powers of π and Gauss sums. Section 7 shows that this distinction is visible internally in the modular theory: the Hecke/Fricke construction survives on the Tate side, but the distinguished target component is already Tate and no longer determines a unique mixed extension. This checkerboard will be summarized in Section 16.

3.2 Two primitive orientations

Put $k = r + 1$. Consider the primitive weight- k Eisenstein orientations

$$E_{r,\chi}^{\text{in}} = E_{k,\chi,1}, \quad E_{r,\chi}^{\text{out}} = E_{k,1,\bar{\chi}}.$$

With the standard newform Eisenstein normalization, their nonconstant Dirichlet series factor as

$$\begin{aligned} L(E_{r,\chi}^{\text{in}}, s) &= L(s, \chi) \zeta(s - r), \\ L(E_{r,\chi}^{\text{out}}, s) &= \zeta(s) L(s - r, \bar{\chi}). \end{aligned} \quad (7)$$

Such factorizations and the character-swapping Atkin–Lehner action are standard in the newform Eisenstein theory; see [11].

Theorem 3.1 (Two-endpoint target theorem). *Assume (6). Then*

$$L(E_{r,\chi}^{\text{in}}, r) = -\frac{1}{2} L(r, \chi).$$

Moreover the regularized reflected endpoint satisfies

$$L(E_{r,\chi}^{\text{out}}, 1) = L'(1 - r, \bar{\chi}) = \kappa_{r,\chi} L(r, \chi),$$

where $\kappa_{r,\chi}$ is an explicit nonzero algebraic/Gauss/ π factor determined by the Dirichlet functional equation. All other finite critical slots are either trivial zeros or Tate-type values.

Proof. The right endpoint follows from $\zeta(0) = -1/2$. At $s = 1$, the pole of $\zeta(s)$ cancels the parity-forced trivial zero of $L(s - r, \bar{\chi})$, leaving $L'(1 - r, \bar{\chi})$. The completed Dirichlet functional equation identifies this derivative with a nonzero explicit multiple of $L(r, \chi)$. For $1 \leq j < r$, either $\zeta(j - r) = 0$ or $L(j, \chi)$ is a critical Dirichlet value of matching parity and hence Tate-type. The outer orientation is the reflected statement. $\square$

Example 3.2 (Catalan). For $r = 2$ and $\chi = \chi_{-4}$,

$$L(E^{\text{in}}, 2) = -\frac{1}{2} G, \quad L(E^{\text{out}}, 1) = \beta'(-1) = \frac{2G}{\pi}.$$

This is the primitive two-orientation period matrix underlying the Catalan theory.

Example 3.3 ($\zeta(3)$). For $r = 3$ and $\chi = 1$, the two orientations coincide. The reflected target identity reduces to the classical relation

$$\zeta'(-2) = -\frac{\zeta(3)}{4\pi^2}.$$

Thus the character-orientation plane collapses to a scalar line.

3.3 Weyl orientation

For nontrivial χ , the two inducing-character placements form a natural finite orientation doublet in the Eisenstein extension sector. After passage to a rational quotient this doublet is represented by deck/normalizer action on the relevant direct image; it is not automatically an extra tensor-factor rank. In the conductor-4 Catalan case, a direct torsion Fourier transform gives a basis (e_x, e_y) with Weyl action

$$e_x \mapsto e_y, \quad e_y \mapsto -e_x,$$

so

$$\rho_{\text{or}}(w) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This is already present in the elliptic-polylogarithmic torsion distribution before any Apéry coordinate is chosen.

4 Hecke projectors, interpolation, and Fricke folding

4.1 Oldforms as finite Hecke correspondences

For $d \geq 1$ write

$$(V_d f)(\tau) = f(d\tau).$$

A finite oldform correspondence

$$C = \sum_{d \in \mathcal{D}} c_d V_d$$

has Mellin eigenvalue

$$P_C(s) = \sum_{d \in \mathcal{D}} c_d d^{-s}.$$

Thus

$$L(C\Phi, s) = P_C(s)L(\Phi, s). \tag{8}$$

Zeros of P_C annihilate selected critical/Tate components.

Theorem 4.1 (Barycentric period interpolation). *Let $d_1, \dots, d_m$ be distinct positive rationals and*

$$P(s) = \sum_{i=1}^m c_i d_i^{-s}.$$

Fix integers α and $q \geq 1$ and impose

$$P(\alpha + 2j) = 0, \quad j = 0, \dots, q-1.$$

Put $x_i = d_i^{-2}$ and $Q(x) = \prod_i (x - x_i)$. Then the complete solution space is

$$\boxed{c_i = d_i^\alpha \frac{R(x_i)}{Q'(x_i)}, \quad \deg R \leq m - q - 1.}$$

In particular the nullspace has dimension $m - q$ when $m \geq q + 1$.

Proof. After writing $w_i = c_i d_i^{-\alpha}$, the equations become

$$\sum_i w_i x_i^j = 0, \quad 0 \leq j < q.$$

For $\deg R \leq m - q - 1$, the partial-fraction identity

$$\sum_i \frac{H(x_i)}{Q'(x_i)} = 0$$

holds for every polynomial H of degree at most $m - 2$. Apply this with $H(x) = R(x)x^j$. Dimension agrees with the Vandermonde nullity, proving exhaustiveness. $\square$

Relation to the full critical kernel. Theorem 4.1 is an explicit divisor-oldform interpolation formula, not a claim that prescribed critical-value vanishing is unique to this framework. Franke gives a substantially broader exact theory of Eisenstein spaces with specified vanishing critical L -values in a nonprincipal-character sector, together with algorithmic Vandermonde kernel constructions [13]. Section 5 explains the precise relationship: our oldform projectors are Hecke-generated spectral slices of Franke-type critical fibres, while the additional structure of this paper begins with boundary completion, isogeny/Fricke organization, companion descent, and arithmetic realization.

4.2 Prime-power projectors

For opposite-parity $L(r, \chi)$, the lower inner-orientation nuisance slots are

$$J_r = \{j : 1 \leq j < r, \ r - j \text{ odd}\}.$$

For one prime p , the minimal critical projector is therefore

$$C_{r,p}^{\text{crit}} = \prod_{j \in J_r} (1 - p^j V_p). \quad (9)$$

This gives

$$r = 2 : 1 - pV_p, \quad r = 3 : 1 - p^2V_p, \quad r = 5 : (1 - p^2V_p)(1 - p^4V_p).$$

Thus $\zeta(5)$ is the first odd-zeta case whose raw critical completion is generically at least two-dimensional.

For trivial character and odd $r = 2m + 1$, the two orientations coincide and full end-point/critical purification gives

$$C_{r,p}^{\text{full}} = \prod_{a=0}^{m+1} (1 - p^{2a} V_p). \quad (10)$$

By the q -binomial theorem its coefficients are Gaussian binomial coefficients:

$$C_{r,p}^{\text{full}}(X) = \sum_{j=0}^{m+2} (-1)^j p^{j(j-1)} \begin{bmatrix} m+2 \\ j \end{bmatrix}_{p^2} X^j.$$

Example 4.2 ($\zeta(3)$). For $r = 3, p = 2$,

$$C_{3,2}^{\text{full}} = (1 - V_2)(1 - 4V_2)(1 - 16V_2) = 1 - 21V_2 + 84V_4 - 64V_8.$$

This is exactly the level-8 ϵ source in the cubic sporadic census.

Example 4.3 ($\zeta(5)$). For $r = 5, p = 2$,

$$C_{5,2}^{\text{full}} = 1 - 85V_2 + 1428V_4 - 5440V_8 + 4096V_{16},$$

recovering the independently found level-16 purified weight-six source in the project calculations.

4.3 Depth and isogeny functoriality

Since

$$DV_d = dV_dD,$$

one has on zero-constant series

$$D^{-r}V_d = d^{-r}V_dD^{-r}. \quad (11)$$

Thus

$$C = \sum_d c_d V_d \quad \mapsto \quad C^{[r]} = \sum_d c_d d^{-r} V_d$$

on the depth- r Eichler primitive. Combined with (8), one operation has three compatible realizations:

Eisenstein source	:	$\sum c_d V_d,$
normal function	:	$\sum c_d d^{-r} V_d,$
target regulator	:	$P_C(r).$

Distribution relations for elliptic-polylogarithmic Eisenstein classes provide the motivic/level-change background for this compatibility [12].

4.4 Mellin-degree shifts by differential conjugation

The relation $DV_d = dV_dD$ has a second useful consequence beyond Eichler integration.

Proposition 4.4 (Mellin-degree shift). *Let $C = \sum_d c_d V_d$ and let $a \in \mathbb{Z}$. Wherever D^a and D^{-a} are defined on the chosen zero-constant sector,*

$$D^a C D^{-a} = \sum_d c_d d^a V_d.$$

Consequently its finite Mellin polynomial is

$$P_{D^a C D^{-a}}(s) = P_C(s - a).$$

Thus differential conjugation shifts every prescribed Mellin zero by a .

Proof. Iterating $DV_d = dV_dD$ gives $D^a V_d D^{-a} = d^a V_d$; the Mellin identity follows immediately. $\square$

This simple operation explains a striking relation in the $\beta(4)$ application. If

$$C_B = 1 - 28V_2 + 63V_3 - 36V_6$$

is the Beukers weight-four annihilator, then

$$DC_B D^{-1} = 1 - 56V_2 + 189V_3 - 216V_6,$$

which is exactly the inner weight-five $\beta(4)$ annihilator. Equivalently

$$P_{\beta(4),\text{in}}(s) = P_B(s - 1).$$

The outer $\beta(4)$ orientation retains C_B itself. Thus the two character orientations are adjacent Mellin-degree realizations of one Hecke correspondence.

4.5 A paired-character Fricke count: the $\beta(4)$ model

The nontrivial-character situation can pair the two Eisenstein orientations rather than diagonalize each separately. The first complete higher-rank example is $r = 4$, $\chi = \chi_{-4}$. On divisor support $d \mid D$ at level $4D$, full inner purification requires

$$P_{\text{in}}(1) = P_{\text{in}}(3) = P_{\text{in}}(5) = 0,$$

while the outer equations are their Fricke-reflected counterparts.

Proposition 4.5 (Level- $4D$ $\beta(4)$ interpolation count). *In the divisor-oldform sector paired by W_{4D} , each $\pm i$ Fricke eigenspace of fully purified $\beta(4)$ sources has dimension*

$$\boxed{\max\{0, \tau(D) - 3\}}.$$

For $D = 4$ the relevant 3×3 Vandermonde determinant is nonzero, so no source exists. For $D = 6$ the space is one-dimensional; with the coefficient at $d = 1$ normalized to 1, its inner vector is

$$(1, -56, 189, -216)$$

on $d = (1, 2, 3, 6)$.

Proof. After Fricke pairing, the three inner conditions are evaluations at the arithmetic progression 1, 3, 5. Theorem 4.1 gives rank 3 on distinct divisor nodes and therefore nullity $\tau(D) - 3$ when nonnegative. The stated $D = 4$ determinant and the $D = 6$ vector are direct evaluations. $\square$

4.6 Fricke folding

For trivial-character weight k and target $\zeta(k - 1)$, the raw even nuisance positions

$$0, 2, 4, \dots, k$$

are reflected by $s \leftrightarrow k - s$. In a fixed Fricke eigenspace only one condition per reflected pair remains.

Theorem 4.6 (Sign-refined Fricke count). *Let k be even and $\varepsilon = \pm 1$. The number of independent even-period conditions in the W_N -eigenspace of sign ε is*

$$q_{\varepsilon}(k) = \begin{cases} (k + 2)/4, & k \equiv 2 \pmod{4}, \\ k/4 + 1, & k \equiv 0 \pmod{4}, \varepsilon = +1, \\ k/4, & k \equiv 0 \pmod{4}, \varepsilon = -1. \end{cases}$$

If $h_{\varepsilon}(N)$ is the dimension of the divisor-oldform Fricke eigenspace, the purification map has maximal rank

$$\min\{q_{\varepsilon}(k), h_{\varepsilon}(N)\}.$$

Hence the purified source dimension is

$$\max\{0, h_{\varepsilon}(N) - q_{\varepsilon}(k)\}.$$

Proof sketch. Fricke symmetry gives a functional relation between $P(s)$ and $P(k - s)$, so reflected conditions are equivalent. When $k \equiv 0 \pmod{4}$, the central point $s = k/2$ is forced to vanish in the anti-Fricke sector. Pairing divisors d and N/d and setting

$$z_d = \frac{d^2}{N} + \frac{N}{d^2}$$

reduces the remaining matrix, after nonzero scalings, to a Chebyshev/Vandermonde evaluation matrix in distinct z_d . $\square$

4.7 The minimal Fricke sign switch

The minimal full odd-zeta prime-power projector has forced sign

$$(-1)^{(r+3)/2}.$$

If the opposite sign is desired, one additional degeneracy suffices.

Theorem 4.7 (One-step sign switch). *Let $r = 2m + 1$, $M = (r + 3)/2$, and*

$$C_{r,p}^{\text{full}}(X) = \prod_{a=0}^{M-1} (1 - p^{2a} X).$$

At degree $M + 1$, the two Fricke eigendirections are

$$C_{r,p}^{(\varepsilon)}(X) = C_{r,p}^{\text{full}}(X) (1 + \varepsilon(-1)^M p^{M-1} X).$$

Thus exactly one additional prime-power degeneracy is necessary and sufficient to switch the Fricke sign.

5 Franke critical kernels and Hecke–Eisenstein purification

The finite critical-value interpolation used throughout this paper has an important general precursor in Franke’s work on Eisenstein series with prescribed vanishing critical values [13]. Franke gives an exact, algorithmic description of such kernels in a broad nonprincipal-character sector. This changes the correct novelty statement of the present theory: the linear act of imposing prescribed Eisenstein critical zeros is not new here. What is added is the organization of explicit Hecke-generated slices of those kernels together with boundary completion, isogeny and Fricke functoriality, affine extension geometry, scalar/vector companion descent, and arithmetic Apéry realization.

5.1 Franke’s critical-coordinate exact sequence

Let p_1, p_2 be odd primes and let

$$3 \leq k \leq \min\{p_1 - 2, p_2 - 2\}.$$

Franke considers the Eisenstein subspace

$$\mathcal{E}_k(\Gamma_1(p_1 p_2))_0^{(p_1, p_2)},$$

spanned by new Eisenstein series whose two primitive inducing characters are nonprincipal and satisfy the weight parity condition. The subscript 0 records vanishing at the cusps 0 and $i\infty$.

For $\mathcal{S} \subset \{0, \dots, k - 2\}$, let

$$\mathcal{L}_{\mathcal{S},k}(f) = (c_{k,\ell} L(f; \ell + 1))_{\ell \in \mathcal{S}},$$

where the $c_{k,\ell}$ are Franke’s nonzero normalizing constants. His Theorem 4.3 constructs a computable weak-function space $U_{\mathcal{S}}$ and a computable map ξ for which

$$0 \longrightarrow U_{\mathcal{S}} \xrightarrow{\xi} \mathcal{E}_k(\Gamma_1(p_1 p_2))_0^{(p_1, p_2)} \xrightarrow{\mathcal{L}_{\mathcal{S},k}} \mathbb{C}^{|\mathcal{S}|} \longrightarrow 0 \quad (12)$$

is exact. In particular, in this small-weight range,

$$\dim \mathcal{E}_k^{\mathcal{S}} = \dim \mathcal{E}_k(\Gamma_1(p_1 p_2))_0^{(p_1, p_2)} - |\mathcal{S}|. \quad (13)$$

The weak-function kernel is algorithmically computed through Vandermonde matrices in cotangent bases [13].

Status. LITERATURE. Equations (12)–(13) summarize Franke’s Theorem 4.3 and Corollary 4.5 in the range stated above. Franke’s paper is a general exact theory of prescribed critical-value vanishing in this nonprincipal Eisenstein sector.

Thus the correct relationship with Theorem 4.1 is inclusion rather than competition: our barycentric formulas give a closed-form divisor-oldform/Hecke slice of a broader critical kernel that Franke constructs by weak-function methods.

5.2 Critical coordinates are diagonal for degeneracy

Franke writes B_d for the degeneracy operator $(f|B_d)(\tau) = f(d\tau)$; this is the operator denoted V_d elsewhere in this paper. He records

$$\Lambda(f|B_d; s) = d^{-s} \Lambda(f; s). \quad (14)$$

Therefore, for

$$C = \sum_d c_d B_d, \quad P_C(s) = \sum_d c_d d^{-s},$$

one has

$$L(Cf; j) = P_C(j) L(f; j) \quad (15)$$

at every critical integer j .

Proposition 5.1 (Hecke spectral form of critical annihilation). *Let f lie in an Eisenstein sector in which its j th critical coordinate is nonzero. For a finite degeneracy correspondence $C = \sum_d c_d B_d$,*

$$L(Cf; j) = 0 \iff P_C(j) = 0.$$

Thus a divisor-oldform annihilator is a polynomial interpolation operator in the degeneracy characters $B_d \mapsto d^{-j}$.

Proof. This is immediate from (15). □

This makes precise the spectral meaning of the projectors in Section 4: the barycentric formulas solve interpolation on arbitrary divisor nodes, the Gaussian-binomial formulas solve it on one prime-power orbit, and Fricke folding quotients the critical coordinate set by a functional-equation symmetry.

5.3 Critical fibres rather than period-killing recipes

The synthesis suggests replacing the informal phrase “period-annihilation principle” by a fibre-theoretic notion.

Definition 5.2 (Hecke critical projector). *A finite degeneracy correspondence $C = \sum_d c_d B_d$ is a Hecke critical projector for a set S if $P_C(s) = 0$ for every critical eigencoordinate $s \in S$ and P_C is nonzero on the desired target coordinates.*

More generally, if $\mathcal{L}_{\text{crit}}$ denotes the full available critical-coordinate map on an Eisenstein space V_{Eis} , define the critical fibre through f_0 by

$$\mathcal{F}_{\text{crit}}(f_0) := \{f \in V_{\text{Eis}} : \mathcal{L}_{\text{crit}}(f) = \mathcal{L}_{\text{crit}}(f_0)\}. \quad (16)$$

Whenever nonempty,

$$\mathcal{F}_{\text{crit}}(f_0) = f_0 + \ker \mathcal{L}_{\text{crit}}.$$

Franke’s theorem shows that these kernels can be large even when the *complete* finite critical vector is fixed. Hence critical values do not determine an Eisenstein extension.

This is exactly the phenomenon encountered independently in our affine sporadic classifications: target-preserving directions may change cusp residues, normalizer fingerprints, companion kernels, or branch geometry while leaving all selected Mellin data fixed. The Apéry problem should therefore be reformulated as

$$\boxed{\begin{array}{c} \text{choose a point in a critical fibre} \\ \text{with optimal boundary, normalizer, companion,} \\ \text{branch, and arithmetic geometry.} \end{array}} \quad (17)$$

Critical annihilation is only the linear entrance to Layer B.

5.4 Where the regulator sector begins: boundary completion

Franke deliberately restricts to nonprincipal inducing characters so that the Eisenstein forms vanish at 0 and $i\infty$. On the weak-function side this corresponds to requiring a removable singularity at $z = 0$. Before imposing that restriction, his weak-function transformation law contains an explicit residue term of the form

$$\vartheta_k\left(\omega \otimes \eta; -\frac{1}{\tau}\right) = -\tau^k \vartheta_k(\eta \otimes \widehat{\omega}; \tau) + 2\pi i \operatorname{Res}_{z=0} \left(z^{k-1} \eta(z) \widehat{\omega}\left(-\frac{z}{\tau}\right) \right), \quad (18)$$

up to the harmless sign convention in the reflected weak function [13, Thm. 2.11].

Our primitive regulator orientations

$$E_{r+1, \chi, 1}, \quad E_{r+1, 1, \bar{\chi}}$$

have a principal inducing character and therefore lie precisely outside the clean nonprincipal sector of (12). These are also the sectors where constant terms, regularized reflected endpoints, Bernoulli cusp residues, and opposite-parity regulator values enter.

This identifies a concrete missing bridge between the two theories.

Problem 5.3 (Boundary-completed Franke sequence). *Extend the weak-function/Eisenstein exact sequence from the removable-singularity space to a controlled space retaining the residue contribution (18). On the modular side, augment the critical map by boundary coordinates*

$$\tilde{\mathcal{P}}_k : \mathcal{E}_k^\partial \longrightarrow \mathbb{C}^{k-1} \oplus B_\partial.$$

Determine conditions under which an exact sequence

$$0 \longrightarrow U^\partial \longrightarrow \mathcal{E}_k^\partial \xrightarrow{\tilde{\mathcal{P}}_k} \mathbb{C}^{k-1} \oplus B_\partial \longrightarrow 0$$

holds, and identify B_∂ with the constant-term/Bernoulli boundary data appearing in Eisenstein-symbol regulators.

Status. **OPEN.** Franke proves the weak-function transformation with its residue term; the boundary-completed exact sequence above is a synthesis problem proposed here. If established, it would place the principal-character regulator fibres of this paper inside an explicit weak-function linear algebra parallel to Franke's nonprincipal theory.

This problem also clarifies the role of Theorem 8.4: the map

$$(\mathfrak{D}_{r, \chi}, \operatorname{Res})$$

should be viewed as the motivic/de Rham form of a boundary-completed critical-coordinate map, rather than as an unrelated second linear algebra.

5.5 Critical-invisible directions and affine extension geometry

Let

$$\mathcal{K}_{\text{crit}} := \ker \mathcal{L}_{\text{crit}}.$$

In the Franke range, whenever the ambient Eisenstein space has dimension exceeding the number of critical coordinates, (13) produces nonzero forms in $\mathcal{K}_{\text{crit}}$. These forms are invisible to every finite critical value in the chosen coordinate system but remain nontrivial modular objects.

In the language of the present paper they are natural *critical-invisible extension directions*. They provide an external structural model for the affine extension fibres of Section 8: fixing the regulator, or even fixing an entire finite critical vector, generally leaves a residual affine space on which boundary and realization data can vary.

5.6 A published deformation direction for the $L(3, \chi_5)$ extension

Franke's Example 4.7 is directly relevant to Section 14. Let χ_5 be the primitive quartic character modulo 5 with $\chi_5(2) = i$. For every even $k \geq 4$, Franke constructs a nonzero form

$$f_k^{(5)} \in M_k(\Gamma_1(25))$$

from the four character Eisenstein series attached to χ_5 and $\bar{\chi}_5$, with

$$L(f_k^{(5)}; 1) = L(f_k^{(5)}; 2) = L(f_k^{(5)}; k-2) = L(f_k^{(5)}; k-1) = 0. \quad (19)$$

For $k = 4$, these positions exhaust the critical set:

$$\boxed{L(f_4^{(5)}; 1) = L(f_4^{(5)}; 2) = L(f_4^{(5)}; 3) = 0.} \quad (20)$$

Our canonical η source satisfies

$$L(\Phi_\eta, 3) = \frac{1}{2}L(3, \chi_5).$$

Raise the level-20 source and Franke's level-25 form to level 100 and set

$$\Phi_{\eta; \mathbf{a}} = \Phi_\eta + (a_0 + a_1 V_2 + a_2 V_4) f_4^{(5)}. \quad (21)$$

Then

$$L(\Phi_{\eta; \mathbf{a}}, j) = L(\Phi_\eta, j), \quad j = 1, 2, 3, \quad (22)$$

and in particular

$$\boxed{L(\Phi_{\eta; \mathbf{a}}, 3) = \frac{1}{2}L(3, \chi_5)}$$

for every parameter vector $\mathbf{a}$.

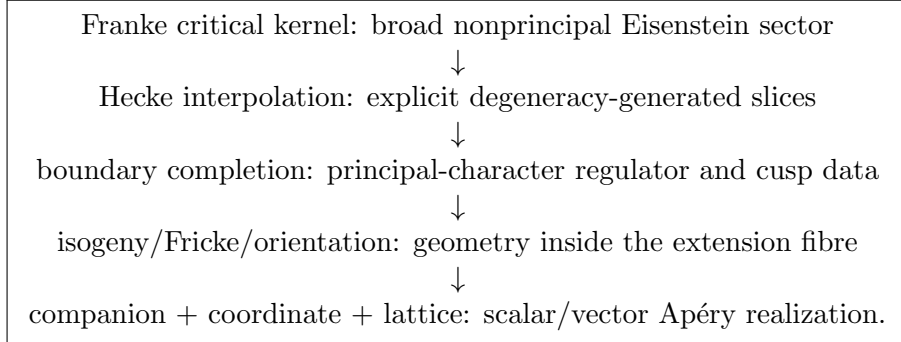
Proposition 5.4 (Franke deformation principle for the η regulator). *After passage to common level, $f_4^{(5)}$ and its degeneracy lifts give explicit critical-invisible deformation directions for the η extension. Every source in (21) has the same complete finite critical vector as Φ_η .*

Proof. Combine (20) with $L(V_d f; j) = d^{-j} L(f; j)$. □

The finite optimization problem suggested by Proposition 5.4 is carried out completely in Section 6. The result is negative for boundary-preserving scalar deformations but structurally decisive: the three-dimensional coarse critical kernel shrinks to the unique orientation-resolved line $(1, -10, 16)$ and then to zero after the complete cusp-boundary map. Thus the worked example identifies exactly where the Franke critical fibre ceases to coincide with the fixed-boundary extension fibre.

5.7 Hierarchy after the Franke synthesis

The linear and geometric stages can now be displayed more precisely:



The first arrow is largely linear and, in Franke’s range, algorithmically understood. The principal conceptual frontier of this paper begins with how critical fibres are organized by boundary and geometric data, and which points admit useful arithmetic realizations.

6 Worked example: the level-100 Franke– η boundary test

The synthesis of Section 5 raises a natural question. Franke’s weight-four level-25 form f_{Fr} has vanishing scalar critical vector, so after passing to common level 100 the family

$$\Phi_\eta + (a + bV_2 + cV_4)f_{\text{Fr}}$$

preserves the scalar critical data of the level-20 η source. Does this affine critical fibre contain a nonzero deformation with improved boundary, normalizer, or companion geometry?

The answer is no for the complete three-oldform family, and the reason gives a useful exact model of the distinction between scalar critical data, orientation-resolved critical data, and cusp boundary.

Status. PROGRAMME THEOREM/CERTIFICATE. Every identity in this section is an exact calculation from the level-100 research certificate accompanying this paper. The use of Franke’s level-25 critical-null form itself is LITERATURE; the diamond decomposition, oldform kernels, cusp matrices, support separation, and normalizer computation are programme calculations.

6.1 Diamond decomposition of Franke’s form

Let ω be the primitive quartic character modulo 5 normalized by

$$\omega(2) = i, \quad \epsilon = \omega^2.$$

Thus ϵ is the real quadratic character modulo 5, the character sector containing the η source.

In the lattice normalization used in the calculation, Franke’s principal-branch constant simplifies to

$$C = -\sqrt{5}(2 + i),$$

and his weight-four form decomposes as

$$f_{\text{Fr}} = f_0 + f_\epsilon, \tag{23}$$

with

$$\begin{aligned} f_\epsilon &= C E_4(\bar{\omega}, \bar{\omega}; 5\tau) + \bar{C} E_4(\omega, \omega; 5\tau), \\ f_0 &= 5 \left(E_4(\bar{\omega}, \omega; 5\tau) + E_4(\omega, \bar{\omega}; 5\tau) \right). \end{aligned}$$

The diagonal terms have nebentypus ϵ , while the cross terms have trivial nebentypus. Hence under the diamond operator $\langle 2 \rangle$,

$$f_0 \mapsto +f_0, \quad f_\epsilon \mapsto -f_\epsilon.$$

The crucial point is that Franke's scalar critical vanishing occurs *after* these sectors are added. The orientation-resolved endpoint values are nonzero:

$$L(f_\epsilon, 1) = +i \frac{448\pi^5}{9375}, \quad L(f_0, 1) = -i \frac{448\pi^5}{9375}, \quad (24)$$

$$L(f_\epsilon, 3) = -i \frac{896\pi^7}{234375}, \quad L(f_0, 3) = +i \frac{896\pi^7}{234375}. \quad (25)$$

At $s = 2$ each sector vanishes separately by the parity trivial zero.

Thus the equality

$$L(f_{\text{Fr}}, 1) = L(f_{\text{Fr}}, 2) = L(f_{\text{Fr}}, 3) = 0$$

does not mean that f_{Fr} is invisible in the symmetry-resolved extension space.

6.2 The unique orientation-resolved critical-null line

Put

$$P(s) = a + b2^{-s} + c4^{-s}.$$

After applying $a + bV_2 + cV_4$, the endpoint vectors in both diamond sectors vanish exactly when

$$P(1) = P(3) = 0.$$

Equivalently,

$$M_{\text{crit}} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0, \quad M_{\text{crit}} = \begin{pmatrix} 1 & 1/2 & 1/4 \\ 1 & 1/8 & 1/64 \end{pmatrix}.$$

This matrix has rank 2, and therefore

$$\ker M_{\text{crit}} = \mathbb{Q}(1, -10, 16). \quad (26)$$

The unique orientation-resolved critical-null direction is

$$f_{\text{pur}} = (1 - 2V_2)(1 - 8V_2)f_{\text{Fr}}. \quad (27)$$

The factorization in (27) is notable: the same Hecke-projector language that organizes the regulator examples isolates the symmetry-resolved kernel inside Franke's coarse critical fibre.

Theorem 6.1 (Strict critical-orientation-boundary filtration). *Let*

$$W_{\text{Fr}} = \text{span}_{\mathbb{Q}}\{f_{\text{Fr}}, V_2 f_{\text{Fr}}, V_4 f_{\text{Fr}}\}.$$

Then

$$\dim K_{\text{sc}}(W_{\text{Fr}}) = 3, \quad \dim K_{\text{or}}(W_{\text{Fr}}) = 1, \quad \dim K_{\partial}(W_{\text{Fr}}) = 0.$$

More precisely,

$$K_{\text{or}}(W_{\text{Fr}}) = \mathbb{Q}f_{\text{pur}},$$

with f_{pur} as in (27).

The first equality follows because each oldform of f_{Fr} has zero scalar critical vector. The second is (26). The third follows from the complete cusp calculation below.

6.3 The complete level-100 cusp map

The 18 cusp classes of $\Gamma_0(100)$ may be represented by

$$1, 1/2, 1/4, 1/5, 2/5, 3/5, 4/5, 1/10, 3/10, 7/10, 9/10, \\ 1/20, 7/20, 3/20, 9/20, 1/25, 1/50, 1/100.$$

For a lattice Eisenstein series $E_k(\chi, \psi; A\tau)$, its constant term at the cusp a/c , after removing the common factor $L(k, \chi\psi)$, is

$$2 \left(\frac{g}{A} \right)^k \chi(c/g) \psi(-Aa/g), \quad g = \gcd(A, c). \quad (28)$$

Applying (28) to the three oldforms gives separate 18×3 matrices in the two diamond sectors. Only the following rows are nonzero.

Trivial diamond sector. With columns (V_1, V_2, V_4) ,

$$\begin{array}{c|ccc|ccc} 1/5 & -20 & 0 & 5/64 & 2/5 & 0 & 5/4 & 0 \\ 3/5 & 0 & -5/4 & 0 & 4/5 & 20 & 0 & -5/64 \\ 1/10 & 0 & -20 & 0 & 3/10 & 20 & 0 & -5/4 \\ 7/10 & -20 & 0 & 5/4 & 9/10 & 0 & 20 & 0 \\ 1/20 & 20 & 0 & -20 & 7/20 & 0 & -20 & 0 \\ 3/20 & 0 & 20 & 0 & 9/20 & -20 & 0 & 20. \end{array}$$

This matrix has rank 3; for example the rows $1/5, 2/5, 3/10$ have determinant

$$\frac{1875}{64} \neq 0.$$

Quadratic diamond sector. After removing the common factor $\sqrt{5}$,

$$\begin{array}{c|ccc|ccc} 1/5 & 8 & 1/4 & -1/32 & 2/5 & 4 & -1/2 & -1/64 \\ 3/5 & -4 & 1/2 & 1/64 & 4/5 & -8 & -1/4 & 1/32 \\ 1/10 & 4 & 8 & 1/4 & 3/10 & 8 & -4 & 1/2 \\ 7/10 & -8 & 4 & -1/2 & 9/10 & -4 & -8 & -1/4 \\ 1/20 & -8 & 4 & 8 & 7/20 & -4 & -8 & 4 \\ 3/20 & 4 & 8 & -4 & 9/20 & 8 & -4 & -8. \end{array}$$

This matrix also has rank 3; the rows $1/5, 2/5, 1/10$ have determinant

$$-\frac{425\sqrt{5}}{64} \neq 0$$

before stripping the common $\sqrt{5}$.

Therefore

$$\boxed{\ker(\text{Res}|_{W_{\text{Fr}}}) = 0.} \quad (29)$$

6.4 Boundary support separation from the η source

The canonical η source is

$$\Phi_\eta = E_{4,\epsilon,1} - 14V_2E_{4,\epsilon,1} - 16V_4E_{4,\epsilon,1}.$$

In the same cusp normalization, its nonzero residues occur only at the denominator 1, 2, 4 cusp classes:

$$\text{Res}_\eta(1) = \frac{1}{8}, \quad \text{Res}_\eta(1/2) = -32, \quad \text{Res}_\eta(1/4) = -2. \quad (30)$$

Every denominator 5, 10, 20 row above vanishes for Φ_η . Conversely, every Franke oldform row vanishes at the three cusp classes in (30). Hence

$$\boxed{\text{supp Res}(\Phi_\eta) \cap \text{supp Res}(W_{\text{Fr}}) = \emptyset.} \quad (31)$$

Combining (29) and (31) with Proposition 8.6 gives:

Corollary 6.2 (Boundary rigidity of the η point). *Inside the level-100 affine deformation family*

$$\Phi_\eta + W_{\text{Fr}},$$

the fixed-regulator, fixed-boundary fibre consists of the single point Φ_η .

This is stronger than the statement that the Franke directions fail to cancel the old boundary. They cannot alter any old boundary component at all; every nonzero Franke deformation necessarily creates boundary data on complementary cusp orbits.

6.5 Normalizer action and vector necessity

On each diamond sector, order the three oldforms as (V_1, V_2, V_4) and define

$$R = \begin{pmatrix} 0 & 0 & 1/16 \\ 0 & 1 & 0 \\ 16 & 0 & 0 \end{pmatrix}, \quad R^2 = I.$$

The level-25 Franke form is W_{25} -odd. On the level-100 oldspace,

$$W_{25} = -I_3, \quad W_4 = R, \quad W_{100} = -R. \quad (32)$$

On the six-dimensional two-sector space,

$$\begin{aligned} \langle 2 \rangle &= \text{diag}(+I_3, -I_3), & W_{25} &= \text{diag}(-I_3, -I_3), \\ W_4 &= \text{diag}(R, R), & W_{100} &= \text{diag}(-R, -R). \end{aligned}$$

The critical-pure vector

$$v = (1, -10, 16)^T$$

satisfies $Rv = v$, and hence

$$f_{\text{pur}}|W_{100} = -f_{\text{pur}}. \quad (33)$$

Thus the unique orientation-critical-null direction is not geometrically featureless. It lies on a distinguished anti-Fricke line. Nevertheless it contains both trivial and quadratic diamond components and has nonzero cusp boundary. It therefore cannot be a scalar deformation of the original η character local system.

Proposition 6.3 (Symmetry-mismatch obstruction). *Within the deformation space W_{Fr} , no nonzero vector defines another scalar extension on the original order-three η host with the same regulator and boundary. Any realization retaining a nonzero Franke direction must enlarge the local system to a higher-rank/vector direct image carrying both diamond sectors.*

Proof. Boundary rigidity gives uniqueness of the scalar fixed-boundary point. Independently, every nonzero Franke direction has nontrivial projection to both diamond sectors, whereas the original η source lies in the quadratic sector. A scalar realization on the original character local system cannot absorb the additional trivial-character component. $\square$

6.6 What the worked example adds to the theory

The level-100 calculation converts the proposed Franke synthesis into a concrete theorem-level stress test.

1. *Scalar critical coordinates can hide representation cancellation.* Franke's complete scalar critical vector vanishes, but the two diamond components have opposite nonzero endpoint values.
2. *Symmetry resolution can be implemented by the same Hecke-projector calculus.* The unique resolved kernel is the factored projector

$$(1 - 2V_2)(1 - 8V_2).$$

3. *Boundary data are strictly finer than symmetry-resolved critical data.* The resolved critical kernel is one-dimensional while the boundary kernel is zero.
4. *Support separation can prove affine rigidity without solving a companion equation.* The η and Franke boundary supports are disjoint, so no nonzero deformation can preserve the old boundary.
5. *A critical-pure direction may still be intrinsically vector-valued.* The surviving anti-Fricke line carries two diamond sectors and therefore requires an enlarged direct-image realization.

The worked example therefore identifies a precise boundary between the theories: Franke's critical kernel describes a coarse linear fibre, while the Hecke–Eisenstein Apéry classification refines that fibre by symmetry, cusp boundary, normalizer action, and companion geometry.

7 The Tate side: central projectors and even-zeta controls

The opposite-parity hypothesis is not merely a convenient source of examples. Its complement behaves differently in a way that can be seen directly in the three-layer theory. For the trivial character, the even zeta values are pure Tate periods,

$$\zeta(2m) \in \mathbb{Q}^\times \pi^{2m},$$

and therefore do not determine a new primitive regulator extension analogous to the odd-zeta classes. Nevertheless the Hecke-projector and Fricke machinery remains meaningful and provides a useful control theory.

7.1 The Tate-side collapse

Proposition 7.1 (Tate/regulator dichotomy). *For primitive χ , the opposite-parity condition*

$$\chi(-1) = (-1)^{r+1}$$

selects the noncritical Dirichlet values to which the Eisenstein-regulator extension theory of Section 2 applies. In the complementary parity, the critical Dirichlet values are algebraic multiples of the expected powers of π and Gauss sums. Consequently the distinguished special-value component in Layer B is Tate-type rather than a new primitive Dirichlet-regulator line.

Remark 7.2. The proposition does *not* say that the corresponding Eisenstein Eichler cocycle is trivial as a modular object. It says that the distinguished special-value coordinate does not define a new non-Tate regulator. This distinction explains why one Tate value can admit several inequivalent modular hosts.

7.2 A universal central Tate projector

For $m \geq 1$ put

$$\mathcal{E}_{4m}(\tau) = \sum_{n \geq 1} \sigma_{4m-1}(n) q^n, \quad L(\mathcal{E}_{4m}, s) = \zeta(s) \zeta(s - 4m + 1).$$

The target $\zeta(2m)$ now sits in the central Fricke position $s = 2m$. Define

$$J_m = \{0, 1, 4m - 1, 4m\} \cup (\{2, 4, \dots, 4m - 2\} \setminus \{2m\}).$$

Thus $|J_m| = 2m + 2$.

Theorem 7.3 (Minimal central Tate projector). *Let p be prime and define*

$$C_{m,p}^T(X) = \prod_{j \in J_m} (1 - p^j X), \quad \Phi_{m,p}^T = C_{m,p}^T(V_p) \mathcal{E}_{4m}.$$

Then:

(i) *the Mellin polynomial is*

$$P_{m,p}^T(s) = \prod_{j \in J_m} (1 - p^{j-s});$$

(ii) *every nonzero endpoint/critical Eisenstein period other than the central $s = 2m$ slot is annihilated; the remaining odd interior slots vanish automatically by the trivial zeros of $\zeta(s - 4m + 1)$;*

(iii) $P_{m,p}^T(2m) \neq 0$;

(iv) $\Phi_{m,p}^T$ *is Fricke-plus at level $N = p^{2m+2}$;*

(v) *up to scale, it is the unique minimum-degree same-prime scalar projector with these properties.*

Proof. The first three assertions follow from $L(V_p f, s) = p^{-s} L(f, s)$ and the definition of J_m . The set J_m is stable under $j \mapsto 4m - j$, which gives the plus reciprocal relation for the coefficient vector at level p^{2m+2} . Finally, $2m + 2$ prescribed distinct Mellin roots require degree at least $2m + 2$, and equality forces the displayed product up to scale. $\square$

The theorem gives a canonical *control realization* of every even zeta value, but unlike the opposite-parity construction it does not claim that weight $4m$ is the uniquely natural host. Indeed $\zeta(2)$ already occurs in the classical order-two Apéry/Zagier D family as well as in the central weight-four construction below.

7.3 $\zeta(2)$: the level-16 control

For $m = 1, p = 2$,

$$J_1 = \{0, 1, 3, 4\},$$

and

$$\boxed{\begin{aligned} C_{1,2}^T &= (1 - V_2)(1 - 2V_2)(1 - 8V_2)(1 - 16V_2) \\ &= 1 - 27V_2 + 202V_4 - 432V_8 + 256V_{16}. \end{aligned}} \tag{34}$$

Its Mellin polynomial satisfies

$$P(0) = P(1) = P(3) = P(4) = 0, \quad P(2) = \frac{9}{8},$$

so

$$L(\Phi_2, 2) = -\frac{3}{32}\zeta(2).$$

At the positive W_{16} fixed point $\tau = i/4$, the standard central Eichler transformation gives

$$D^{-3}\Phi_2(i/4) = \frac{3\pi}{128}\zeta(2) = \frac{\pi^3}{256}.$$

Equivalently, for

$$S_3(y) = \sum_{n \geq 1} \frac{1}{n^3(e^{ny} - 1)},$$

one obtains

$$\begin{aligned} \frac{\pi^3}{256} &= S_3(\pi/2) - \frac{27}{8}S_3(\pi) + \frac{101}{32}S_3(2\pi) \\ &\quad - \frac{27}{32}S_3(4\pi) + \frac{1}{16}S_3(8\pi). \end{aligned}$$

This recovers the earlier Tate-control calculation in the programme.

7.4 $\zeta(4)$: a Fricke obstruction and the weight-8 cure

A naive weight-six realization has

$$L(\mathcal{E}_6, s) = \zeta(s)\zeta(s-5).$$

Although $s = 4$ contains $\zeta(4)$, Fricke reflection exchanges $s = 4$ and $s = 2$. Hence a scalar Fricke eigen-oldform obeys a reciprocal relation forcing

$$P(2) = 0 \implies P(4) = 0.$$

Thus one cannot scalarly kill the reflected Tate slot while retaining $\zeta(4)$ in weight six.

Proposition 7.4 (Weight-six scalar obstruction). *There is no scalar Fricke eigen-oldform in the trivial-character weight-six old-Eisenstein sector whose Mellin polynomial vanishes at the reflected $s = 2$ slot but is nonzero at $s = 4$.*

The natural scalar control therefore places $\zeta(4)$ in the central slot of weight eight. For $m = 2, p = 2$,

$$J_2 = \{0, 1, 2, 6, 7, 8\},$$

and Theorem 7.3 gives the level-64 source

$$\begin{aligned} C_{2,2}^T &= 1 - 455V_2 + 60494V_4 - 2504840V_8 \\ &\quad + 15486464V_{16} - 29818880V_{32} + 16777216V_{64}. \end{aligned} \tag{35}$$

Its Mellin polynomial satisfies

$$P(0) = P(1) = P(2) = P(6) = P(7) = P(8) = 0, \quad P(4) = -\frac{99225}{512}.$$

Since $\zeta(-3) = 1/120$,

$$L(\Phi_4, 4) = -\frac{6615}{4096}\zeta(4).$$

At $\tau = i/8$,

$$D^{-7}\Phi_4(i/8) = \frac{2205}{2^{20}}\pi^3\zeta(4) = \frac{49}{2^{21}}\pi^7.$$

Thus, with $S_7(y) = \sum_{n \geq 1} n^{-7}(e^{ny} - 1)^{-1}$,

$$\begin{aligned} \frac{49\pi^7}{2^{21}} = & S_7(\pi/4) - \frac{455}{128}S_7(\pi/2) + \frac{30247}{8192}S_7(\pi) \\ & - \frac{313105}{262144}S_7(2\pi) + \frac{30247}{524288}S_7(4\pi) \\ & - \frac{455}{524288}S_7(8\pi) + \frac{1}{262144}S_7(16\pi). \end{aligned}$$

We treat the exact coefficient pattern as a programme-derived identity; no novelty claim is made without a separate historical audit.

7.5 What the controls teach the three-layer theory

The even-zeta examples sharpen rather than weaken the three-layer picture:

opposite-parity/regulator square critical/Tate square	Layer B is the essential new extension datum, the distinguished target component is already Tate.
--	--

On the Tate side, Layer C can still be rich—Hecke projectors, Fricke folds, CM values, Lambert identities, recurrences, and integral lattices survive—but the target itself does not determine a unique local-system host.

8 Affine extension geometry and boundary purification

8.1 Critical fibres and regulator fibres

Section 5 shows that a useful linear precursor to the regulator fibre is the full finite critical-value fibre. On any finite Eisenstein space for which the relevant critical values are defined, write

$$\mathcal{L}_{\text{crit}} : V_{\text{Eis}} \longrightarrow C_{\text{crit}}$$

for the critical-coordinate map. For $f_0 \in V_{\text{Eis}}$,

$$\mathcal{F}_{\text{crit}}(f_0) = f_0 + \ker \mathcal{L}_{\text{crit}}.$$

Franke’s dimension theorem gives broad sectors where this kernel is explicitly computable and nontrivial [13]. Thus even the complete finite critical vector need not determine an Eisenstein extension.

The fixed-regulator fibre below is generally a coarser or differently normalized affine slice, especially in the principal-character regulator sectors where boundary/regularized endpoint data enter. The hierarchy to keep in mind is

$$\text{critical fibre} \supseteq \text{fixed regulator/boundary slice} \supseteq \text{realization-compatible locus}.$$

This is why target-zero or critical-invisible translations are not nuisances: they are the natural deformation directions on which cusp, normalizer, branch, and arithmetic geometry can be optimized.

8.2 Fixed-regulator fibres

Huber–Kings construct Dirichlet motivic classes from Eisenstein symbols on powers of the universal elliptic curve and compute their realizations in terms of Dirichlet L -values [7]. We use this only on a finite Eisenstein-symbol subspace where the specialization is defined. If the chosen realization base is a rational quotient, all extension and boundary maps in this section are understood after the finite direct image of Layer A; this does not change their linear-fibre character.

Let

$$\mathfrak{D}_{r,\chi} : V_{\text{Eis}}(Y, r, \chi) \longrightarrow M_{r,\chi}$$

be the Dirichlet specialization and choose a target class $\xi_{r,\chi}$ with regulator proportional to $L(r, \chi)$.

Definition 8.1 (Fixed-regulator fibre).

$$\mathcal{A}_{r,\chi}(Y) := \mathfrak{D}_{r,\chi}^{-1}(\xi_{r,\chi}).$$

Proposition 8.2 (Affine fibre). *Whenever nonempty, $\mathcal{A}_{r,\chi}(Y)$ is an affine torsor over $\ker \mathfrak{D}_{r,\chi}$.*

Proof. This is the fibre of a linear map over a fixed target. Choosing one lift ν_0 gives

$$\mathcal{A}_{r,\chi}(Y) = \nu_0 + \ker \mathfrak{D}_{r,\chi}.$$

□

This elementary observation is conceptually important. Different modular forms with the same normalized regulator need not be the same extension. Target-zero Eisenstein classes change cusp residues, normalizer fingerprints, local cocycles, or branch geometry while preserving the special value.

Proposition 8.3 (Affine functoriality under algebraic pullback). *Let $f : Y' \rightarrow Y$ be a finite algebraic pullback on which the relevant Eisenstein extension is defined. Then pullback sends a fixed-regulator affine fibre to the corresponding pulled-back fibre and sends target-zero translations to target-zero translations:*

$$f^*(\nu_0 + \ker \mathfrak{D}_{r,\chi}) = f^*\nu_0 + f^*(\ker \mathfrak{D}_{r,\chi}).$$

In particular, transport of Layer B is functorial independently of the coefficient coordinate chosen later in Layer C.

Proof. Functoriality of the Eisenstein/Dirichlet specialization gives

$$\mathfrak{D}(f^*\nu) = f^*\mathfrak{D}(\nu),$$

so the pullback of the target fibre is again a target fibre, and the pullback of a target-zero class remains target-zero. □

The distinction in the last sentence is essential. If the coefficient coordinate is changed nonlinearly after pullback, the transported extension remains the same Layer-B object, whereas the rigid exact representative $\Phi = FDt$ may acquire the ramification defect of Proposition 9.4. Thus affine extension functoriality belongs to Layer B, while coordinate re-canonicalization belongs to Layer C.

8.3 The boundary map

Eisenstein symbols have explicit cusp residues governed by Bernoulli polynomials; for the index $r - 1$ class the relevant polynomial is B_{r+1} [10, 8]. On a finite basis, collect the unwanted cusp/orientation residues into a linear map

$$\text{Res} : V_{\text{Eis}}(Y, r, \chi) \longrightarrow B_{\text{cusp}}.$$

Theorem 8.4 (Affine purification). *Define the combined map*

$$\Psi = (\mathfrak{D}_{r,\chi}, \text{Res}).$$

Then the completely purified fixed-regulator space is

$$\boxed{\mathcal{A}_{r,\chi}^{\text{pur}}(Y) = \Psi^{-1}(\xi_{r,\chi}, 0).}$$

Whenever nonempty it is an affine torsor over

$$\ker \mathfrak{D}_{r,\chi} \cap \ker \text{Res}.$$

Thus regulator fixing and cusp-period purification form a single finite linear stage.

8.4 Critical–orientation–boundary kernel filtration

Let $W \subset V_{\text{Eis}}$ be a finite deformation space. Suppose a symmetry group G (diamond, Galois, or normalizer) refines the scalar critical-coordinate map into a representation-resolved map

$$\mathcal{L}_{\text{or}} : W \longrightarrow C_{\text{crit}}^{\text{or}},$$

and let $\mathcal{L}_{\text{sc}}$ be the corresponding scalar/aggregated critical map. Define

$$K_{\text{sc}} := \ker \mathcal{L}_{\text{sc}}, \quad K_{\text{or}} := \ker \mathcal{L}_{\text{or}}, \quad K_{\partial} := K_{\text{or}} \cap \ker \text{Res}.$$

Proposition 8.5 (Kernel filtration principle). *One has*

$$\boxed{K_{\partial} \subseteq K_{\text{or}} \subseteq K_{\text{sc}}.}$$

Both inclusions may be strict. In particular, scalar critical vanishing does not imply orientation-resolved critical vanishing, and orientation-resolved critical vanishing does not imply preservation of the cusp boundary.

Proof. The scalar critical coordinates are obtained from the resolved coordinates by forgetting or summing symmetry sectors, so $\mathcal{L}_{\text{sc}}$ factors through $\mathcal{L}_{\text{or}}$. Hence $\ker \mathcal{L}_{\text{or}} \subseteq \ker \mathcal{L}_{\text{sc}}$. The second inclusion is the definition of K_{∂} . $\square$

Proposition 8.6 (Support-separation rigidity). *Let Φ_0 be a base source and W a deformation space. Assume:*

1. $\text{Res}|_W$ is injective;
2. the cusp support of $\text{Res}(W)$ is disjoint from the support of $\text{Res}(\Phi_0)$.

Then the fixed-boundary affine fibre meets $\Phi_0 + W$ only at Φ_0 :

$$\boxed{\{\Phi \in \Phi_0 + W : \text{Res}(\Phi) = \text{Res}(\Phi_0)\} = \{\Phi_0\}.$$

The same conclusion holds a fortiori in any fixed-regulator, fixed-boundary slice.

Proof. If $\Phi_0 + w$ has the same residue vector as Φ_0 , then $\text{Res}(w) = 0$. Injectivity of $\text{Res}|_W$ gives $w = 0$. The disjoint-support hypothesis makes the geometric reason transparent: a nonzero deformation cannot cancel an old boundary component and can only create new ones. $\square$

Section 6 realizes the strict chain

$$\dim K_{\text{sc}} = 3, \quad \dim K_{\text{or}} = 1, \quad \dim K_{\partial} = 0$$

inside the level-100 Franke oldspace, and simultaneously satisfies the support-separation criterion relative to the η source.

8.5 Completion cubes

Let

$$J_r = \{j : 1 \leq j < r, \ r - j \text{ odd}\}$$

be the lower nuisance set. Choose an isogeny degree p_j for each $j \in J_r$ and put

$$C_j = 1 - p_j^j V_{p_j}.$$

The C_j commute. For a subset $S \subseteq J_r$, define

$$C_S = \prod_{j \in S} C_j.$$

Proposition 8.7 (Completion cube). *The family $S \mapsto C_S$ is a Boolean cubical diagram of dimension $|J_r| = \lfloor r/2 \rfloor$. Every edge inserts exactly one Mellin zero at its corresponding nuisance slot. At the target $s = r$, that edge multiplies the regulator by $1 - p_j^{j-r}$; after r Eichler integrations it acts as $1 - p_j^{j-r} V_{p_j}$.*

For $r = 2, 3$ the completion cube is a line. For $r = 5$ it is a square. For $r = 7$ it is a cube. This supplies a simple structural reason why higher odd-zeta problems naturally invite multi-row or Hermite–Padé constructions.

8.6 Cusp boundary versus realization-fold functionals

The residue map above belongs intrinsically to Layer B. By contrast, a branch or fold encountered after choosing an Apéry coordinate is generally a Layer-C object: it depends on the quotient and on the chosen companion. The two should not be conflated. Nevertheless modular involutions can relate the two. In the $L(3, \chi_5)$ application, an order-two elliptic fixed point has a branch-cancellation constant equal to an Eichler value at the rational boundary point attached to the involution. Thus an elliptic fold functional can sometimes be computed from the same extension cocycle that controls cusp periods, even though it is not itself a cusp residue.

This distinction is useful in applications. Complete cusp purification can be solved linearly in Layer B, while the remaining elliptic/branch functionals form a finite analytic matrix in Layer C. Their rank can then prove scalar no-go theorems, as in Theorem 9.7 and Section 14.

9 Companion descent, exactness, and vector-valued realizations

9.1 Companion descent as a cohomological kernel

Let $\mathbb{V}$ be the orientation-twisted local system and let

$$[\nu] \in H^1(Y, \mathbb{V})$$

represent the extension. Let $\mathbb{L}$ be a target automorphy local system and let

$$a : \mathbb{V} \longrightarrow \mathbb{L}$$

be a candidate companion morphism. Pushout sends $[\nu]$ to

$$a_*[\nu] \in H^1(Y, \mathbb{L}).$$

Theorem 9.1 (Companion-descent criterion). *A corrected scalar contraction associated to a descends as a single-valued section precisely when*

$$a_*[\nu] = 0.$$

Consequently the admissible scalar companion space is

$$\boxed{\text{Comp}_\nu(\mathbb{L}) = \ker(\text{Hom}(\mathbb{V}, \mathbb{L}) \rightarrow H^1(Y, \mathbb{L}), \ a \mapsto a_*[\nu]) .}$$

The vector-valued version is obtained by replacing $\mathbb{L}$ by a higher-rank target system.

Proof. The pushout extension splits if and only if its extension class vanishes. A splitting is exactly the single-valued corrected companion contraction. $\square$

This theorem is the correct general replacement for the informal instruction “find a modular companion.” On a fixed host it is a finite cohomological kernel problem. Hermite–Padé constructions are naturally interpreted as left kernels of the corresponding vector cocycle matrix.

9.2 The rigid exactness locus

Many especially elegant modular Apéry systems satisfy the stronger identity

$$\Phi = F D t,$$

where F is a distinguished homogeneous period. Since Φ has weight $r + 1$ and F has weight $r - 1$, the quotient has weight 2. Define

$$\omega_{\Phi, F} = \frac{\Phi}{F} \frac{dq}{q}.$$

Proposition 9.2 (Exactness criterion).

$$\Phi = F D t \quad \Longleftrightarrow \quad \omega_{\Phi, F} = dt.$$

On a genus-zero modular curve this reduces to vanishing of all residues of $\omega_{\Phi, F}$; on positive genus one must additionally kill its H_{dR}^1 class.

This exactness locus is a rigid subspace of the broader companion-descent space. It contains the canonical modular Zagier systems and the simplest Catalan examples, but the level-12 $\zeta(5)$ construction illustrates that a successful companion need not arise from $\Phi = F D t$. The level-24 $\beta(4)$ application gives a higher-rank nontrivial-character confirmation: the naive source $F^3 D t$ is not the purified Eisenstein source, whereas an independent Sym^3 isogeny companion nevertheless kills the corrected Eichler cocycle.

9.3 Marked de Rham sections and scalar presentations

The pure local system does not determine a unique scalar recurrence. A realization also chooses a cyclic/de Rham section inside that local system. Contiguous hypergeometric identities, derivatives, algebraic gauges, and changes of cyclic vector can alter the scalar operator and its recurrence without changing the underlying differential module.

Definition 9.3 (Marked de Rham section). *A Layer-C realization contains a marked local de Rham/cyclic section*

$$\sigma_{\text{dR}} \in \mathbb{V}_{\text{dR}}$$

(or a finite tuple of such sections in a vector realization) whose horizontal translates generate the scalar or block differential presentation used for coefficient extraction. Two realizations differing only by a rational change of de Rham basis have the same Layer-A local system but need not have the same scalar recurrence.

This datum is essential in the Shvets application of Section 15: one of the eleven Belyi rows is a contiguous derivative section of the Legendre system rather than a new rank-two motive.

9.4 Extension transport versus coordinate re-canonicalization

Rigid exactness is also not functorial under an arbitrary change of coefficient coordinate in the naive sense. Suppose an old realization uses a coordinate y and exact source $\Phi_y = F_y Dy$, and let $y = R(x)$ be a nonconstant algebraic pullback. Put $F_x(x) = F_y(R(x))$. There are two different operations:

1. transport the existing extension, represented by $\Phi_{\text{tr}} := F_x Dy$;
2. declare x to be a new exact coordinate and form $\Phi_x := F_x Dx$.

Proposition 9.4 (Pullback transport versus coordinate re-canonicalization). *With the notation above,*

$$\boxed{\Phi_{\text{tr}} = R'(x)\Phi_x.} \tag{36}$$

If x_0 is a critical point of R lying over an ordinary point of the old differential system, and if Φ_{tr} is nonzero there, then the re-canonicalized source Φ_x is meromorphic at x_0 . A zero of order $e - 1$ of R' contributes a pole of order $e - 1$ unless cancelled by the transported source.

Proof. The chain rule gives $Dy = R'(x)Dx$, hence

$$\Phi_{\text{tr}} = F_x Dy = R'(x)F_x Dx = R'(x)\Phi_x.$$

The local statement follows by taking valuations. □

Definition 9.5 (Ramification defect and exactness-compatible pullback). *The ramification defect of the pullback R relative to a transported exact source is the divisor contribution of R' not cancelled by the divisor of Φ_{tr} . The pullback is exactness-compatible when this defect is nonnegative in the holomorphic direction, so that the re-canonicalized source remains holomorphic. Otherwise the realization belongs naturally to a meromorphic/principal-part exact-source stratum.*

This distinction separates two functorialities that are often conflated. Pullback of the differential module and of an existing extension is always the natural operation. Re-canonicalizing the identity $\Phi = FDx$ after a ramified coordinate change is an additional choice and can create apparent poles above ordinary points.

9.5 Holomorphic and meromorphic exact-source strata

The Shvets signature-three square, as well as the generalized Cooper analysis, shows that meromorphic exact sources are not exceptional pathologies. Orbifold and ramification points can force controlled principal parts even when the homogeneous local system is modular and geometric. It is therefore useful to enlarge the exactness locus to

$$\mathcal{X}_{\text{exact}}^{\text{mer}} = \{(F, t, \Phi) : \Phi = F D t \text{ with prescribed principal-part divisor}\},$$

with the holomorphic exact locus as the zero-principal-part stratum. The principal-part divisor is Layer-C realization data: it records the interaction between the chosen section, coordinate, and extension representative, not a new pure motive.

9.6 Purification depends on the arithmetic task

A second distinction is equally important. The maximally period-pure extension is not automatically the optimal arithmetic realization. If the goal is to extract one scalar CM value, killing all nuisance Tate directions is natural. For a joint linear-independence or arithmetic-holonomy problem, however, retaining a controlled Tate coordinate can enlarge the useful realization space.

Proposition 9.6 (Task-dependent purification). *The Hecke projector should be regarded as part of Layer C rather than as an absolute canonical choice inside Layer B. Full purification canonically isolates a regulator coordinate, but vector and arithmetic-holonomy realizations may retain selected Tate directions when the intended Diophantine statement concerns several period coordinates simultaneously.*

Section 12 gives a decisive example: Calegari–Dimitrov–Tang retain both $L(2, \chi_{-3})$ and $\zeta(2)$ directions and use their joint overconvergence and arithmetic holonomy to prove linear independence. Section 14 gives the complementary phenomenon: after centering by the regulator, a tensor realization can remove its nearest non-Tate branch while retaining only Tate branch data at the same depth-three denominator cost.

9.7 Centered tensor reduction and scalarization obstruction

Tensor, Hadamard, and diagonal constructions enlarge Layer C without changing the underlying regulator target. A recurrent local situation is a conjugate pair of branch constants

$$C_+ = \frac{\Lambda}{2} + \delta, \quad C_- = \frac{\Lambda}{2} - \delta, \quad \delta \neq 0,$$

where Λ is the non-Tate regulator and δ is Tate. Let A be a homogeneous row and B its companion, and suppose the natural quadratic tensor/Hadamard rows

$$X = A \star A, \quad Y = A \star B, \quad Z = B \star B$$

have side-plus, cross, and side-minus branch functionals given by evaluation at C_+, C_- in the standard product manner.

Theorem 9.7 (Centered tensor reduction and three-branch obstruction). *Define*

$$U_0 = X, \quad U_1 = Y - \frac{\Lambda}{2}X, \quad U_2 = Z - \Lambda Y + \frac{\Lambda^2}{4}X.$$

Then the three branch-functional rows are

$$\begin{pmatrix} 1 & \delta & \delta^2 \\ 2 & 0 & -2\delta^2 \\ 1 & -\delta & \delta^2 \end{pmatrix}.$$

In particular U_1 kills the cross branch exactly. If δ is Tate, every residual branch constant in the centered basis is Tate. Moreover the uncentered branch matrix

$$M = \begin{pmatrix} 1 & C_+ & C_+^2 \\ 2 & C_+ + C_- & 2C_+C_- \\ 1 & C_- & C_-^2 \end{pmatrix}$$

satisfies

$$\det M = (C_- - C_+)^3 = -8\delta^3 \neq 0.$$

Hence no nonzero scalar combination of X, Y, Z can be holomorphic through all three branch orbits simultaneously.

Proof. Substitute $C_{\pm} = \Lambda/2 \pm \delta$ into the three product branch functionals. Centering B by $(\Lambda/2)A$ gives the displayed matrix by direct expansion. The determinant of the uncentered matrix is the cubic Vandermonde factor $(C_- - C_+)^3$. $\square$

The theorem gives a precise sense in which target projection can simplify monodromy without producing scalar overconvergence. It also identifies a mechanism for arithmetic weight drop: if U_1 is linear in the original companion B , then it can retain the original Eichler denominator depth even though it lives in a tensor realization. The $L(3, \chi_5)$ application realizes exactly this pattern with

$$\delta = \frac{2i\pi^3}{125\sqrt{5}}.$$

Thus a failed scalarization can itself be a positive structural result: it certifies the need for a larger vector/holonomic or arithmetic-holonomy realization.

9.8 Scalar versus vector descent

Let

$$\rho_{\text{or}} : \Gamma_{\text{geom}} \longrightarrow GL(\mathcal{O})$$

be the finite orientation/normalizer representation. For an orientation vector v , define its orbit span

$$\mathcal{O}(v) = \text{span}\{\rho_{\text{or}}(\gamma)v : \gamma \in \Gamma_{\text{geom}}\}.$$

Theorem 9.8 (Minimal orientation multiplicity). *On a fixed orientation-resolved pure host, the minimal dimension of an invariant orientation subspace containing v is*

$$\dim \mathcal{O}(v).$$

In particular an orientation-scalar realization on that fixed host exists if and only if the line spanned by v is invariant under the full orientation representation.

Proof. Any invariant orientation subspace containing v must contain the full orbit span. Conversely the orbit span is stable and is minimal by construction. This counts orientation multiplicity only; the rank after passage to a quotient coordinate is governed by Proposition 2.1. $\square$

9.9 Orientation multiplicity versus direct-image rank

Theorem 9.8 is deliberately not a formula for the scalar Picard–Fuchs order in an arbitrary rational coordinate. If the coordinate map forgets a finite set of modular/isogeny sheets, the pure system must first be pushed forward as in Proposition 2.1. The deck action on this direct image may already encode the character orientation.

Hence three ranks should be distinguished:

upstairs pure rank	$= r,$
orientation orbit multiplicity	$= \dim \mathcal{O}(v),$
downstairs rational rank	$= \text{rank } f_* \widetilde{\mathbb{V}}.$

There is no universal rule multiplying the last two. In the $\beta(4)$ level-24 trace coordinate, Sym^3 has upstairs rank 4 and the modular field extension has degree 4, giving rational rank 16; the two character orientations are already visible in the deck action of that degree-four cover.

9.10 Sources of non-scalar complexity

The theory separates three phenomena that should not be conflated.

1. **Character/orientation vectoriality.** For nontrivial χ , Weyl can exchange the two inducing-character orientations. The quintic Catalan system realizes an irreducible dihedral orientation representation and therefore has no scalar descent.
2. **Higher-extension vectoriality.** Even when $\chi = 1$, $\text{Sym}^{r-1} \mathbb{H}$ has rank r . Starting at $\zeta(5)$ there are multiple independent completion directions, and vector/Hermite–Padé realizations can be natural even though the character orientation itself is scalar.
3. **Direct-image complexity.** A quotient coordinate can forget modular or isogeny sheets. The resulting rational differential module is a finite direct image and may have rank strictly larger than any naive orientation count; $\beta(4)$ gives the first explicit rank-16 example in the present theory.

This distinction is one of the main benefits of the three-layer formulation.

9.11 Vectoriality can lower arithmetic cost

Vector-valuedness is not only an analytic or representation-theoretic burden. Once the realization is endowed with a Frobenius/free-depth flag, a denominator layer that is absent on a u -dimensional subspace lowers the arithmetic-holonomy penalty by u^2/m^2 times its scale; see Theorem 10.2. Hence a larger orientation, direct-image, or Hermite–Padé module can be arithmetically superior to a scalar row when it contains a substantial Frobenius-stable subspace with cheaper denominators. This changes the optimization problem: one should choose vector rank and finite-place integrality simultaneously rather than minimize rank first and study denominators afterward.

10 Adelic Layer C: arithmetic holonomy and Frobenius filtration

The realization layer has two genuinely different arithmetic faces. Complex continuation determines how many holonomic functions can coexist on a large common multivalent leaf, while p -adic/Frobenius structure determines how cheaply those functions can live in an integral lattice. Calegari–Dimitrov–Tang’s arithmetic holonomy bounds make the first statement quantitative; Lucas, Dwork, Cartier, and crystalline/syntomic methods govern the second [16, 17]. Their combination suggests that Layer C should be treated adelicly rather than by a single scalar denominator exponent; the synthesis and project-level test programme are recorded in the accompanying research note [28].

10.1 The arithmetic-holonomy penalty

We use the quantitative form recalled by Calegari–Dimitrov–Tang. Let $f_1, \dots, f_m \in \mathbb{Q}[[x]]$ be $\mathbb{Q}(x)$ -linearly independent holonomic functions. Suppose their coefficient denominators are described by an $m \times r$ array $\mathbf{b} = (b_{ij})$ whose j th column has the form

$$b_{ij} = \begin{cases} 0, & i \leq u_j, \\ b_j, & i > u_j, \end{cases}$$

for some $0 \leq u_j \leq m$ and $b_j \geq 0$. In CDT's notation the corresponding coefficient type is

$$f_i(x) = \sum_{n \geq 0} a_{i,n} \frac{x^n}{\prod_{j=1}^r [1, \dots, b_{ij}n]}, \quad a_{i,n} \in \mathbb{Z},$$

where $[1, \dots, y]$ denotes the relevant LCM truncation. Put

$$\tau(\mathbf{b}) = \sum_{j=1}^r b_j \left(1 - \frac{u_j^2}{m^2} \right).$$

Let $\varphi : (\overline{\mathbf{D}}, 0) \rightarrow (\mathbb{C}, 0)$ be a holomorphic pullback such that every $f_i \circ \varphi$ extends meromorphically to the unit disc. Write

$$E(\varphi) := \iint_{\mathbf{T}^2} \log |\varphi(z) - \varphi(w)| d\mu(z) d\mu(w).$$

Then the arithmetic holonomy theorem gives, when $\log |\varphi'(0)| > \tau(\mathbf{b})$,

$$m \leq \frac{E(\varphi)}{\log |\varphi'(0)| - \tau(\mathbf{b})}. \quad (37)$$

This is exactly the setting in which a vector-valued Apéry construction can outperform a scalar one: the analytic continuation is shared, but different basis vectors can have different denominator columns [17].

10.2 Frobenius-free depth

A nominal depth- r Eichler companion often carries r LCM layers. The project examples show that some layers can be arithmetically free: a primitive may already be integral, or a Frobenius/Lucas congruence may force extra divisibility. We package that phenomenon as a filtration.

Definition 10.1 (Frobenius/free-depth filtration). *Let V be an m -dimensional $\mathbb{Q}(x)$ -space of holonomic functions whose generic denominator type has r LCM layers with scales $b_1, \dots, b_r$. A Frobenius/free-depth filtration is a nested flag*

$$V = U^0 \supseteq U^1 \supseteq \dots \supseteq U^r$$

with the property that functions in U^a admit an all- n denominator bound in which the first a nominal LCM layers are absent. Put

$$u_a := \dim U^a.$$

The filtration is required to be witnessed by uniform arithmetic input, for example Lucas/Dwork congruences, Cartier/Frobenius divisibility, factorial-ratio valuation theorems, an excellent Frobenius lift, or an explicit all- n p -adic valuation argument.

The word “Frobenius” is intentionally broad here: the filtration records a finite-place improvement of the lattice, while an actual crystalline Frobenius structure gives a conceptual mechanism for such improvements in geometric situations.

Theorem 10.2 (Frobenius-filtered holonomy). *Assume the hypotheses of the arithmetic holonomy theorem and suppose the basis is adapted to a Frobenius/free-depth filtration so that the j th denominator layer is absent on exactly u_j basis vectors. Then*

$$\tau_F = \sum_{j=1}^r b_j \left(1 - \frac{u_j^2}{m^2} \right). \quad (38)$$

In the ordinary depth- r case $b_1 = \dots = b_r = 1$,

$$\tau_F = r - \frac{1}{m^2} \sum_{j=1}^r u_j^2. \quad (39)$$

Define the Frobenius-holonomy score

$$\text{Score}_F := m(\log |\varphi'(0)| - \tau_F) - E(\varphi). \quad (40)$$

If a rationality hypothesis would produce m independent functions satisfying the stated denominator and continuation hypotheses, then

$$\text{Score}_F > 0$$

contradicts that rationality hypothesis.

Proof. Choose a basis adapted to the nested flag. In the j th denominator column the first u_j rows are zero and the remaining rows carry the common scale b_j . The identity (38) is therefore exactly the Calegari–Dimitrov–Tang formula for $\tau(\mathbf{b})$. Inequality (37) is equivalent to $\text{Score}_F \leq 0$ for every such independent m -tuple. Hence a rationality assumption producing a tuple with positive score is impossible. $\square$

This theorem is a rigorous vector-valued replacement for the rough scalar comparison

analytic gain – denominator entropy.

Its most important feature is the quadratic saving u_j^2/m^2 . If a denominator layer is free on a positive-dimensional Frobenius-stable subspace, vector-valuedness can improve the arithmetic side as well as the analytic side.

Example 10.3 (A depth-three numerical scale). For $m = 14$ and three unit LCM layers, the unfiltered penalty is 3. If one layer is free on a 7-dimensional subspace, then

$$\tau_F = 3 - \frac{49}{196} = 2.75.$$

If the first two layers are free on the same 7-dimensional flag, the penalty is 2.5; if all three are free there, it is 2.25.

10.3 The Cooper weight drop as a first arithmetic check

The three Cooper cubic systems s_7, s_{10}, s_{18} provide an exact finite-place test of the new language. The project proves

$$(n+1) \mid A_n \quad (n \geq 0),$$

so the first nominal Eichler integration is integral and the normalized companion has depth two rather than the generic cubic depth three:

$$d_n^2 B_n \in \mathbb{Z}.$$

For a scalar realization this is exactly one free denominator layer, hence the holonomy penalty drops from 3 to 2. More generally, any holonomy tuple all of whose rows inherit the same free primitive has $u_1 = m$ and receives the identical one-unit saving. The Cooper systems therefore validate the arithmetic half of the adelic synthesis even though their complex branch geometry is not strong enough for a new irrationality theorem.

10.4 The Shvets symmetric cube as an independent depth-four check

A second, externally obtained finite-place check occurs in Shvets's symmetric-cube sequence. The homogeneous modular object is a Sym^3 elliptic system, while Section 15 identifies its canonical exact source with $E_{5,\chi_{-3},1}$ and hence its regulator depth with $r = 4$. Independently, Shvets proves the supercongruence

$$A_{mp} \equiv A_m \pmod{p^4} \quad (p \geq 5)$$

for the homogeneous coefficient sequence [24]. Thus this single object exhibits the alignment

$$\boxed{\text{Sym}^3 \mathbb{H} \longleftrightarrow \text{Eichler depth } 4 \longleftrightarrow L(4, \chi_{-3}) \longleftrightarrow \text{mod-}p^4 \text{ supercongruence.}}$$

We emphasize that this is an example, not a universal implication from depth to congruence exponent. Its value for the adelic theory is that the archimedean regulator depth and an independently proved finite-place Frobenius divisibility scale coincide in a nontrivial character-valued symmetric-power system.

10.5 Local Frobenius of the extension

The finite-place structure is richer than congruences of coefficient sequences. Motivic elliptic Eisenstein classes have syntomic realizations on the ordinary locus, described explicitly by p -adic Eisenstein–Kronecker series [18, 19]. Thus for a p -adic realization of

$$0 \longrightarrow \mathbb{V} \longrightarrow \mathbb{E}_\nu \longrightarrow \mathbf{1} \longrightarrow 0,$$

one expects, after choosing a local splitting, a triangular Frobenius matrix of the schematic form

$$\Phi_p(\mathbb{E}_\nu) = \begin{pmatrix} \Phi_p(\mathbb{V}) & \kappa_p(\nu) \\ 0 & 1 \end{pmatrix}. \quad (41)$$

The off-diagonal entry $\kappa_p(\nu)$ is a finite-place extension/regulator coordinate. Beukers–Vlasenko prove in broad Calabi–Yau families that p -adic zeta values occur in Frobenius-matrix entries near maximally unipotent points [20]. This strongly motivates recording κ_p , not merely Lucas congruences, as part of the arithmetic fingerprint of a Hecke–Eisenstein extension.

Problem 10.4 (Syntomic Dirichlet regulator coordinate). *For the Eisenstein extensions in the present classification, identify $\kappa_p(\nu)$ explicitly with the appropriate normalized p -adic Dirichlet L -value, including the character, Euler-factor, and Hecke-degeneracy normalizations.*

The existence of syntomic Eisenstein classes is literature; the explicit identification required in Problem 10.4 is not asserted here.

10.6 Isogeny and almost-everywhere Frobenius

Layer A and the finite-place part of Layer C interact cleanly. If two elliptic families are isogenous of degree N , then away from bad primes and auxiliary primes dividing N , the associated ℓ -adic/crystalline homogeneous local systems have the same Frobenius data up to isomorphism. A fixed finite-index change of integral lattice changes denominators by only a bounded factor and therefore does not change the exponential LCM entropy. Thus the homogeneous Frobenius profile is an almost-everywhere isogeny invariant.

Proposition 10.5 (Almost-everywhere Frobenius isogeny principle). *Let two elliptic families be related by an isogeny of degree N on a common base. At every good auxiliary prime $\ell \nmid N$, their rational ℓ -adic/crystalline H^1 local systems are isomorphic after pullback, and so are all symmetric powers. Consequently the characteristic polynomials of geometric Frobenius on the homogeneous Layer-A fibres agree. Replacing one stable integral lattice by a fixed finite-index lattice changes coefficient denominators by only a bounded factor and does not alter exponential LCM entropy.*

The free-depth filtration is finer. It can change with the Eisenstein extension, the companion, the coordinate, or the chosen integral lattice. This is the finite-place analogue of the complex principle

isogeny controls the skeleton; the extension and realization control the special arithmetic.

For a degeneracy correspondence $C = \sum_d c_d V_d$, the complex Eichler realization satisfies

$$D^{-r}C = C^{[r]}D^{-r}, \quad C^{[r]} = \sum_d c_d d^{-r} V_d.$$

Motivic/syntomic functoriality suggests that the same correspondence transports the finite-place extension class and scales its target by the Mellin eigenvalue $P_C(r)$. We isolate the explicit form as an open problem.

Problem 10.6 (p -adic Hecke–Eisenstein functoriality). *Prove, with complete Euler-factor and integral-lattice normalizations, that the syntomic/crystalline extension coordinates of a Hecke-degeneracy transform $C\nu$ are obtained from those of ν by the same motivic correspondence and that the distinguished p -adic regulator component is multiplied by $P_C(r)$.*

10.7 Adelic Apéry realization

We can now state the enriched form of Layer C.

Definition 10.7 (Adelic Apéry realization). *An adelic Apéry realization of a Layer-B extension is a package*

$$\mathcal{R}_{\text{ad}} = (\sigma_{\text{dR}}, a, (t, \varphi), \Lambda, U^\bullet, \{\Phi_p, \kappa_p\}_{p \notin S}),$$

where

- σ_{dR} is the marked cyclic/de Rham section (or marked tuple) used to present the pure local system as a scalar or block differential system;
- a is a scalar or vector companion-descent morphism;
- t is the algebraic/modular coefficient coordinate and φ specifies the common complex continuation leaf or pullback used in holonomy;
- Λ is the integral lattice of coefficient functions;

- $U^\bullet$ is a Frobenius/free-depth filtration of the relevant holonomy space;
- at good finite places, Φ_p records the Frobenius structure of the pure realization and κ_p the extension coordinate when available.

The coarse quadruple $(\sigma_{\text{dR}}, a, t, \Lambda)$ is the archimedean-integral shadow obtained by forgetting the continuation and finite-place structures. The marked-section entry is necessary because contiguous or derivative presentations can alter the scalar recurrence without changing the Layer-A local system; see Theorem 15.1.

The quantitative Diophantine invariant attached to such a realization is not a single recurrence merit. It is the interaction of complex potential-theoretic capacity and Frobenius-filtered denominator entropy, measured in the simplest CDT setting by (40).

10.8 Known p -adic Apéry phenomena

The adelic formulation is not merely formal analogy. Calegari adapted Beukers' modular method to the p -adic setting, replacing complex Eichler functional equations by overconvergent p -adic modular forms and obtaining irrationality of the p -adic analogues of $\zeta(3)$ for $p = 2, 3$ and of Catalan's constant for $p = 2$ [21]. More recent arithmetic-holonomy work includes the 2-adic value $\zeta_2(5)$ among its applications [17]. These results support treating the archimedean and finite-place realizations as complementary parts of one Layer C rather than unrelated techniques.

11 Applications portfolio

A structural theory should earn its abstraction by explaining existing phenomena, proving classification or obstruction statements, and predicting new examples. We therefore record a portfolio of applications. Status labels are part of the statement.

11.1 Application A: Beukers and the prototype $\zeta(3)$ projector

LITERATURE. Beukers' modular interpretation of Apéry's proof uses the weight-four level-six Eisenstein source

$$E_4(\tau) - 28E_4(2\tau) + 63E_4(3\tau) - 36E_4(6\tau).$$

Its finite Dirichlet polynomial vanishes at the unwanted even positions and retains the $s = 3$ target [2].

PROGRAMME THEOREM/CERTIFICATE. In the present language this is not an isolated coefficient miracle: it is a point in the sign-refined interpolation space of Theorem 4.6. The recurrence appears only after the extension is paired with the elliptic Sym^2 host and an integral modular coordinate.

11.2 Application B: the modular Zagier sporadic universe

PROGRAMME THEOREM/CERTIFICATE. For the twelve modularly identified sporadic families

$$A, B, C, D, E, F, \quad \alpha, \gamma, \delta, \epsilon, \zeta, \eta,$$

the canonical companion source satisfies

$$\boxed{\Phi = F D t}$$

and is a pure Eisenstein series of weight $r + 1$, with explicit divisor-sum coefficients and zero cuspidal projection. The resulting companion is

$$y_B(t(q)) = F(q) D^{-r} \Phi(q).$$

The identities are supported in the programme by eta-quotient, Wronskian/Schwarzian, and Sturm certificates.

The three Cooper cubic systems s_7, s_{10}, s_{18} fit the same generalized symmetric-square mechanism; their first Eichler integration is arithmetically free because $(n+1) \mid A_n$ in the certified project analysis. The two infinite Zagier hypergeometric families HG_d and Leg_d exhibit a complementary phenomenon: their companions can be described explicitly, but the resulting convergence is degenerate or merely polynomial. These examples show that the source theorem is stronger than a list of attractive limits: it explains which parts of the Zagier universe can and cannot support exponential Apéry phenomena.

11.3 Application C: a finite $\zeta(3)$ extension classification

PROGRAMME THEOREM/CERTIFICATE. Among the six canonical cubic sporadic families, four untwisted $\zeta(3)$ -type sources lie over one homogeneous elliptic correspondence class. After normalizing the target period, they form four affinely independent extension points, hence an affine tetrahedron of dimension three. Three are full projector sources; one retains extra Tate/Eisenstein critical data. This is a concrete demonstration that the isogeny/homogeneous class is not the full classification invariant: one pure local system supports several distinct mixed extensions.

11.4 Application D: Ramanujan-Machine identities

This application requires careful attribution.

LITERATURE. Shvets proved the Domb Apéry limit and the Ramanujan-Machine conjecture Z_2 , and separately the level-8 Apéry limit and conjecture Z_1 [25, 26]. Yamamoto proved 38 Ramanujan-Machine conjectures and strongly generalized 31 of them by hypergeometric and difference-equation methods [22]. Thus this paper does not claim the first proof of those identities.

PROGRAMME THEOREM/CERTIFICATE. The classification explains why the Domb and level-8 identities belong to the same $\zeta(3)$ Eisenstein-extension landscape and why many Catalan continued fractions collapse to a much smaller collection of underlying hypergeometric/normal-function directions. Conservative Matrix Field coboundaries provide a complementary discrete equivalence language [27].

The payoff is conceptual rather than chronological: identities discovered one-by-one become orbit representatives of a smaller modular/extension structure.

11.5 Application E: new prime-isogeny $\zeta(3)$ completions

PROGRAMME THEOREM/CERTIFICATE. A level-6 parent source admits the prime completion

$$\Phi_p = (1 - p^2 V_p) \Phi_{\text{parent}}.$$

For $p = 5, 7, 11$, the project calculations land on genus-zero elliptic Apéry systems with exact targets

$$\frac{7}{20}\zeta(3), \quad \frac{3}{8}\zeta(3), \quad \frac{35}{88}\zeta(3),$$

respectively, together with certified Picard–Fuchs operators and all- n arithmetic. The construction is an example of the desired implication

classification $\rightarrow$ prediction $\rightarrow$ new systems.

The adelic theory adds a finite-place prediction. Away from the levels and the completion prime p , the homogeneous elliptic Frobenius polynomials, hence their Sym^2 polynomials, should

agree after the same isogeny pullback. The depth-three extension transforms by the integrated degeneracy factor

$$1 - p^{-1}V_p.$$

Thus any differences in Lucas or supercongruence quality among the $p = 5, 7, 11$ coefficient realizations should be attributed to the extension/lattice/coordinate rather than to a new homogeneous motive. This is one of the cleanest finite tests of Problems 10.4 and 10.6.

11.6 Application F: Catalan and intrinsic vector-valuedness

For Catalan's constant $G = L(2, \chi_{-4})$, the primitive Eisenstein sector has two Weyl orientations. At low level one obtains scalar exact systems such as the level-16 period-killed source. The quintic completion is qualitatively different.

PROGRAMME THEOREM/CERTIFICATE. On $X_0(20)$ the homogeneous elliptic system is rational rank two with quadratic isogeny algebra

$$\mathbb{Q}[J], \quad J^2 = 5I.$$

Yet the Catalan extension is intrinsically vector-valued: translation and Fricke generate a dihedral group of order eight on the orientation space, and no common invariant line exists. The same 5-isogeny diagonalizes the Picard–Fuchs system and the polynomial block recurrence, and an explicit Conservative Matrix Field coboundary transports the same decomposition. This is a theorem-level example in which isogeny, differential gauge, and discrete coboundary are three realizations of one homogeneous operation, while the extension orientation prevents scalar descent.

11.7 Application G: $\zeta(5)$ as the first multi-completion problem

The critical nuisance set for $r = 5$ has two interior directions. Thus the natural completion geometry is already two-dimensional:

$$(1 - p^2V_p)(1 - q^4V_q).$$

The minimal same-prime full projector is Fricke-even, while the successful level-12 scalar construction lies in the anti-Fricke sector and uses an independent weight-four companion. The sign-refined rank theorem predicts the uniqueness of that source in its divisor-oldform eigenspace. This is a clean example of the distinction between period purification and companion descent.

11.8 Application H: the hidden $\zeta(7)$ parent

PROGRAMME THEOREM/CERTIFICATE. The sign-refined classification exposed a fully purified anti-Fricke level-12 weight-eight source

$$\begin{aligned} \Phi_{12}^- = \frac{1}{480} & (E_8 - 572E_8(2\tau) + 11583E_8(3\tau) - 36608E_8(4\tau) \\ & + 46332E_8(6\tau) - 20736E_8(12\tau)), \end{aligned}$$

with

$$L(\Phi_{12}^-, 7) = \frac{209}{1728}\zeta(7).$$

A natural weight-six companion realizes it on the same level-12 elliptic host used by the $\zeta(5)$ construction, now through $\text{Sym}^6 \mathbb{H}$. The $p = 2$ completion

$$(1 - 16V_2)\Phi_{12}^-$$

recovers the previously known level-24 source, while the $p = 3$ completion produces a level-36 scalar system with target

$$\frac{2717}{23328}\zeta(7).$$

This is a particularly strong predictive application: the classification found a hidden parent and explained an existing higher-level object as an isogeny completion.

11.9 Application I: Tate controls $\zeta(2)$ and $\zeta(4)$

PROGRAMME THEOREM/CERTIFICATE. Section 7 tests the checkerboard on values that are already Tate. The theory produces a universal central projector in weight $4m$, recovers the level-16 $\zeta(2)$ Lambert identity, and predicts a level-64 $\zeta(4)$ projector. It also proves a useful no-go statement: a scalar weight-six Fricke eigensource cannot kill the $s = 2$ reflection while retaining $s = 4$. These controls show that the classification distinguishes the regulator and Tate sides rather than forcing all special values into one symmetric-power template.

11.10 Application J: the known-irrational $L(2, \chi_{-3})$ benchmark

LITERATURE. Calegari–Dimitrov–Tang prove the $\mathbb{Q}$ -linear independence of $1, \zeta(2), L(2, \chi_{-3})$ by arithmetic holonomy [16].

PROGRAMME THEOREM/CERTIFICATE. Section 12 shows that their two weight-three source forms are exactly

$$(1 - 8V_2)E_{3, \chi_{-3}, 1} \quad \text{and} \quad (1 - V_2)E_{3, 1, \chi_{-3}},$$

the inner and outer orientations predicted by the classification. The same section finds the hidden fully purified source

$$(1 - 2V_2)(1 - 8V_2)E_{3, \chi_{-3}, 1} = 3\Phi_C - 2\Phi_F$$

and an exact level-12 factorization $\Phi_{\text{pur}} = F_{12}D(u/3)$. CDT’s proof then demonstrates a crucial realization-level lesson: full purification is not universally optimal, because their successful holonomy argument deliberately retains both a non-Tate and a Tate period coordinate.

11.11 Application K: $\beta(4)$, the first higher-rank character stress test

PROGRAMME THEOREM/CERTIFICATE. Section 13 treats $r = 4$, $\chi = \chi_{-4}$. The sign-refined paired-orientation interpolation theorem forces the first fully purified divisor-oldform Fricke source to occur at level 24, with inner vector

$$(1, -56, 189, -216)$$

and target $\beta(4)/6$. The outer vector is exactly the Beukers annihilator $(1, -28, 63, -36)$, while the inner vector is its differential conjugate $DC_B D^{-1}$.

The same application constructs an independent Sym^3 isogeny companion, a pure Fricke-fixed-point formula with no extra π^4 correction, and a rational Galois-trace Apéry pair with

$$a_n \in \mathbb{Z}, \quad 4d_n^4 b_n \in \mathbb{Z}, \quad \frac{b_n}{a_n} \longrightarrow \frac{\beta(4)}{6}.$$

Most importantly, the rational trace coordinate forgets a degree-four modular cover. Hence the natural rational Picard–Fuchs module has rank $4 \cdot 4 = 16$, not the earlier heuristic value 8. This forces the direct-image refinement of Layer A used throughout the present formulation.

11.12 Application L: $L(3, \chi_5)$ and exact conjugate-fold geometry

PROGRAMME THEOREM/CERTIFICATE. Section 14 closes the analytic geometry of the η sporadic target $\Lambda = L(3, \chi_5)$. A half-translation identifies the level-20 presentation with the level-5 host, and the two dominant folds are the order-two elliptic points with

$$t_{\pm} = \frac{11 \pm 2i}{125}.$$

Their exact branch constants are

$$C_{\pm} = \frac{\Lambda}{2} \pm \frac{2i\pi^3}{125\sqrt{5}}.$$

Thus regulator centering separates the branch data into a symmetric non-Tate part and an antisymmetric Tate part. In the natural quadratic Hadamard realization, the centered row

$$U_1 = A \star B - \frac{\Lambda}{2}(A \star A)$$

kills the cross branch exactly while paying only the original depth-three denominator cost. However the exact branch matrix satisfies

$$\det M = (C_- - C_+)^3 \neq 0,$$

so no scalar combination of $A \star A, A \star B, B \star B$ can cancel all three nearest branch orbits. This gives both a positive reduction theorem and an exact scalar no-go theorem, and points naturally toward a larger holonomic/arithmetical-holonomy realization.

11.13 Application M: the level-100 Franke- η boundary test

LITERATURE. Franke's weight-four level-25 example gives a nonzero character Eisenstein form $f_4^{(5)}$ with vanishing complete scalar critical vector [13].

PROGRAMME THEOREM/CERTIFICATE. Sections 5 and 6 raise the Franke form and the level-20 η source to level 100 and completely test the three-oldform deformation space

$$\Phi_{\eta} + (a + bV_2 + cV_4)f_4^{(5)}.$$

The scalar critical kernel is all of $\mathbb{Q}^3$, but the Franke form splits into two diamond sectors; requiring the endpoint critical values to vanish in each sector leaves the unique line

$$\mathbb{Q}(1, -10, 16) = \mathbb{Q}(1 - 2V_2)(1 - 8V_2).$$

The full cusp-constant matrix has rank 3, so the boundary kernel is zero. The η and Franke residue supports are disjoint, and therefore the fixed-regulator, fixed-boundary affine fibre meets this deformation family only at the original η point. This exact worked example proves that scalar critical-null, orientation-null, and boundary-null are genuinely different notions.

11.14 Application N: adelic Layer C and the Cooper weight drop

The new Layer-C synthesis makes an old arithmetic phenomenon quantitatively compatible with modern arithmetic holonomy. For each Cooper cubic s_7, s_{10}, s_{18} , the project proves

$$(n+1) \mid A_n,$$

so the first nominal Eichler primitive is arithmetically free and the companion satisfies

$$d_n^2 B_n \in \mathbb{Z}$$

rather than the generic cubic bound $d_n^3 B_n \in \mathbb{Z}$. In the scalar case this is exactly a one-unit reduction of the CDT denominator penalty,

$$\tau_F : 3 \mapsto 2.$$

Thus the Cooper examples supply a clean finite-place validation of Definition 10.1 and Theorem 10.2, even though their archimedean branch geometry is still too weak for an irrationality theorem.

The same perspective turns several existing project constructions into concrete adelic testbeds. The prime-isogeny $\zeta(3)$ systems at $p = 5, 7, 11$ should have the same good-prime homogeneous Frobenius profile after pullback, while differences in Lucas/supercongruence quality should belong to the extension/lattice realization. In the centered $L(3, \chi_5)$ Hadamard system, the arithmetic task is no longer merely to prove a common depth-three denominator bound: one should compute the entire nested flag $U^1 \supseteq U^2 \supseteq U^3$ and optimize $\sum_j u_j^2$. Analogously, the most promising Catalan vector/direct-image systems should be evaluated by a two-axis search: maximize the common complex continuation leaf and simultaneously maximize the Frobenius-free arithmetic rank.

11.15 Application O: Shvets scan compression and a new $L(4, \chi_{-3})$ realization

LITERATURE. Shvets's eleven-row Belyi scan proves integrality and Sym^2 pullback structure for eleven additional sequences, while his neighboring symmetric-cube paper proves a mod- p^4 supercongruence for the level-three signature-three cube [23, 24].

PROGRAMME THEOREM/CERTIFICATE. Section 15 compresses the eleven rows to six finite-monodromy algebraic objects and five elliptic objects on only two Gauss progenitors. Row #15 is exactly a quadratic pullback of A036917, while row #2 is merely a contiguous de Rham section of the Legendre system. The Legendre mother itself has canonical source

$$(1 - V_2)(1 - 16V_2)\mathcal{E}_4,$$

exactly the programme's #11 $\zeta(3)$ parent, and its 3-isogeny completion is the Domb source.

The neighboring symmetric cube then supplies an independent depth-four character test: its canonical source is

$$E_{5, \chi_{-3}, 1}, \quad L(E_{5, \chi_{-3}, 1}, 4) = -\frac{1}{2}L(4, \chi_{-3}).$$

The classification manufactures a fully purified level-24 $\pm i$ Fricke doublet with target

$$\frac{7}{32}L(4, \chi_{-3})$$

and a rational Galois-trace Apéry pair satisfying

$$A_n \in \mathbb{Z}, \quad d_n^4 B_n \in \mathbb{Z}, \quad \frac{B_n}{A_n} \longrightarrow \frac{7}{32}L(4, \chi_{-3}).$$

The same example yields the general pullback-defect principle $\Phi_{\text{tr}} = R'(x)\Phi_x$: transported extensions and new coordinate-canonical exact sources are distinct under ramified algebraic pullback. Shvets's mod- p^4 theorem for the homogeneous cube provides a clean external finite-place check of the adelic Layer-C organization.

11.16 Application P: no-go theorems

The same theory proves negative statements. Entire candidate families can be excluded because

- the Fricke eigenspace is too small for the required period conditions;

- the full cusp-residue matrix has full rank;
- the orientation representation has no invariant line;
- the scalar companion kernel is trivial;
- or the arithmetic denominator entropy dominates the surviving analytic decay.

These are structural obstructions, not failed numerical searches.

11.17 What this portfolio demonstrates

No new irrationality theorem is required for the framework to have concrete utility. The applications already realize three logically different payoffs:

recover/explain known formulas,
 prove finite classification and impossibility results,
 predict and construct previously hidden modular systems.

An irrationality theorem would be a fourth and much deeper kind of payoff rather than the first evidence that the theory is useful.

12 A known-irrational regulator benchmark: $L(2, \chi_{-3})$

The value $L(2, \chi_{-3})$ is the most informative external benchmark currently available for the classification. It lies in the same opposite-parity depth-2 square as Catalan's constant, but Calegari–Dimitrov–Tang (CDT) prove the stronger statement that

$$1, \quad \zeta(2), \quad L(2, \chi_{-3})$$

are $\mathbb{Q}$ -linearly independent [16]. The project calculations reveal that the modular input of their arithmetic-holonomy proof is exactly the two-orientation Eisenstein geometry predicted here.

12.1 Primitive orientations and Zagier sources

Write $\chi = \chi_{-3}$ and

$$S = E_{3,\chi,1}, \quad T = E_{3,1,\chi}.$$

Then

$$L(S, s) = L(s, \chi)\zeta(s-2), \quad L(T, s) = \zeta(s)L(s-2, \chi).$$

At the target,

$$L(S, 2) = -\frac{1}{2}L(2, \chi).$$

The canonical Zagier source theorem in the programme gives, among others,

$$\Phi_A = (1 - V_2)T, \tag{42}$$

$$\Phi_C = (1 - 8V_2)S, \tag{43}$$

$$\Phi_F = (1 - 7V_2 - 8V_4)S. \tag{44}$$

Their natural source periods include

$$L(\Phi_A, 2) = \frac{1}{4}\zeta(2), \quad L(\Phi_C, 2) = \frac{1}{2}L(2, \chi), \quad L(\Phi_F, 2) = \frac{5}{8}L(2, \chi).$$

12.2 A hidden fully purified source

For $r = 2$, the complete primitive inner-orientation nuisance set is $s = 1, 3$. Hence the minimal power-of-2 projector is

$$(1 - 2V_2)(1 - 8V_2).$$

The resulting source is already hidden in the Zagier list.

Proposition 12.1 (Purified χ_{-3} combination).

$$\Phi_{\text{pur}} = (1 - 2V_2)(1 - 8V_2)S = 3\Phi_{\text{C}} - 2\Phi_{\text{F}}.$$

Moreover

$$L(\Phi_{\text{pur}}, 2) = \frac{1}{4}L(2, \chi_{-3}).$$

Proof. The source identity is the polynomial calculation

$$3(1 - 8X) - 2(1 - 7X - 8X^2) = 1 - 10X + 16X^2 = (1 - 2X)(1 - 8X).$$

At $s = 2$ the projector eigenvalue is $-1/2$, which multiplied by $L(S, 2) = -L(2, \chi)/2$ gives the target. $\square$

After normalizing both sources to the same target, one obtains the explicit target-zero translation

$$(2\Phi_{\text{pur}}) - \Phi_{\text{C}} = (1 - 4V_2)(1 - 8V_2)S,$$

which is a concrete point in the affine kernel of the regulator map.

12.3 Exactness on the level-12 elliptic host

Define

$$u(\tau) = 3 \frac{\eta(2\tau)^2 \eta(12\tau)^4}{\eta(4\tau)^4 \eta(6\tau)^2}, \quad \psi(\tau) = \frac{\eta(2\tau)^6 \eta(3\tau)}{\eta(\tau)^3 \eta(6\tau)^2}.$$

The exact project identities

$$\frac{D \log u}{\psi^2} = \frac{(1 - u)(9 - u^2)}{9(1 + u)}$$

and

$$\frac{\Phi_{\text{pur}}}{\psi^3} = -\frac{u(u - 3)(u - 1)(u + 3)(3u^2 + 4u - 3)}{81(1 + u)^3}$$

imply the following.

Theorem 12.2 (Scalar exactness of the purified benchmark). *Put*

$$F_{12} = \psi \frac{3 - 4u - 3u^2}{3(1 + u)^2}.$$

Then

$$\Phi_{\text{pur}} = F_{12} D(u/3).$$

Thus the fully inner-purified $L(2, \chi_{-3})$ extension lies on the rigid exactness locus of Section 2.

12.4 CDT use exactly the two predicted orientations

CDT work on $Y_0(6)$ with a weight-one period $A(\tau)$ and two double Eichler integrals $B_{\text{CDT}}, C_{\text{CDT}}$. Their weight-three source forms are exactly our two orientation representatives.

Theorem 12.3 (Source-level identification with CDT). *In the above notation,*

$$D^2 B_{\text{CDT}} = (1 - 8V_2)S = \Phi_C, \quad D^2 C_{\text{CDT}} = (1 - V_2)T = \Phi_A.$$

Consequently their two distinguished period coordinates are

$$\frac{1}{2}L(2, \chi_{-3}), \quad \frac{1}{4}\zeta(2).$$

Source-algebra proof. For the inner orientation, changing the divisor weight by $(-1)^{d-1}$ subtracts twice the even-divisor contribution; that contribution is 4 times the coefficient at $n/2$, giving $1 - 8V_2$. CDT's second source is the outer divisor sum minus its 2-dilate, giving $1 - V_2$. The period values then follow from the two Dirichlet-series factorizations. $\square$

This correspondence is a significant external validation of the classification. The successful CDT argument does not merely use vaguely analogous modular forms: it begins with precisely the inner and outer Eisenstein orientations singled out by the theory.

12.5 The arithmetic-holonomy realization

CDT form three functions H_A, H_B, H_C with

$$H_B/H_A = B_{\text{CDT}}, \quad H_C/H_A = C_{\text{CDT}}.$$

They establish d_n^2 -type arithmetic for the Taylor coefficients and, crucially, the corrected extensions

$$H_B - \frac{1}{2}L(2, \chi_{-3})H_A, \quad H_C - \frac{1}{4}\zeta(2)H_A$$

overconverge through the nearest singularity. Under a hypothetical rational relation among $1, \zeta(2), L(2, \chi_{-3})$, a rational linear combination of these corrected extensions would therefore produce a rational target-zero object with strong arithmetic and analytic properties. CDT then pass through a normalizer involution and an invariant quotient coordinate and prove a 14-function arithmetic-holonomy contradiction [16].

In the present language:

Layer A : one rank-two elliptic period system,
 Layer B : one non-Tate inner extension plus one Tate outer extension,
 Layer C : overconvergent corrections, normalizer descent, and a d_n^2 lattice.

Thus a successful modern irrationality proof acts directly on the kind of extension data classified here.

The adelic refinement of Layer C makes this benchmark even more useful. CDT's denominator array is not just a scalar depth-two bound: it is an adapted arithmetic profile across a 14-dimensional holonomy space. Recasting that profile as a Frobenius/free-depth flag in the sense of Definition 10.1 would provide the first calibration of the exact penalty

$$\tau_F = \sum_j b_j \left(1 - \frac{u_j^2}{m^2} \right)$$

on a theorem that is already known to succeed. This computation is therefore the highest-priority quantitative test of the adelic synthesis.

12.6 Task-dependent purification

The benchmark also corrects a possible overinterpretation of the period-annihilator philosophy. Full purification is natural if the goal is a single scalar CM value, but it need not be optimal for a multivariate Diophantine argument. CDT deliberately retain both

$$\frac{1}{2}L(2, \chi_{-3}) \quad \text{and} \quad \frac{1}{4}\zeta(2),$$

because their theorem proves joint linear independence with 1.

Proposition 12.4 (Task-dependent purification principle). *Within a fixed Eisenstein extension space, the preferred Hecke projector depends on the intended realization. Full period purification is canonical for isolating one regulator coordinate, but a vector or arithmetic-holonomy realization may profit from retaining controlled Tate directions and proving joint linear independence of the resulting period coordinates.*

The proposition is a structural lesson rather than an abstract optimality theorem. The CDT construction is the decisive concrete example.

12.7 Implication for Catalan

Catalan's constant $L(2, \chi_{-4})$ occupies the same depth-2 opposite-parity position but has a richer orientation representation. The CDT benchmark therefore suggests that a future Catalan arithmetic-holonomy attack should not insist on eliminating every Tate direction at the outset. A more faithful analogue would keep a controlled Catalan/Tate extension vector, analyze its normalizer orbit, and seek an arithmetic contradiction on the resulting vector-valued realization. The quintic D_4 orientation module suggests that such an analogue may be intrinsically vector-valued.

13 $\beta(4)$ at level 24: higher symmetric power, character orientation, and direct image

This section records the first battle test in which all three complications of the regulator side occur simultaneously: a nontrivial character, a higher odd symmetric power, and a nontrivial finite modular direct image. The source is the programme dossier [34]. We preserve its evidence distinction: the Eisenstein interpolation, Fricke exchange, Eichler fixed-point calculation, eta-transformation companion, and arithmetic consequences below are formal or exact consequences of the displayed formulas; the degree-four elimination and full rank-16 cyclicity are exact-computational certificates whose routine modular-function/divisor upgrade remains to be written.

13.1 Prediction and the minimal purified source

Take

$$r = 4, \quad \chi = \chi_{-4}.$$

On the orientation-resolved elliptic cover the predicted pure geometry is

$$\widetilde{\mathbb{V}}_{4, \chi_{-4}} \sim \text{Sym}^3 \widetilde{\mathbb{H}},$$

with the two character placements appearing as the Eisenstein orientation doublet

$$S = E_{5, \chi_{-4}, 1}, \quad T = E_{5, 1, \chi_{-4}}.$$

Their nonconstant Mellin transforms are

$$L(S, s) = \beta(s)\zeta(s-4), \quad L(T, s) = \zeta(s)\beta(s-4).$$

On the divisor support $(1, 2, 3, 6)$ define

$$\Phi_{\text{in}} = S - 56S(2\tau) + 189S(3\tau) - 216S(6\tau), \quad (45)$$

$$\Phi_{\text{out}} = T - 28T(2\tau) + 63T(3\tau) - 36T(6\tau). \quad (46)$$

The associated finite polynomials are

$$\begin{aligned} P_{\text{in}}(s) &= 1 - 56 \cdot 2^{-s} + 189 \cdot 3^{-s} - 216 \cdot 6^{-s}, \\ P_{\text{out}}(s) &= 1 - 28 \cdot 2^{-s} + 63 \cdot 3^{-s} - 36 \cdot 6^{-s}. \end{aligned}$$

Exact evaluation gives

$$P_{\text{in}}(1) = P_{\text{in}}(3) = P_{\text{in}}(5) = 0, \quad P_{\text{out}}(0) = P_{\text{out}}(2) = P_{\text{out}}(4) = 0,$$

and

$$P_{\text{in}}(4) = -\frac{1}{3}.$$

Since $\zeta(0) = -1/2$,

$$\boxed{L(\Phi_{\text{in}}, 4) = \frac{1}{6}\beta(4).} \quad (47)$$

The outer orientation has no $s = 4$ target term.

The source is also the first clean application of Proposition 4.4. If

$$C_B = 1 - 28V_2 + 63V_3 - 36V_6$$

is the Beukers annihilator, then

$$C_{\text{out}} = C_B, \quad C_{\text{in}} = DC_B D^{-1}.$$

Equivalently

$$P_{\text{out}}(s) = P_B(s), \quad P_{\text{in}}(s) = P_B(s - 1).$$

Thus the two $\beta(4)$ orientations are adjacent Mellin-degree realizations of the same finite Hecke correspondence.

13.2 Level-24 minimality

Proposition 4.5 gives, in the divisor-oldform sector at level $4D$,

$$\dim \mathcal{A}_{4D, \beta(4)}^{\pm i} = \max\{0, \tau(D) - 3\}.$$

At level 16 ($D = 4$), the obstruction determinant is

$$\det \begin{pmatrix} 1 & 1/2 & 1/4 \\ 1 & 1/8 & 1/64 \\ 1 & 1/32 & 1/1024 \end{pmatrix} = -\frac{135}{8192} \neq 0,$$

so no fully purified Fricke-eigen source exists on that support. At level 24 ($D = 6$), $\tau(D) = 4$ and the purified eigenspace is one-dimensional. The normalized inner vector is precisely

$$(1, -56, 189, -216).$$

Hence level 24 is the first divisor-oldform Fricke realization in this sector.

13.3 The exact orientation doublet and pure CM value

With the Fourier normalizations above, the programme calculation gives

$$\Phi_{\text{in}}|W_{24} = \frac{i\sqrt{6}}{16}\Phi_{\text{out}}, \quad \Phi_{\text{out}}|W_{24} = \frac{8i\sqrt{6}}{3}\Phi_{\text{in}}. \quad (48)$$

The product of the exchange constants is -1 . Consequently

$$\Phi_{\pm} = \Phi_{\text{in}} \pm \frac{\sqrt{6}}{16}\Phi_{\text{out}}$$

satisfy

$$\Phi_{\pm}|W_{24} = \pm i\Phi_{\pm}.$$

This is the two-orientation character geometry predicted by the general Eisenstein theory.

Let

$$\Theta_+ = D^{-4}\Phi_+, \quad C_4 = \frac{1}{6}\beta(4).$$

Because the six unwanted critical/endpoint slots have been killed, the standard Eichler transformation collapses to

$$\Theta_+(W_{24}\tau) - C_4 = \frac{i}{(\sqrt{24}\tau)^3}(\Theta_+(\tau) - C_4). \quad (49)$$

At the positive fixed point

$$\tau_* = \frac{i}{\sqrt{24}},$$

the multiplier is -1 , and hence

$$\boxed{\Theta_+(\tau_*) = \frac{1}{6}\beta(4).} \quad (50)$$

There is no separate π^4 correction term.

For reference, if

$$J_4(y) = \sum_{n \geq 1} \frac{\chi_{-4}(n)}{n^4(e^{ny} - 1)}, \quad K_4(y) = \sum_{n \geq 1} \frac{1}{n^4 \cosh(ny)}, \quad y = \frac{\pi}{\sqrt{6}},$$

then (50) yields the pure Lambert identity

$$\begin{aligned} \beta(4) &= 6J_4(y) - 21J_4(2y) + 14J_4(3y) - J_4(6y) \\ &\quad + \frac{3\sqrt{6}}{16}K_4(y) - \frac{21\sqrt{6}}{64}K_4(2y) + \frac{7\sqrt{6}}{48}K_4(3y) - \frac{\sqrt{6}}{192}K_4(6y). \end{aligned}$$

The dossier records a high-precision numerical audit; the proof route is the short Eichler derivation above.

13.4 Independent Sym^3 companion: exactness is not universal

Let

$$F(\tau) = \frac{\eta(2\tau)^{10}}{\eta(\tau)^4\eta(4\tau)^4},$$

the rank-two Catalan elliptic period. Direct eta transformation gives

$$F(W_{24}\tau) = -12i\tau F(6\tau), \quad F(6W_{24}\tau) = -2i\tau F(\tau).$$

Cubing yields the weight-three companion

$$A_{24}^- = F(\tau)^3 - 6\sqrt{6}F(6\tau)^3, \quad A_{24}^-|W_{24} = -iA_{24}^-. \quad (51)$$

Combining (49) and (51),

$$A_{24}^- \left(\Theta_+ - \frac{1}{6}\beta(4) \right)$$

is W_{24} -invariant.

This is conceptually decisive. The naive exact source F^3Dt is not the purified Eisenstein source. Nevertheless the predicted Sym^3 homogeneous geometry supports the extension through an independent companion. Thus the invariant Layer-C notion is the cohomological companion condition of Theorem 9.1, while $\Phi = FDt$ remains a special rigid locus.

13.5 Rational trace and arithmetic realization

Write

$$\Theta_{\text{in}} = D^{-4}\Phi_{\text{in}}, \quad \Theta_{\text{out}} = D^{-4}\Phi_{\text{out}}.$$

Taking the rational Galois trace of the companion construction gives

$$A(\tau) = F(\tau)^3,$$

and

$$B(\tau) = F(\tau)^3\Theta_{\text{in}}(\tau) - \frac{9}{4}F(6\tau)^3\Theta_{\text{out}}(\tau).$$

Use the rational coordinate

$$r = \left(\frac{\eta(2\tau)\eta(12\tau)}{\eta(4\tau)\eta(6\tau)} \right)^6, \quad z = \frac{r}{1+r} = q - q^2 - 5q^3 + 11q^4 - 2q^5 - \dots.$$

Write

$$A = \sum_{n \geq 0} a_n z^n, \quad B = \sum_{n \geq 0} b_n z^n.$$

The programme dossier proves from the integral q -series and integral mirror map that

$$\boxed{a_n \in \mathbb{Z}, \quad 4d_n^4 b_n \in \mathbb{Z}, \quad \frac{b_n}{a_n} \longrightarrow \frac{1}{6}\beta(4).} \quad (52)$$

The denominator sequence begins

$$1, 12, 72, 364, 1860, 9840, 53804, 302328, 1735644, \dots$$

The first Fricke branch is cancelled in the corrected trace, and the recorded level-24 geometry gives

$$\limsup_{n \rightarrow \infty} \left| \frac{b_n}{a_n} - \frac{1}{6}\beta(4) \right|^{1/n} \leq \frac{1}{\sqrt{2}}.$$

This is far from sufficient to beat the generic depth-four denominator cost, so no irrationality claim results.

13.6 Why the rational Picard–Fuchs rank is 16

The most important correction supplied by the battle test concerns Layer A. Let

$$t = \frac{\eta(\tau)^4 \eta(4\tau)^2 \eta(8\tau)^4}{\eta(2\tau)^{10}}$$

be the oriented Catalan elliptic coordinate. The dossier records an exact-computational elimination, checked through q^{70} , giving a degree-four algebraic relation

$$\begin{aligned} 0 = & 1024t^4z^4 - 1024t^4z^3 - 1024t^3z^4 + 1536t^3z^3 - 384t^3z^2 - 128t^3z + 32t^3 \\ & + 320t^2z^4 - 640t^2z^3 + 336t^2z^2 - 16t^2z - 12t^2 \\ & - 32tz^4 + 80tz^3 - 60tz^2 + 12tz + t + z^4 - 3z^3 + 3z^2 - z. \end{aligned} \quad (53)$$

The formal modular-function/divisor certificate for (53) remains to be written.

The elliptic period satisfies the rank-two equation

$$(1 - 12t + 32t^2)\theta_t^2 F + (-12t + 64t^2)\theta_t F + (-4t + 32t^2)F = 0.$$

Hence Sym^3 has rank 4 over $\mathbb{Q}(t)$. Since (53) has degree four in the oriented field over $\mathbb{Q}(z)$, Proposition 2.1 predicts

$$\boxed{\text{rank}_{\mathbb{Q}(z)} f_* \text{Sym}^3 \mathbb{H} = 4 \cdot 4 = 16.} \quad (54)$$

The dossier further records a finite-field cyclic-vector certificate modulo 1,000,003 at $z = 2$ with successive ranks $1, 2, \dots, 16$. Subject to the stated certificate status, F^3 is therefore generically cyclic of full order 16; there is no hidden order-8 scalar reduction.

After clearing denominators, the recorded rank-16 system has the form

$$Q(z)\theta_z Y = B(z)Y,$$

where $B(z)$ is an integral 16×16 matrix polynomial of degree at most 10 and

$$\begin{aligned} Q(z) = & 32(z-1)(2z-1)(4z^2-4z-1)(8z^2-8z+1) \\ & \times (16z^4-32z^3+20z^2-4z+1). \end{aligned}$$

The factorization records the cusps and several oriented-isogeny branch orbits forgotten by z .

13.7 Lessons for the general classification

The $\beta(4)$ experiment closes several conceptual loops at once.

1. **Layer A.** The predicted Sym^3 elliptic geometry is correct upstairs, but a rational quotient can require a higher-rank finite direct image. Orientation is deck/descent data inside that direct image, not a universally independent multiplicative rank.
2. **Layer B.** Full character-period purification is finite interpolation. The first possible divisor-oldform level is 24, and the purified source is unique up to scale.
3. **Layer C.** An independent isogeny companion realizes the extension even though the rigid exactness identity fails. Rational Galois trace then produces an integral scalar Apéry pair.
4. **Hecke functoriality.** Differential conjugation of the Beukers annihilator shifts the Mellin zeros by one and produces the inner $\beta(4)$ correspondence.
5. **Diophantine separation.** The construction is structurally complete but arithmetically too weak for a classical irrationality argument, confirming that Diophantine merit belongs to the realization layer rather than to the existence of the extension itself.

Thus $\beta(4)$ is the first application in the paper that simultaneously validates the symmetric-power prediction, the two-orientation extension theory, the non-exact companion formalism, and the direct-image refinement of Layer A.

14 $L(3, \chi_5)$: elliptic folds, centered tensor descent, and a scalar no-go theorem

This section records the $L(3, \chi_5)$ stress test supplied by the programme dossier [35]. It is especially useful because it closes an analytic branch-connection problem that had remained open in earlier drafts, and because it produces a rigorous obstruction to naive scalar tensorization. No irrationality theorem is claimed.

Put

$$\Lambda = L(3, \chi_5).$$

The canonical η sporadic source is a weight-four inner Eisenstein series with natural period

$$L(\Phi_\eta, 3) = \frac{1}{2}\Lambda.$$

The homogeneous geometry is the predicted rank-three Sym^2 elliptic system; the novelty of the example lies in the nontrivial character/fold geometry and in what happens after tensor realization.

14.1 Half-translation to the level-five host

The decisive simplification is the half-translation

$$z = \tau + \frac{1}{2}.$$

In this coordinate the level-20 η presentation becomes

$$t_\eta(\tau) = -\left(\frac{\eta(5z)}{\eta(z)}\right)^6, \quad F_\eta(\tau) = \frac{\eta(z)^5}{\eta(5z)}, \quad \Phi_\eta(\tau) = -E_{4, \chi_5, 1}(z). \quad (55)$$

Thus the two dominant folds are ordinary order-two elliptic points of $\Gamma_0(5)$ rather than rational cusps. They lie at

$$z_2 = \frac{2+i}{5}, \quad z_3 = \frac{3+i}{5},$$

fixed respectively by

$$\begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix}, \quad \begin{pmatrix} 3 & -2 \\ 5 & -3 \end{pmatrix}.$$

In the original τ coordinate these are

$$-\frac{1}{10} + \frac{i}{5}, \quad \frac{1}{10} + \frac{i}{5},$$

and their t -values are the conjugate roots

$$t_\pm = \frac{11 \pm 2i}{125} \quad (56)$$

of $1 - 22t + 125t^2$.

14.2 An exact elliptic fold/cusp formula

Let

$$\gamma = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in \Gamma_0(5)$$

be an order-two element with $\chi_5(-a) = -1$. Let $S = E_{4, \chi_5, 1}$ be the primitive inner Eisenstein series, put

$$E = D^{-3}S, \quad r = \frac{a}{c}, \quad z_0 = \frac{a+i}{c},$$

and take the canonical companion with $\Theta = -E$.

Proposition 14.1 (Elliptic fold/cusp formula). *The local branch-cancellation constant at the elliptic fixed point z_0 is*

$$\boxed{C(z_0) = -E(r)}.$$

Proof. The Eichler cocycle

$$P_\gamma(z) = (cz - a)^2 E(\gamma z) + E(z)$$

is a degree-two polynomial satisfying

$$P_\gamma(\gamma z) = (cz - a)^{-2} P_\gamma(z).$$

The corresponding polynomial eigenspace is one-dimensional and is spanned by

$$Q_\gamma(z) = cz^2 - 2az - b.$$

Since $Q_\gamma(r) = 1/c$ and $P_\gamma(r) = E(r)$,

$$P_\gamma = cE(r)Q_\gamma.$$

Differentiating at z_0 gives

$$E'(z_0) = ci(E(r) - E(z_0)).$$

For the weight-two period f one has $f'(z_0)/f(z_0) = ci$. Therefore

$$\Theta(z_0) + \frac{f(z_0)\Theta'(z_0)}{f'(z_0)} = -E(r),$$

as claimed. □

This proposition is a useful local refinement of the general companion-descent picture: a branch constant at an elliptic fold is controlled by an Eichler value at the rational boundary point naturally attached to the involution.

14.3 Exact branch constants

Abel continuation at $r = 2/5, 3/5$ gives

$$E(r) = -\frac{\Lambda}{2} + \frac{i}{2} \sum_{m \geq 1} \frac{\chi_5(m) \cot(\pi mr)}{m^3}.$$

The finite reflection calculation yields

$$\sum_{m \geq 1} \frac{\chi_5(m) \cot(2\pi m/5)}{m^3} = \frac{4\pi^3}{125\sqrt{5}},$$

with the conjugate negative value at $3/5$. Hence, up to relabeling,

$$\boxed{C_\pm = \frac{\Lambda}{2} \pm \frac{2i\pi^3}{125\sqrt{5}}}. \tag{57}$$

Thus the symmetric part of the branch data is the non-Tate target $\Lambda/2$, while the antisymmetric part is purely Tate.

14.4 Centered Hadamard normal form

Let A denote the homogeneous coefficient row and B its canonical companion row, and write $\star$ for coefficientwise Hadamard product. Set

$$X = A \star A, \quad Y = A \star B, \quad Z = B \star B.$$

Put

$$a = \frac{2\pi^3}{125\sqrt{5}},$$

so $C_{\pm} = \Lambda/2 \pm ia$. The target-centered basis is

$$U_0 = X, \quad U_1 = Y - \frac{\Lambda}{2}X, \quad U_2 = Z - \Lambda Y + \frac{\Lambda^2}{4}X. \quad (58)$$

At the side-plus, cross, and side-minus Hadamard punctures, the branch-functional matrix is exactly

$$\begin{pmatrix} 1 & ia & -a^2 \\ 2 & 0 & 2a^2 \\ 1 & -ia & -a^2 \end{pmatrix}. \quad (59)$$

Consequently U_1 kills the cross branch exactly, and U_2 has only pure π^6 branch constants. Equivalently, for

$$H_{\Lambda} = \Lambda(A \star A) - 2(A \star B) = -2U_1,$$

the cross puncture $t_+t_- = 1/125$ has zero logarithmic coefficient, while the side punctures retain only

$$\mp \frac{4i\pi^3}{125\sqrt{5}}.$$

Thus after projection away from the L -value direction, the entire nearest residual Hadamard monodromy is Tate.

This is the concrete model for Theorem 9.7: centering a tensor realization by the regulator can turn all remaining connection constants into Tate data even when scalar overconvergence is impossible.

14.5 Exact three-branch scalarization obstruction

In the uncentered basis (X, Y, Z) the three branch functionals are

$$M = \begin{pmatrix} 1 & C_+ & C_+^2 \\ 2 & C_+ + C_- & 2C_+C_- \\ 1 & C_- & C_-^2 \end{pmatrix}.$$

A direct determinant calculation gives

$$\boxed{\det M = (C_- - C_+)^3 = \left(-\frac{4i\pi^3}{125\sqrt{5}}\right)^3 \neq 0.} \quad (60)$$

Hence no nonzero scalar combination

$$\alpha X + \beta Y + \gamma Z$$

over any coefficient field can be holomorphic through all three nearest Hadamard branch orbits. This is an exact no-go theorem for the entire natural rank-three scalar tensorization. Any further improvement must enlarge the realization space, for example by derivatives/extensions with a full-germ argument, target-zero normal-function directions, modular/Galois traces, or arithmetic holonomy.

14.6 Order-six module and arithmetic lattice

The programme constructs an all- n order-six Hadamard differential certificate. In its centered basis the operator L_6 satisfies

$$\begin{aligned} L_6 U_0 &= 0, \\ L_6 U_1 &= x(x-1)(5x^2 - 56x + 5), \\ L_6 U_2 &= -x(1+x)^4 - \Lambda x(x-1)(5x^2 - 56x + 5). \end{aligned}$$

Under the hypothesis $\Lambda \in \mathbb{Q}$, the right-hand sides are rational polynomials and all three centered rows have rational Taylor coefficients.

The local exponents are

$$\begin{array}{c|c} 0 & 0, 0, 0, 0, 0, 0 \\ 1, \alpha, \bar{\alpha} & 0, 1, 2, 2, 3, 4 \\ -1 & 0, 1, 2, 4, 5, 6 \\ \infty & 1, 1, 1, 1, 1, 1 \end{array}$$

where $125x^2 - 234x + 125 = 0$ at $\alpha, \bar{\alpha}$. A complete subset-sum audit rules out every proper factor that introduces no new apparent singularities. This is a rigorous singularity-preserving minimality statement, not yet full irreducibility. The good-prime p -curvature computation at 7 is regular nilpotent with ranks 5, 4, 3, 2, 1, 0, consistent with global nilpotence/MUM geometry but not by itself an irreducibility proof.

For the original η rows,

$$A_0 = 1, \quad A_1 = 5, \quad B_0 = 0, \quad B_1 = 1,$$

and the discrete Wronskian is exactly

$$A_n B_{n-1} - A_{n-1} B_n = -\frac{125^{n-1}}{n^3}. \quad (61)$$

The modular companion theorem gives

$$A_n \in \mathbb{Z}, \quad d_n^3 B_n \in \mathbb{Z}.$$

Consequently

$$X_n \in \mathbb{Z}, \quad d_n^3 Y_n \in \mathbb{Z}, \quad d_n^6 Z_n \in \mathbb{Z}.$$

If $\Lambda = a/b \in \mathbb{Q}$, then

$$b d_n^3 [z^n] H_\Lambda \in \mathbb{Z}. \quad (62)$$

The important point is that the cross-purified centered row U_1 removes the non-Tate cross branch while paying only the original depth-three arithmetic weight, rather than the naive tensor weight six. In the adelic language of Section 10, the next arithmetic object is therefore not a single improved denominator exponent but the full flag

$$U^1 \supseteq U^2 \supseteq U^3$$

inside the chosen holonomy module. Even a proper nonzero U^1 lowers the CDT penalty; the quantitative target is to maximize $u_1^2 + u_2^2 + u_3^2$ while preserving the common continuation leaf.

14.7 The level-100 Franke deformation test

The critical-invisible family from Section 5 has now been subjected to the full Layer-B test. The result is summarized in the worked example of Section 6. Although every vector

$$(a + bV_2 + cV_4)f_4^{(5)}$$

is invisible to Franke's *scalar* critical coordinates, the form splits into trivial and quadratic diamond sectors whose endpoint critical values cancel only after the sectors are added. Requiring vanishing in each sector separately leaves exactly

$$(a, b, c) \in \mathbb{Q}(1, -10, 16),$$

that is,

$$f_{\text{pur}} = (1 - 2V_2)(1 - 8V_2)f_4^{(5)}.$$

The complete level-100 cusp-constant matrix has rank three on the Franke three-oldform space, so no nonzero direction survives the boundary map.

Moreover the η residues are supported only at denominator 1, 2, 4 cusp classes, while the Franke residues are supported on denominator 5, 10, 20 classes. Thus the two supports are disjoint. Proposition 8.6 gives

$$\boxed{(\Phi_\eta + \text{span}\{f_{\text{Fr}}, V_2 f_{\text{Fr}}, V_4 f_{\text{Fr}}\}) \cap \{\text{fixed regulator and fixed boundary}\} = \{\Phi_\eta\}.$$

The unique orientation-critical-null direction is nevertheless geometrically structured: it is W_{100} -anti-Fricke. It also contains both diamond characters, so any realization retaining it must be a higher-rank/vector direct image rather than another scalar order-three η recurrence.

This closes the deformation experiment suggested in Version 5. The failure is useful: it proves in a concrete externally motivated family that

$$\boxed{\text{Franke critical-null} \neq \text{orientation-null} \neq \text{boundary-null}.$$

14.8 What the example changes in the general theory

The $L(3, \chi_5)$ application supplies six general lessons.

1. *Critical fibres must be symmetry-resolved before boundary purification.* The Franke deformation test exhibits the strict kernel dimensions $3 \rightarrow 1 \rightarrow 0$.
2. *Elliptic folds need not be cusp folds.* Their connection constants can nevertheless be reduced to boundary Eichler values by the elliptic involution, as Proposition 14.1 demonstrates.
3. *Centering by the regulator is a natural realization operation.* It can convert residual connection data into pure Tate data and lower the relevant arithmetic weight.
4. *Scalar/vector necessity can be proved by local branch functionals.* The determinant (60) proves that no scalar miracle exists inside the natural tensor span.
5. *The arithmetic frontier can begin only after the analytic geometry is essentially closed.* Here the remaining tasks are full irreducibility, the exact prime-by-prime lattice of U_1 , a suitable pure-Tate auxiliary or arithmetic-holonomy tuple, linear independence, and the final energy/capacity inequality.
6. *The arithmetic frontier is adelic.* Once the complex branch functionals are known exactly, the finite-place problem is to compute a Frobenius-adapted free-depth flag on the same holonomy space and use Theorem 10.2, rather than treating d_n^3 as a final scalar invariant.

The example therefore complements $\beta(4)$: $\beta(4)$ forces the direct-image correction in Layer A, whereas $L(3, \chi_5)$ forces a centered tensor/vector refinement in Layer C.

15 Shvets Belyi pullbacks and symmetric-cube cross-pollination

This section records an external stress test of all three layers. Shvets's 2026 work on order-three π -formulas and conservative matrix fields produces eleven additional integral Belyi-pullback sequences of the form

$$A(x) = {}_2F_1(a, b; c; \phi(\lambda x))^2,$$

proves coefficient integrality, and places every row in a Sym^2 pullback/twist framework [23]. A neighboring paper studies the symmetric cube of the signature-three Gauss system and proves a mod- p^4 supercongruence for its homogeneous coefficients [24]. The project calculation summarized here asks a different question: what survives after the raw recurrence list is passed through the three-layer classification? The answer gives a substantial compression at Layer A, an exact identification of an existing $\zeta(3)$ extension at Layer B, and a new $L(4, \chi_{-3})$ extension and rational Apéry realization at Layer C. The detailed project derivations are recorded in [36].

15.1 Layer-A compression of the eleven-row scan

Theorem 15.1 (Shvets scan compression). *Among the eleven integral Belyi-pullback rows in Shvets's scan:*

1. *six rows, namely #3, #5, #9, #10, #11, #13, have finite projective monodromy and algebraic generating functions;*
2. *the remaining five rows, #2, #4, #12, #14, #15, are elliptic Sym^2 pullbacks supported on only two Gauss progenitor types:*

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right) \quad \text{and} \quad {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; z\right);$$

3. *row #15 is exactly the quadratic pullback*

$$A_{15}(x) = A_{AZ}(x(1 - 4x))$$

of the classical A036917 period;

4. *row #2 is a contiguous de Rham section of the Legendre system, since*

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 2; z\right) = 4(1 - z) \frac{d}{dz} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right);$$

5. *rows #4 and #14 are algebraic Kummer gauges of the same pulled-back signature-three local system, while row #12 belongs to the same signature-three progenitor after the standard Pfaff permutation of hypergeometric singular points.*

Proof sketch. Rows #5, #9, #11, #13 are algebraic in Shvets's treatment. For rows #3 and #10 one applies the classical special-value identities

$${}_2F_1\left(\frac{1}{6}, \frac{2}{3}; \frac{3}{2}; z\right) = \frac{3}{4\sqrt{z}} \left((1 + \sqrt{z})^{2/3} - (1 - \sqrt{z})^{2/3} \right)$$

and

$$4zY^3 - 27Y + 27 = 0, \quad Y = {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; \frac{3}{2}; z\right).$$

The remaining assertions follow from the derivative identity above together with Euler and Pfaff transformations. The exact row formulas and pullbacks are those of Shvets's scan. These operations preserve the differential local system up to algebraic gauge/twist, which gives the stated Layer-A compression. $\square$

Remark 15.2 (Progenitor equivalence versus explicit isogeny). Theorem 15.1 is deliberately a local-system/progenitor statement. For the signature-three triple it does not, by itself, assert that every pair of pulled-back x -families is connected by a specified elliptic isogeny over the displayed base. Such a stronger statement requires a common modular-cover normalization. This is exactly the distinction between hypergeometric progenitor equivalence and a certified isogeny correspondence in Layer A.

The compression is already substantial:

$$\begin{aligned} 11 \text{ integral recurrences} &\longrightarrow 6 \text{ algebraic} + 5 \text{ elliptic} \\ &\longrightarrow 2 \text{ elliptic progenitor types.} \end{aligned}$$

It is therefore an example in which recurrence-level novelty greatly overcounts geometric novelty.

15.2 The Legendre mother is exactly the project $\zeta(3)$ parent

The Layer-B identification is stronger. Put

$$\begin{aligned} z &= \lambda(2\tau), & f(z) &= {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; z\right) = \theta_3(2\tau)^2, \\ F &= f^2, & t &= \frac{z}{16}. \end{aligned}$$

The hypergeometric Wronskian gives

$$Dz = z(1-z)f^2.$$

Hence the canonical cubic exact source is

$$\Phi_L = F D t = \frac{1}{16} z(1-z)f^4. \quad (63)$$

Its Fourier expansion is

$$q - 8q^2 + 28q^3 - 64q^4 + 126q^5 - 224q^6 + \cdots.$$

If

$$\mathcal{E}_4 = \sum_{n \geq 1} \sigma_3(n) q^n,$$

then the half-translation identity yields the exact Eisenstein decomposition

$$\boxed{\Phi_L = (1 - V_2)(1 - 16V_2)\mathcal{E}_4.} \quad (64)$$

Thus

$$L(\Phi_L, 3) = \frac{7}{16} \zeta(3).$$

In the notation of the project, (64) is precisely the canonical #11 $\zeta(3)$ parent.

This gives an exact bridge from Shvets's Legendre Gauss mother to the project's isogeny ladder. In particular

$$\Phi_p = (1 - p^2 V_p) \Phi_L, \quad L(\Phi_p, 3) = \frac{7(p-1)}{16p} \zeta(3),$$

and the case $p = 3$ gives the Domb/ α source

$$\boxed{\Phi_\alpha = (1 - 9V_3) \Phi_L, \quad L(\Phi_\alpha, 3) = \frac{7}{24} \zeta(3).} \quad (65)$$

The relation between Shvets's A036917 kernel and Domb kernel therefore persists beyond Sym^2/CMF functoriality: their canonical Eisenstein normal functions are connected by the same Hecke/isogeny correspondence.

15.3 Nonlinear pullback: transported extension versus canonical coordinate source

The Shvets rows make a general distinction visible. Let $y = R(x)$ be an algebraic pullback, let $F_y(y)$ be a homogeneous period, and put

$$F_x(x) = F_y(R(x)).$$

Assume the old realization has the rigid exact source

$$\Phi_y = F_y Dy.$$

There are then two natural source constructions downstairs in the x -presentation:

$$\Phi_{\text{tr}} := F_x Dy \quad \text{and} \quad \Phi_x := F_x Dx.$$

The first transports the existing Layer-B extension. The second re-canonicalizes the rigid exactness condition using the new coefficient coordinate.

Proposition 9.4 gives the general relation

$$\boxed{\Phi_{\text{tr}} = R'(x)\Phi_x.}$$

Thus the natural pullback of Layer B and the new rigid exact-source normalization agree only when the ramification factor is harmless. Definition 9.5 records the resulting principal-part defect.

Row #15 gives the simplest example:

$$y = x(1 - 4x), \quad R'(x) = 1 - 8x.$$

The point $x = 1/8$ lies over the ordinary point $y = 1/16$ of the old A036917 realization. The transported $\zeta(3)$ extension is regular there, while the source obtained by declaring x to be a new exact coordinate carries the reciprocal ramification factor $(1 - 8x)^{-1}$.

For the signature-three Belyi map

$$z = \frac{27x}{(1 - 4x)^3},$$

one has

$$\frac{dz}{dx} = \frac{27(1 + 8x)}{(1 - 4x)^4}.$$

The critical point $x = -1/8$ maps to the ordinary Gauss point $z = -1$. Hence the x -canonical source again acquires a ramification denominator, even though the transported local system is perfectly regular. This explains why arbitrary Belyi pullback and canonical modular exactness are not equivalent notions.

15.4 A meromorphic signature-three square source

The same phenomenon can occur even before an additional Belyi pullback. Let

$$a(q) = \sum_{m,n \in \mathbb{Z}} q^{m^2 + mn + n^2}, \quad \alpha = \frac{c(q)^3}{a(q)^3},$$

where

$$b(q) = \frac{\eta(\tau)^3}{\eta(3\tau)}, \quad c(q) = 3 \frac{\eta(3\tau)^3}{\eta(\tau)}, \quad a^3 = b^3 + c^3.$$

Then

$${}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; \alpha\right) = a, \quad D\alpha = \alpha(1 - \alpha)a^2.$$

For the square period $F = a^2$ and $t = \alpha/27$,

$$\Phi_{\text{sq}} = FDt = \frac{1}{27}\alpha(1 - \alpha)a^4 = \frac{\eta(\tau)^6\eta(3\tau)^6}{a(q)^2}. \quad (66)$$

Thus the rigid source is weight four but meromorphic at the order-three elliptic point where a vanishes. The signature-three Shvets square rows therefore belong naturally to a meromorphic exact-source stratum with controlled principal parts rather than to the holomorphic Eisenstein locus. This is an independent reason that the foundational theory should distinguish extension transport from a holomorphic exact-source normalization.

15.5 A symmetric-cube regulator: $L(4, \chi_{-3})$

Shvets's symmetric-cube paper provides an especially clean neighboring test. Set

$$t_0 = \frac{\eta(3\tau)^{12}}{\eta(\tau)^{12}}, \quad F_3 = \frac{\eta(\tau)^9}{\eta(3\tau)^3}.$$

Shvets writes

$$C = F_3 D \log t_0 = 3E_{5,1,\chi_{-3}}$$

in his Fourier normalization [24]. The canonical exact source of the present theory uses Dt_0 , not $D \log t_0$:

$$\boxed{\Phi_{\text{cube}} = F_3 Dt_0 = t_0 C = E_{5,\chi_{-3},1}.} \quad (67)$$

Consequently

$$\boxed{L(\Phi_{\text{cube}}, 4) = -\frac{1}{2}L(4, \chi_{-3}).} \quad (68)$$

This is exactly the checkerboard prediction

$$r = 4, \quad \text{Sym}^{r-1} \mathbb{H} = \text{Sym}^3 \mathbb{H}, \quad \chi = \chi_{-3}.$$

Moreover the two inducing-character orientations are exchanged by the level-three Fricke involution, exactly as required by the orientation-module formalism.

Remark 15.3 (Logarithmic versus coordinate source). Equation (67) is a useful normalization warning. The logarithmic source $FD \log t$ and the coordinate source FDt differ by a factor of t . The former is natural in coefficient extraction and Hecke arguments; the latter is the rigid exact source attached to the differential dt . They should not be identified when discussing Layer-B regulator normalization.

15.6 A new fully purified level-24 doublet

Let

$$S = E_{5,\chi_{-3},1}, \quad T = 3E_{5,1,\chi_{-3}}.$$

Define the same-prime purifications

$$\begin{aligned} S_{\text{pur}} &= (1 - 2V_2)(1 - 8V_2)(1 - 32V_2)S, \\ T_{\text{pur}} &= (1 - V_2)(1 - 4V_2)(1 - 16V_2)T. \end{aligned}$$

Their finite Mellin polynomials are

$$\begin{aligned} P_S(s) &= (1 - 2^{1-s})(1 - 2^{3-s})(1 - 2^{5-s}), \\ P_T(s) &= (1 - 2^{-s})(1 - 2^{2-s})(1 - 2^{4-s}). \end{aligned}$$

Thus

$$P_S(1) = P_S(3) = P_S(5) = 0, \quad P_T(0) = P_T(2) = P_T(4) = 0,$$

and

$$P_S(4) = -\frac{7}{16}.$$

Using (68),

$$\boxed{L(S_{\text{pur}}, 4) = \frac{7}{32} L(4, \chi_{-3}).} \quad (69)$$

The level-24 Fricke exchange is

$$S_{\text{pur}}|W_{24} = \frac{2i\sqrt{2}}{27} T_{\text{pur}}, \quad T_{\text{pur}}|W_{24} = \frac{27i\sqrt{2}}{4} S_{\text{pur}}.$$

Therefore

$$\Phi_{\pm} = S_{\text{pur}} \pm \frac{2\sqrt{2}}{27} T_{\text{pur}} \quad (70)$$

satisfy

$$\Phi_+|W_{24} = +i\Phi_+, \quad \Phi_-|W_{24} = -i\Phi_-.$$

The outer orientation has zero $s = 4$ target, so both eigenlines retain the value in (69). This is the χ_{-3} analogue of the level-24 $\beta(4)$ orientation doublet.

At the positive Fricke fixed point $\tau_* = i/\sqrt{24}$, the standard depth-four Eichler transformation gives

$$\boxed{D^{-4}\Phi_+(\tau_*) = \frac{7}{32} L(4, \chi_{-3}).} \quad (71)$$

Equivalently one obtains a pure Lambert/polylogarithm identity with no additional non-target period. This construction is project-derived; no novelty claim for the explicit CM identity is made here.

15.7 A rational vector-descended Apéry realization

The purified doublet admits a rational trace realization. Put

$$U(\tau) = \frac{\eta(\tau)^9}{\eta(3\tau)^3}, \quad V(\tau) = t_0(\tau)U(\tau) = \frac{\eta(3\tau)^9}{\eta(\tau)^3}.$$

The level-24 combinations

$$A_- = U(2\tau) - 54\sqrt{2}V(4\tau), \quad A_+ = U(2\tau) + 54\sqrt{2}V(4\tau)$$

have opposite $\mp i$ Fricke characters. Pairing them with the two Eisenstein eigenlines and taking the $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ trace gives

$$\boxed{A_{\mathbb{Q}}(\tau) = U(2\tau), \quad B_{\mathbb{Q}}(\tau) = U(2\tau)\Theta_S(\tau) - 8V(4\tau)\Theta_T(\tau),} \quad (72)$$

where

$$\Theta_S = D^{-4}S_{\text{pur}}, \quad \Theta_T = D^{-4}T_{\text{pur}}.$$

The rational coefficient -8 is forced by the product of the two quadratic orientation constants.

A convenient W_{24} -invariant Hauptmodul is

$$x(\tau) = \frac{\eta(\tau)^2\eta(24\tau)^2\eta(4\tau)\eta(6\tau)}{\eta(2\tau)\eta(12\tau)\eta(3\tau)^2\eta(8\tau)^2} = q - 2q^2 + 2q^4 - 3q^5 + \dots \quad (73)$$

With

$$g = \eta(2\tau)\eta(4\tau)\eta(6\tau)\eta(12\tau), \quad Y = -\frac{Dx}{g},$$

the project calculation gives the exact elliptic relation

$$\boxed{Y^2 = x^4 + 8x^3 + 2x^2 - 8x + 1.} \quad (74)$$

The positive fixed point maps to

$$x_1 = -2 - \sqrt{3} + 2\sqrt{2 + \sqrt{3}},$$

and the next positive branch value is

$$x_2 = -2 + \sqrt{3} + 2\sqrt{2 - \sqrt{3}},$$

with the exact ratio

$$\boxed{\frac{x_1}{x_2} = 3 - 2\sqrt{2}.} \quad (75)$$

Write

$$A_{\mathbb{Q}} = \sum_{n \geq 0} A_n x^n, \quad B_{\mathbb{Q}} = \sum_{n \geq 0} B_n x^n.$$

Then the project calculation proves

$$\boxed{A_n \in \mathbb{Z}, \quad d_n^4 B_n \in \mathbb{Z}.} \quad (76)$$

The Fricke branch cancellation and the level-24 singularity model give

$$\boxed{\frac{B_n}{A_n} \longrightarrow \frac{7}{32} L(4, \chi_{-3}),} \quad (77)$$

and, after the standard cusp-value audit on the same modular curve,

$$\limsup_{n \rightarrow \infty} \left| \frac{B_n}{A_n} - \frac{7}{32} L(4, \chi_{-3}) \right|^{1/n} \leq 3 - 2\sqrt{2}. \quad (78)$$

This is not an irrationality proof: the depth-four denominator cost remains too large. Its significance is structural. Starting from an external symmetric-cube object, the classification identifies its regulator, manufactures the unique natural purified orientation doublet, and constructs a rational vector-descended Apéry realization.

15.8 An adelic validation: Sym^3 , depth four, and a mod- p^4 congruence

Shvets independently proves that the homogeneous symmetric-cube coefficient sequence satisfies

$$A_{mp} \equiv A_m \pmod{p^4} \quad (p \geq 5),$$

using the level-three modular realization and Fricke–Hecke descent [24]. In the present language the same object has

$$\text{Sym}^3 \mathbb{H} \longleftrightarrow \text{Eichler depth 4} \longleftrightarrow L(4, \chi_{-3}).$$

The additional mod- p^4 supercongruence is therefore an unusually clean finite-place confirmation of the adelic organization of Layer C. We do *not* infer a general theorem saying that depth r forces a mod- p^r congruence; the point is that in this independently studied example the motivic symmetric-power index, regulator depth, and Frobenius congruence depth align exactly.

15.9 What the Shvets cross-pollination teaches the general theory

The application gives four general lessons.

1. **A marked de Rham section belongs to the realization.** Row #2 changes the scalar recurrence by choosing a contiguous derivative section inside the same Legendre H^1 . Hence Layer C should record the cyclic/de Rham section used to present the pure local system, not only its coordinate and lattice.
2. **Extension transport and coordinate re-canonicalization are different.** Proposition 9.4 isolates the exact defect factor $R'(x)$. Nonlinear pullback can preserve Layers A and B while pushing the new coordinate source out of the holomorphic exact locus.
3. **Meromorphic exact-source strata are unavoidable.** Signature-three squares naturally have principal parts at elliptic points. A complete theory must allow exact sources with controlled ramification/elliptic principal parts rather than treating holomorphic Eisenstein exactness as universal.
4. **External Layer-A data can predict a new Layer-B target.** The symmetric-cube modular dictionary immediately identifies the opposite-parity value $L(4, \chi_{-3})$, and the Hecke/Fricke classification then produces a new fully purified level-24 doublet and a rational Apéry realization.

Thus the Shvets corpus supplies one of the clearest demonstrations so far of the purpose of the classification:

compress raw recurrences $\longrightarrow$ identify/transport extensions $\longrightarrow$ predict new regulator realizations
--

16 The parity checkerboard and a periodic table of tests

The three-layer theory becomes easier to visualize when organized by depth r and character parity.

16.1 Tate versus regulator parity

For a primitive character χ , the condition

$$\chi(-1) = (-1)^{r+1}$$

selects the opposite-parity values that enter the Eisenstein-regulator story developed here. The complementary parity gives the classical critical/Tate side.

For the trivial character this is simply the familiar distinction

$$\zeta(2), \zeta(4), \dots \quad \text{Tate}, \quad \zeta(3), \zeta(5), \dots \quad \text{opposite parity}.$$

For χ_{-4} it is shifted:

$$\beta(2), \beta(4), \dots \quad \text{opposite parity}, \quad \beta(1), \beta(3), \dots \quad \text{critical/Tate-type}.$$

16.2 A first periodic table

	$r = 2$	$r = 3$	$r = 4$	$r = 5$
$\chi = 1$	$\zeta(2)$ (Tate control)	$\zeta(3)$	$\zeta(4)$ (Tate control)	$\zeta(5)$
χ_{-4}	$\beta(2) = G$	$\beta(3)$ (Tate)	$\beta(4)$	$\beta(5)$ (Tate)
χ_{-3}	$L(2, \chi_{-3})$	critical/Tate	$L(4, \chi_{-3})$	critical/Tate
χ_5	parity-dependent control	$L(3, \chi_5)$	parity-dependent control	$L(5, \chi_5)$

The table is not meant as a claim that every entry has a useful Apéry realization. It is a test grid for the classification.

16.3 The controls are now part of the theorem

Section 7 proves the expected behavior. The modular period-annihilation and CM machinery survives on the even-zeta side, but the target component is already Tate. The universal central projector of Theorem 7.3 gives the explicit controls

$$\zeta(2) : 1 - 27V_2 + 202V_4 - 432V_8 + 256V_{16},$$

and

$$\begin{aligned} \zeta(4) : 1 - 455V_2 + 60494V_4 - 2504840V_8 \\ + 15486464V_{16} - 29818880V_{32} + 16777216V_{64}. \end{aligned}$$

The weight-six obstruction for $\zeta(4)$ also shows that a Tate target need not determine a unique preferred elliptic host: the central scalar realization moves to weight eight.

16.4 Regulator-side benchmark and next character tests

Three experiments are especially informative.

1. $L(2, \chi_{-3})$: now treated in Section 12. The exact CDT source pair is identified with the two Eisenstein orientations of the classification, providing an external arithmetic benchmark [16].
2. $L(3, \chi_5)$: now treated in Section 14. It has the same $\text{Sym}^2 \mathbb{H}$ homogeneous rank as $\zeta(3)$ but nontrivial character/fold geometry. Its conjugate branch constants split exactly as $\Lambda/2 \pm 2i\pi^3/(125\sqrt{5})$; after regulator centering, the nearest residual Hadamard monodromy is pure Tate, while an exact cubic determinant rules out complete scalar cancellation in the natural tensor span.
3. $\beta(4) = L(4, \chi_{-4})$: now treated in Section 13. It is the first small case combining nontrivial character orientation with a higher symmetric power Sym^3 , and it also exposes the necessity of finite direct image after quotient descent.
4. $L(4, \chi_{-3})$: Section 15 gives an independent signature-three Sym^3 test coming from Shvets's symmetric cube. Its canonical source is exactly $E_{5, \chi_{-3}, 1}$, purification produces a level-24 Fricke doublet with target $7L(4, \chi_{-3})/32$, and Shvets independently proves a mod- p^4 supercongruence for the homogeneous coefficients.

The completed $L(2, \chi_{-3})$, $L(3, \chi_5)$, $\beta(4)$, and Shvets-derived $L(4, \chi_{-3})$ benchmarks support the extension language at depths two, three, and four. The two depth-four character tests are complementary: $\beta(4)$ stresses direct-image rank and Galois trace, while $L(4, \chi_{-3})$ independently aligns Sym^3 geometry, depth-four Eichler arithmetic, a purified orientation doublet, and a

mod- p^4 homogeneous supercongruence. The level-100 Franke- η test further shows that the depth-three character square carries a nontrivial Layer-B filtration: scalar critical invisibility can disappear after diamond resolution and disappear completely after cusp-boundary refinement. In particular $\beta(4)$ shows that orientation should be treated as descent data on the actual direct image rather than as a universally independent rank factor. The $L(3, \chi_5)$ test adds a complementary realization principle: centering by a non-Tate regulator can leave a purely Tate branch defect without producing a scalar solution. The remaining tests should determine the correct Galois/deck orientation descent datum beyond real quadratic characters and whether the centered-tensor mechanism persists in genuinely complex-character sectors.

16.5 Each square now has an adelic Layer-C profile

The checkerboard organizes the motivic/regulator parity, but the Diophantine quality of an entry is a second, independent datum. Once a Layer-B extension has been fixed, a useful realization should be assigned an adelic profile

$$(\text{complex continuation leaf, holonomy rank } m, \text{free-depth dimensions } u_1, \dots, u_r, \{\Phi_p, \kappa_p\}).$$

Thus two entries with similar symmetric-power geometry can have very different arithmetic merit, and two realizations of the same entry can differ substantially. The Cooper weight drop is the first exact example: its depth-three geometry carries only depth-two denominator cost. The $L(2, \chi_{-3})$ theorem shows that a larger vector holonomy space can convert such arithmetic structure into an irrationality statement. The $\zeta(3)$ isogeny ladder, centered $L(3, \chi_5)$ system, Shvets symmetric cube, and Catalan direct images are therefore not only geometric tests of the checkerboard; they are finite experiments in whether isogeny, orientation, and Frobenius-free rank can be optimized simultaneously.

17 Research programme and publication architecture

17.1 What is essentially linear and what remains geometric

The theory separates into four operational stages, although only three are foundational layers.

Stage I: critical-fibre and boundary linear algebra. Start from the full available Eisenstein critical-coordinate map. In Franke's nonprincipal small-weight sector its kernels and dimension counts are already algorithmically controlled by weak-function/Vandermonde methods [13]; our Hecke projectors give explicit degeneracy-generated slices. Before interpreting a scalar kernel geometrically, resolve its diamond/Galois/normalizer sectors. Then add the principal-character boundary/regulator coordinates, impose the desired cusp data, and quotient by the surviving critical-invisible or target-zero translations. The level-100 worked example shows that these successive kernels can have dimensions $3 \rightarrow 1 \rightarrow 0$. The proposed boundary-completed Franke exact sequence, Problem 5.3, is the natural bridge between these two linear descriptions.

Stage II: representation geometry. Determine Fricke/Atkin-Lehner signs, Weyl folding, Galois orientation, and the orbit rank of the extension. This decides whether scalar descent is representation-theoretically possible.

Stage III: realization geometry. Choose the elliptic host, compute the companion-descent kernel, identify the exactness sublocus when present, and choose a quotient coordinate with controlled branch geometry.

Stage IV: adelic arithmetic. Determine the integral lattice and then treat the two arithmetic faces of the realization simultaneously. On the complex side, choose a common continuation leaf/pullback and compute the arithmetic-holonomy energy. At finite places, determine the Frobenius/free-depth flag, Lucas/Dwork laws, supercongruences, unit-root behavior, and exterior-power divisibility. The resulting Diophantine inequality is governed by the Frobenius-filtered penalty of Theorem 10.2, not merely by a single scalar denominator exponent.

The linear critical-kernel stage is already broadly understood in Franke’s nonprincipal range, while the boundary-completed regulator sector and its geometric organization are not. After that completion, the first two stages are close to a classification in several scalar Eisenstein sectors. The difficult frontier begins with host/companion geometry and arithmetic.

17.2 Next additions after Version 8

The Tate controls, the $L(2, \chi_{-3})$ arithmetic benchmark, $\beta(4)$, the analytic/geometric $L(3, \chi_5)$ application, the Franke boundary test, the adelic Layer-C synthesis, and the Shvets Belyi/symmetric-cube cross-pollination are now incorporated. The next decisive tests are:

1. compute the Frobenius/free-depth filtration of the Calegari–Dimitrov–Tang $L(2, \chi_{-3})$ holonomy basis, using a basis adapted to their successful denominator array; this is the natural calibration case because the final irrationality theorem is already known;
2. compute Lucas/Dwork/Frobenius fingerprints for the new $p = 5, 7, 11$ prime-isogeny $\zeta(3)$ systems and test the predicted almost-everywhere invariance of the homogeneous Sym^2 Frobenius data;
3. build the exact free-depth table for the centered $L(3, \chi_5)$ Hadamard/holonomy module, with the objective of maximizing $u_1^2 + u_2^2 + u_3^2$ rather than merely proving a uniform d_n^3 bound;
4. compute the corresponding filtration on the most promising Catalan direct-image/normalizer modules, optimizing the complex continuation leaf and the finite-place flag simultaneously;
5. formulate and prove the syntomic Dirichlet regulator identification of Problem 10.4;
6. prove the p -adic Hecke–Eisenstein extension functoriality of Problem 10.6, including the target factor $P_C(r)$ and integral-lattice normalizations;
7. prove or sharply delimit the boundary-completed Franke exact sequence of Problem 5.3, including its relation to Bernoulli/Eisenstein-symbol residues;
8. generalize the exact level-100 Franke– η kernel filtration beyond the three-oldform slice and determine when symmetry-resolved critical-null directions can survive the full boundary map;
9. complete the common-cover modular/isogeny audit of Shvets rows #4, #12, #14, and compute the Layer-B sources attached to the five elliptic survivors beyond the transported classes already identified;
10. compute the Frobenius/free-depth filtration of the new rational level-24 $L(4, \chi_{-3})$ Apéry pair and compare it with Shvets’s mod- p^4 homogeneous congruence;
11. continue the odd-zeta ladder to $\zeta(9)$ and test whether the isogeny/Frobenius pattern remains predictive;
12. complete the remaining geometric arithmetic frontier of $L(3, \chi_5)$ and the publication-level direct-image certificate for $\beta(4)$.

17.3 Suggested paper series

We recommend keeping the foundational theory separate from detailed application papers.

1. *Hecke–Eisenstein Extensions and Modular Apéry Systems*: the present foundational paper.
2. A $\zeta(3)$ application paper: finite cubic classification and prime-isogeny completions.
3. A Catalan application paper: character orientations, affine lifts, scalar rigidity, and the quintic vector system.
4. A higher odd-zeta paper: $\zeta(5)$ completion geometry, independent companions, $\zeta(7)$ hidden parent, and vector/Hermite–Padé extensions.
5. A character symmetric-power application paper beginning with $\beta(4)$, $L(3, \chi_5)$, and the Shvets-derived $L(4, \chi_{-3})$ system, then extending to higher beta values and genuinely complex characters.

This architecture keeps the explicit theorems independently valuable while allowing the general framework to evolve.

17.4 Diophantine merit is an adelic Layer-C invariant

The scalar estimate remains useful as a first approximation. If

$$L_n = B_n - CA_n,$$

the denominator-clearing lattice costs $\exp(\kappa n + o(n))$, and

$$|L_n| = \exp(-\lambda n + o(n)),$$

then a classical one-row Apéry contradiction requires $\lambda > \kappa$.

The vector-valued theory is sharper. Let V be an m -dimensional holonomy space with common pullback φ and a Frobenius/free-depth filtration giving penalty

$$\tau_F = \sum_j b_j \left(1 - \frac{u_j^2}{m^2} \right).$$

The natural quantitative invariant is

$$\boxed{\text{Score}_F = m(\log |\varphi'(0)| - \tau_F) - E(\varphi).}$$

Under the CDT hypotheses, positivity contradicts the existence of the corresponding rational m -tuple. Hence the search problem has two coupled axes:

$$\boxed{\begin{array}{l} \text{maximize complex continuation/capacity,} \\ \text{maximize Frobenius-free arithmetic rank.} \end{array}}$$

This is the correct adelic replacement for the earlier scalar “Apéry score.” A modular construction may be structurally perfect yet Diophantinely insufficient because either side of this inequality is poor; conversely a vector realization can win through a combination of extra complex branches and a lower Frobenius-filtered denominator penalty.

A Further proof details and useful formulas

A.1 The reflected Dirichlet target

Let $a \in \{0, 1\}$ satisfy $\chi(-1) = (-1)^a$ and assume $a \equiv r + 1 \pmod{2}$. The completed Dirichlet function

$$\Lambda(s, \chi) = \left(\frac{f}{\pi}\right)^{(s+a)/2} \Gamma\left(\frac{s+a}{2}\right) L(s, \chi)$$

satisfies the usual functional equation relating (s, χ) and $(1-s, \bar{\chi})$. At $s = 1-r$, the gamma factor has a simple pole and $L(s, \bar{\chi})$ has a simple trivial zero. Comparing the finite value with the completed value at r yields the explicit proportionality in Theorem 3.1. In the quadratic real cases the Gauss-sum factor simplifies to a rational multiple of a power of π .

A.2 The completion cube as a Koszul face diagram

The commuting operators $C_j = 1 - p_j^j V_{p_j}$ define a Koszul complex. We use only its face diagram, not a general exactness assertion. The useful point is combinatorial: every nuisance slot corresponds to one generator, and partial completions correspond canonically to subsets of slots. Thus the familiar $\zeta(3)$ prime ladder is the one-generator case of a higher-dimensional completion geometry.

A.3 Fricke reciprocal polynomials

For prime-power support $1, p, \dots, p^M$, the weight- k Fricke operation acts on the finite coefficient polynomial by a reciprocal involution of the shape

$$J_M P(X) = p^{Mk/2} X^M P(p^{-k}/X)$$

up to the fixed normalization of the raw Eisenstein basis. Applying this to the Gaussian-binomial full projector gives its forced sign. On the next degree, the purified space is $C^{\text{full}}(X)(A + BX)$, and diagonalizing the induced two-dimensional reciprocal action gives Theorem 4.7.

A.4 Integrality under mirror-map composition

Suppose $t(q) \in q\mathbb{Z}[[q]]$ and its inverse $q(t) \in t\mathbb{Z}[[t]]$. If a modular source has integral q -coefficients, then one application of D^{-1} introduces denominators dividing $d_n = \text{lcm}(1, \dots, n)$ in the t -coefficients. Iterating gives the generic bound

$$d_n^r B_n \in \mathbb{Z}$$

for a depth- r Eichler primitive after an integral companion multiplication. Special congruence/divisibility phenomena can make one or more integrations arithmetically free; the Cooper cubic systems furnish examples in the project calculations.

B A compact dictionary

Classical/modular language	Three-layer interpretation
Elliptic period	section of the resolved pure local system $\widetilde{\mathbb{H}}$ or $\text{Sym}^{r-1} \widetilde{\mathbb{H}}$
Marked cyclic/de Rham section	Layer-C choice of scalar/block presentation inside a fixed pure differential module

Finite quotient coordinate	direct image $f_*\tilde{V}$; deck/orientation action is encoded in this descended system
Algebraic pullback $y = R(x)$	functorial transport of Layers A/B; re-canonicalized exact source differs by the ramification factor $R'(x)$
Ramification defect	principal-part cost of declaring a ramified pullback coordinate to be newly exact
Eisenstein source Φ	de Rham representative of the extension $[\nu]$
Special value $L(r, \chi)$	distinguished regulator/connection period of $[\nu]$
Oldform coefficients	finite Hecke/isogeny correspondence acting on $[\nu]$
Period polynomial zeros	annihilation of unwanted Tate/endpoint components
Cusp corrections	translation inside the fixed-regulator affine fibre
Fricke/Weyl sign	representation-theoretic fingerprint of the extension/descent data
Companion modular form $\Phi = F D t$	functional whose pushout kills the extension cocycle
Recurrence	rigid exactness sublocus of companion descent
CMF coboundary	connection matrix expressed in a local coordinate
Denominator theorem	discrete gauge equivalence of realization data
Lucas/Dwork data	choice/properties of the arithmetic lattice Λ
	Frobenius/free-depth filtration, including supercongruence information
Irrationality inequality	complex holonomy/capacity versus Frobenius-filtered nonarchimedean height

References

- [1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$* , Astérisque **61** (1979), 11–13.
- [2] F. Beukers, *Irrationality proofs using modular forms*, Astérisque **147–148** (1987), 271–283.
- [3] D. Zagier, *Integral solutions of Apéry-like recurrence equations*, in *Groups and Symmetries: From Neolithic Scots to John McKay*, CRM Proc. Lecture Notes **47**, Amer. Math. Soc., 2009, 349–366.
- [4] Y. Yang, *Apéry limits and special values of L -functions*, J. Math. Anal. Appl. **343** (2008), 492–513; arXiv:0709.1968.
- [5] V. Golyshev, *Deresonating a Tate period*, arXiv:0908.1458.
- [6] V. Golyshev, M. Kerr and T. Sasaki, *Apéry extensions*, J. London Math. Soc.; arXiv:2009.14762.
- [7] A. Huber and G. Kings, *Dirichlet motives via modular curves*, Ann. Sci. École Norm. Sup. (4) **32** (1999), 313–345.
- [8] G. Kings, *Eisenstein classes, elliptic Soulé elements and the ℓ -adic elliptic polylogarithm*, London Math. Soc. Lecture Note Ser. 418, Cambridge Univ. Press, 2015; arXiv:1304.7161.
- [9] J. Sprang, *The algebraic de Rham realization of the elliptic polylogarithm via the Poincaré bundle*, Algebra Number Theory **14** (2020), no. 3, 545–585; arXiv:1802.04996.
- [10] N. Schappacher and A. J. Scholl, *The boundary of the Eisenstein symbol*, Math. Ann. **290** (1991), 303–321.
- [11] M. P. Young, *Explicit calculations with Eisenstein series*, J. Number Theory **199** (2019), 1–48; arXiv:1710.03624.

- [12] S. W. A. Shah, *On distribution relations of polylogarithmic Eisenstein classes*, arXiv:2309.10938.
- [13] J. Franke, *L-series of Eisenstein series vanishing at critical values*, arXiv:2401.08230v2, 2024.
- [14] R. S. Maier, *Algebraic hypergeometric transformations of modular origin*, arXiv:math/0501425.
- [15] S. Cooper, *Sporadic sequences, modular forms and new series for $1/\pi$* , Ramanujan J. **29** (2012), 163–183.
- [16] F. Calegari, V. Dimitrov and Y. Tang, *The linear independence of 1 , $\zeta(2)$, and $L(2, \chi_{-3})$* , arXiv:2408.15403.
- [17] F. Calegari, V. Dimitrov and Y. Tang, *Arithmetic holonomy bounds and effective Diophantine approximation*, arXiv:2510.04156, 2025.
- [18] K. Bannai and G. Kings, *p-adic elliptic polylogarithm, p-adic Eisenstein series and Katz measure*, Amer. J. Math. **132** (2010), no. 6, 1609–1654; arXiv:0707.3747.
- [19] J. Sprang, *The syntomic realization of the elliptic polylogarithm via the Poincaré bundle*, Doc. Math. **24** (2019), 1099–1134; arXiv:1802.04999.
- [20] F. Beukers and M. Vlasenko, *Frobenius structure and p-adic zeta values*, Adv. Math. **480** (2025), 110512; arXiv:2302.09603.
- [21] F. Calegari, *Irrationality of certain p-adic periods for small p*, Int. Math. Res. Not. 2005, no. 20, 1235–1249; arXiv:math/0408214.
- [22] S. Yamamoto, *Proof and generalization of conjectures of Ramanujan Machine*, arXiv:2403.09729.
- [23] A. Shvets, *Order-3 pi-formulas, Apéry-like kernels, and Clausen functoriality for Conservative Matrix Fields*, arXiv:2604.09723, 2026.
- [24] A. Shvets, *Order drop, Hecke descent, and a mod p^4 supercongruence for symmetric-cube hypergeometric coefficients*, arXiv:2604.06238, 2026.
- [25] A. Shvets, *The Domb Apéry-limit and a proof of the Ramanujan Machine conjecture Z2*, arXiv:2604.06239, 2026.
- [26] A. Shvets, *The level-8 Apéry-limit and a proof of the Ramanujan Machine conjecture Z1*, arXiv:2604.14219, 2026.
- [27] T. Raz, M. Shalyt, E. Leibtag, R. Kalisch, S. Weinbaum, Y. Hadad and I. Kaminer, *From Euler to AI: Unifying formulas for mathematical constants*, arXiv:2502.17533, 2026 version.
- [28] Research note of this programme, *Arithmetic holonomy + Frobenius: an adelic Layer-C synthesis*, 2026.
- [29] Working paper of this programme, *Eisenstein sources for the modular sporadic Apéry companions*, 2026.
- [30] Working paper of this programme, *Prime-isogeny completions of modular $\zeta(3)$ Apéry systems*, 2026.
- [31] Working paper of this programme, *Catalan Eisenstein lifts: Dirichlet regulators, elliptic polylogarithms, and orientation monodromy*, 2026.

- [32] Working paper of this programme, *The quintic Catalan isogeny: vector-valued Eisenstein normal functions and an explicit CMF coboundary*, 2026.
- [33] Research dossier of this programme, *$\zeta(7)$ level-12 parent and prime-isogeny completion ladder*, 2026.
- [34] Research dossier of this programme, *$\beta(4)$ as a Hecke–Eisenstein Apéry application: level 24, Sym^3 companion descent, and rank-16 direct image*, 2026.
- [35] Research dossier of this programme, *$L(3, \chi_5)$: elliptic folds, centered Hadamard/Tate normal form, and scalar no-go theorem*, 2026.
- [36] Research treatment of this programme, *Shvets’s 11 Belyi-pullback sequences through the three-layer classification*, 2026.